Vertical Structure of the Atmosphere

Complete lab companion: expanded derivations, plotting workflow and worked answers

Original questions first; every symbol, equation, plotted point and inference explained step by step, with live cross-references

How this edition is organised

The practical is taught here in the same order in which you encounter it on the worksheet: **Question 1 first, then Question 2, and so on through Question 14**. Each question begins with the wording and data of the supplied handout. Only after you have seen the problem do the notes open out the concepts that the problem requires. Definitions, units, mathematical notation, first-principles derivations, physical interpretation, worked numerical examples, graphs and model answers therefore appear *at the point of use*, rather than in a detached theory section at the front.

Reading rule

Whenever a reusable equation, table or figure is introduced, it is numbered. Later references are live hyperlinks. In a digital PDF reader, click a coloured reference such as 'Equation (2.14)', 'Table 6.2' or 'Figure 12.1' to return to the original definition, derivation, dataset or diagram; use the reader's Back command to return to your place.

Source and scientific treatment

The question wording, worksheet datasets and plotting tasks follow the supplied blank practical. Worked values are cross-checked against the supplied answer version. Where the answer sheet contains a transcription/sign inconsistency or a useful teaching simplification, this edition identifies it explicitly rather than silently reproducing it. In particular, the Florida data give $-75.6\,^{\circ}\mathrm{C}$ at 16,000 m even though one answer line prints $-75.4\,^{\circ}\mathrm{C}$; and the physical explanation of Question 13 is developed with local buoyancy and potential temperature rather than treating 'cold air sinks' as an absolute rule.

Prepared by Ahmad Ammar, Practical demonstrator. Built as a self-contained teaching companion to EGM305 Practical 1.

How to work through the practical

Read the blue Question box first. Immediately below it, a green **In plain language** box restates what the question actually wants you to find. Only then do we introduce the technical words, symbols and equations needed to answer it. The rule throughout is: *ordinary meaning first, technical notation second*. Do not memorise a formula before you can say in words what every symbol represents.

Every major relationship is built from a definition or a stated physical law, its units are checked, and numerical values are substituted explicitly. When a compact scientific phrase could be misunderstood—for example 'percentage above', 'density at a level', 'pressure falls', or 'cold air sinks'—the notes spell out exactly what object, amount, location and comparison are meant.

The document deliberately distinguishes **source material** from **expanded explanation**. The blue Question boxes reproduce the supplied practical; the surrounding teaching text explains, derives and contextualises it. Graph values read by eye can differ slightly from calculated values, so the notes show both the graphical method and the exact mathematical check where useful.

Four questions to ask whenever you meet an equation

1. What physical quantity does each symbol represent?
2. What are the units, and do both sides have the same dimensions?
3. Which quantity is changing, and with respect to what?
4. Is this equation a definition, an empirical approximation, a mathematical consequence of more basic laws, or a convention chosen for convenience?

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1 Question 1: the 5.6 km halving rule

Question 1

Unlike the oceans, the atmosphere is highly compressible. As a consequence of gravity, the molecules that make up the atmosphere are being pulled down to the Earth's surface. The result is that the density of the atmosphere decreases with altitude. The atmosphere thins out with increasing height until there are very few gas molecules left beyond 100 km.

A simple rule describing the decrease of atmospheric density with height is as follows: 'For every 5.6 km you go up, there is half the atmosphere above you as when you started.' Keeping this rule in mind, calculate and draw the percentage of atmosphere that is above each point (x-axis = Percentage of the Atmosphere above (%); y-axis = Height above sea level (km)).

Height (km)	% above	Pressure (hPa)
Sea level	100	1000
5.6	50	500
11.2	25	250
16.8	12.5	125
22.4	6.25	62.5
28.0	3.125	31.25
33.6	1.56	15.6
39.2	0.781	7.81
44.8	0.391	3.91
50.4	0.195	1.95

Table 1 of the handout. The pressure column is completed in Question 2.

What Question 1 means in ordinary words

At each stated height, imagine a horizontal level cutting through the atmosphere. **Percentage above** means: *what percentage of the mass of the entire atmosphere is still above that level?* It does not mean the local air is made of that percentage, and it does not mean the local density has that numerical percentage.

At sea level, essentially all atmospheric mass is above the level, so the value is 100 %. At 5.6 km, the practical's rule says half of the total atmospheric mass is still above you and half is below you, so the value is 50 %. At 11.2 km, only one quarter is still above you, so the value is 25 %. Question 1 asks you to continue that repeated halving and plot the result.

1.1 Build the ideas before doing the arithmetic

The handout gives the 5.6 km halving rule as the working rule for this question. Before using it, we need to make four ideas precise: what it means for the atmosphere to be thin, what *density* means, why a volume can be written as area times thickness, and why repeated halving produces a curved rather than linear graph. The complete mechanical derivation of the pressure law and the physical origin of the 5.6 km value are developed in Question 2, once pressure itself has been introduced.

Density versus percentage of atmosphere above

Do not confuse these two ideas.

Local density ρ: choose a small cube of space at one particular height, for example a cube 1 m wide, 1 m deep and 1 m high. Density tells you how many kilograms of air are inside that 1 m^3 cube. Near sea level, a typical value is about 1.2 kg/m^3; near 10 km, the same-size cube contains much less air mass.

Percentage above: now forget the small cube and imagine the entire atmospheric column from your chosen height all the way upward. This percentage tells you what fraction of the *total mass of the whole atmosphere* is still above that height. For example, 25 % above means one quarter of the atmosphere's total mass is above you and three quarters is below you.

Density is therefore a *local amount per unit volume*; percentage above is a *whole-column fraction*. They are related, but they are not the same quantity.

How thick is the atmosphere?

The atmosphere has no sharp top; it simply thins out with height. Roughly half of all the air lies below about 5.5 km, about 99 % lies below about 30 km, and very little remains above 100 km. A height of 100 km, known as the Kármán line, is often used as a conventional boundary between the atmosphere and space.

Compared with the size of the planet this is astonishingly thin. The Earth's radius is about 6,371 km, so 99 % of the air lies within less than one two-hundredth of the radius. If the Earth were shrunk to the size of a football (about 22 cm across), 99 % of its air would fit into a layer about half a millimetre thick. This is why vertical changes in the atmosphere are so rapid: the whole weather-producing part of the atmosphere is squeezed into a layer only a few kilometres deep.

The quantities and units you will use

Table 1.1 summarises the physical quantities that appear in the practical.

Table 1.1: Quantities used in this practical.

Quantity	Symbol	Units used here	Typical sea-level value	What it means
Height (altitude)	z	m or km	0	Vertical distance above mean sea level. 1 km = 1,000 m.
Pressure	p	hPa	1,013.25 hPa (standard); 1,000 hPa in this practical	How much force the air exerts on each unit area; in a hydrostatic atmosphere it also measures the weight of air above per unit area.
Temperature	T	°C (K in equations)	15 °C (standard)	How warm the air is. $T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$.
Density	ρ	kg/m^3	1.225 kg/m^3 (standard)	Mass of air contained in each cubic metre of space at that height.
Lapse rate	Γ	°C/km	6.5 °C/km (troposphere average)	How many degrees the temperature falls for each kilometre you climb.

Definition: pressure and the hectopascal

Pressure is *force per unit area*. Its SI unit is the pascal: 1 Pa is a force of one newton acting on one square metre. Meteorologists use the **hectopascal**, 1 hPa = 100 Pa. One hectopascal is exactly one **millibar** (mb), the older unit that you will still see on many barometers and weather charts, so the two can be used interchangeably.

Standard sea-level pressure is 1,013.25 hPa. On any particular day, sea-level pressure in the British Isles usually lies somewhere between about 970 and 1,040 hPa. The handout rounds it to 1,000 hPa to keep the arithmetic simple, and we will do the same.

Key idea: kelvin versus degrees Celsius

A *change* of 1 °C is exactly the same size as a change of 1 K, so a lapse rate of 6.5 °C/km is identical to 6.5 K/km. However, in any equation that contains the temperature T on its own (such as the gas law in Section 2.2), T must be in kelvin. For example, -56.5 °C is $-56.5 + 273.15 = 216.65$ K.

Geometry of the atmospheric slab: why $V = A\Delta z$?

The hydrostatic derivation imagines a very thin box-shaped slab of air. Let its horizontal cross-sectional area be A and its vertical thickness be Δz.

For a rectangular box,

$$V = (\text{length})(\text{width})(\text{height}). \tag{1.1}$$

But

$$A = (\text{length})(\text{width}), \tag{1.2}$$

so the volume can be written as

$$\boxed{V = A\,\Delta z}. \tag{1.3}$$

This is not a special atmospheric formula. It is the ordinary geometry of a prism: *base area times perpendicular height*. The units confirm it:

$$[V] = \mathrm{m}^2 \times \mathrm{m} = \mathrm{m}^3. \tag{1.4}$$

For example, a slab with $A = 2\,\mathrm{m}^2$ and $\Delta z = 0.5\,\mathrm{m}$ has

$$V = (2)(0.5) = 1\,\mathrm{m}^3. \tag{1.5}$$

Density: from volume to mass

Density answers a very literal question: *how much mass is packed into a given volume of space?* For air, we normally express this as kilograms of air per cubic metre of space. Density is defined as mass per unit volume:

$$\boxed{\rho = \frac{m}{V}}. \tag{1.6}$$

For example, if $5\,\mathrm{m}^3$ of air has a total mass of $6\,\mathrm{kg}$, then its density is $6/5 = 1.2\,\mathrm{kg/m}^3$. Conversely, if the density is $1.2\,\mathrm{kg/m}^3$, every $1\,\mathrm{m}^3$ of that air contains $1.2\,\mathrm{kg}$ of mass, so $5\,\mathrm{m}^3$ contains $6\,\mathrm{kg}$. Algebraically, multiply both sides of Equation (1.6) by V to obtain

$$\boxed{m = \rho V}. \tag{1.7}$$

For the atmospheric slab, substitute the volume from Equation (1.3) into Equation (1.7):

$$\boxed{m = \rho A \Delta z}. \tag{1.8}$$

The dimensions show that the expression really is a mass:

$$\frac{\mathrm{kg}}{\mathrm{m}^3} \times \mathrm{m}^2 \times \mathrm{m} = \mathrm{kg}. \tag{1.9}$$

This is the origin of the label $\rho A \Delta z$ on the slab diagram later in the notes.

1.2 The exponential atmosphere and the 5.6 km halving rule

The halving rule

The handout describes the decrease of the atmosphere with height using a simple rule:

> *'For every* 5.6 km *you go up, there is half the atmosphere above you as when you started.'*

Notice that the rule does *not* say that a fixed number of kilograms disappears every 5.6 km. It says that when you climb another 5.6 km, the atmospheric mass that is still above your new height is half the mass that was above your old height. Start at sea level with 100 % of the atmosphere above you. After 5.6 km you have climbed through half of it, and 50 % remains above. The next 5.6 km removes half of that 50 %, leaving 25 %; the next leaves 12.5 %, and so on. The amount above you keeps halving but never quite reaches zero.

If you have met radioactive decay, this will feel familiar: 5.6 km plays exactly the role of a *half-life*, except that the thing being halved is the amount of air above you and the 'clock' is height rather than time. We can call 5.6 km the **half-height** of the atmosphere.

Written as a formula, if z is the height in kilometres, the number of halvings you have gone through is $n = z/5.6$, and

$$\text{fraction of the atmosphere above } z = \left(\tfrac{1}{2}\right)^{n} = \left(\tfrac{1}{2}\right)^{z/5.6}, \tag{1.10}$$

$$\text{percentage above} = 100 \times \left(\tfrac{1}{2}\right)^{z/5.6}, \qquad p(z) = 1{,}000\,\text{hPa} \times \left(\tfrac{1}{2}\right)^{z/5.6}. \tag{1.11}$$

The number of halvings, n, does not have to be a whole number. At 2.8 km you have gone through half a halving, and $\left(\tfrac{1}{2}\right)^{0.5} = 0.707$, so about 70.7 % of the atmosphere is above you.

Why is there multiplication, and where exactly does the formula for P come from?

The handout says that after each 5.6 km rise you keep *half of whatever amount was still above you before*. That wording is a multiplication rule, not a subtraction rule.

At sea level, call the percentage above P_0:

$$P_0 = 100. \tag{1.12}$$

After one half-height,

$$P_1 = P_0 \times \frac{1}{2} = 100 \times \frac{1}{2} = 50. \tag{1.13}$$

After a second half-height, we halve the *new* amount:

$$P_2 = P_1 \times \frac{1}{2} = 100 \times \frac{1}{2} \times \frac{1}{2} = 100 \left(\frac{1}{2}\right)^{2} = 25. \tag{1.14}$$

After n equal half-height steps, the same factor has been multiplied n times:

$$\boxed{P_n = 100 \left(\frac{1}{2}\right)^{n}.} \tag{1.15}$$

The number of completed 5.6 km steps at a general height z is

$$\boxed{n = \frac{z}{5.6}.} \tag{1.16}$$

Substituting Equation (1.16) into Equation (1.15) gives

$$\boxed{P(z) = 100 \left(\frac{1}{2}\right)^{z/5.6}.} \tag{1.17}$$

This is the formula used in Question 3. The exponent $z/5.6$ literally means 'how many atmospheric halvings have occurred by this height?'

Why not subtract 50 percentage points each time? Because that would give $100 \to 50 \to 0$ and would falsely make the atmosphere end after 11.2 km. The rule says *half of what remains*, so the amount removed becomes smaller at every step.

Method: using the halving rule with a calculator

1. Write the height in **kilometres** (divide metres by 1000).
2. Work out the number of halvings, $n = z/5.6$.
3. Work out the fraction remaining, 0.5^n. On most scientific calculators, type `0.5`, then the power key (x^y, y^x or `^`), then n. An equivalent form is 2^{-n}.
4. Multiply by 100 to get the percentage of the atmosphere above you, or by 1,000 hPa to get the pressure.

Worked example: Slieve Donard, 850 m

Slieve Donard in the Mourne Mountains is the highest point in Northern Ireland, at 850 m. What fraction of the atmosphere is above its summit, and what is the pressure there?

1. Height in km: $z = 850/1000 = 0.85$ km.
2. Number of halvings: $n = 0.85/5.6 = 0.1518$.
3. Fraction remaining: $0.5^{0.1518} = 0.900$.
4. Percentage above $= 100 \times 0.900 = 90.0\,\%$; pressure $= 1000 \times 0.900 = 900$ hPa.

So a walk up Slieve Donard takes you above one tenth of the entire mass of the atmosphere.

The rule can also be run backwards to find the height at which a given percentage, P, of the atmosphere remains above you:

$$z = 5.6 \times \log_2\left(\frac{100}{P}\right) = 5.6 \times \frac{\ln(100/P)}{\ln 2} \quad \text{km}. \tag{1.18}$$

Deriving the inverse formula: why logarithms appear

Suppose the percentage P is known and the height z is unknown. Begin with Equation (1.17):

$$P = 100\left(\frac{1}{2}\right)^{z/5.6}. \tag{1.19}$$

Divide by 100:

$$\frac{P}{100} = \left(\frac{1}{2}\right)^{z/5.6} = 2^{-z/5.6}. \tag{1.20}$$

A logarithm is useful because it brings an unknown exponent down in front. Taking $\log_2$ of both sides,

$$\log_2\left(\frac{P}{100}\right) = -\frac{z}{5.6}. \tag{1.21}$$

Multiply by -5.6:

$$z = -5.6\log_2\left(\frac{P}{100}\right). \tag{1.22}$$

Using $-\log_2(a) = \log_2(1/a)$,

$$\boxed{z = 5.6\log_2\left(\frac{100}{P}\right).} \tag{1.23}$$

If your calculator has only ln or $\log_{10}$, use the change-of-base identity

$$\log_2 x = \frac{\ln x}{\ln 2}, \tag{1.24}$$

which turns Equation (1.23) into Equation (1.18). Physically, $\log_2(100/P)$ is simply the number of halvings needed to get from 100 % down to P; multiplying by 5.6 km/halving converts that number of halvings into a height.

For example, 10 % of the atmosphere is above you at $z = 5.6 \times \ln(10)/\ln(2) = 5.6 \times 3.32 = 18.6$ km, and 1 % remains above 37.2 km.

Where this equation comes from

Where it comes from. Unlike the hydrostatic equation and the gas law, this rule is not a fundamental physical law at all — it is a rounded, teachable approximation to the true, exact solution of those two equations together, worked out in full in the 'Going further' box just below. In short: combining Newton's second law (hydrostatic balance) with the gas law shows mathematically that pressure *must* decay exponentially with height (proportional to $\left(\frac{1}{2}\right)^{z/H}$ for some constant H); the number 5.6 km is then chosen, for round, easy classroom arithmetic, to be close to the true value of that constant in the lowest few kilometres of the real atmosphere.

What each symbol means.

Symbol	Unit (here)	Meaning
z	km	Height above sea level.
$n = z/5.6$	dimensionless	Number of 5.6 km 'halvings' completed; need not be a whole number.
$\left(\frac{1}{2}\right)^n$	dimensionless	Fraction of the atmosphere remaining above you; multiply by 100 for a percentage.
$p(z)$	hPa	Pressure at height z.
P (Equation (1.18))	%	A chosen percentage of the atmosphere, used to solve the rule *backwards* for the height z at which that percentage remains above you.
$\log_2$, ln	—	Logarithm to base 2, and the natural logarithm (base e); either can be used, since $\log_2 x = \ln x / \ln 2$.

Where the number 5.6 comes from in this practical. The practical *gives* 5.6 km as a convenient approximate half-height. It is not a fundamental constant like g or R_d. The clean logical direction is therefore to start from the stated 5.6 km rule and ask what simple isothermal model would reproduce it. The derivation below gives $z_{1/2} = H \ln 2$, so setting $z_{1/2} = 5.6$ km gives $H = 8.08$ km. Using $H = R_d T/g$ then gives an *effective constant model temperature* of about 276 K, or 3 °C. That 3 °C is therefore a fitted parameter of the simplified model; it is not the actual temperature at 5.6 km, nor a claim that the real lower atmosphere is everywhere 3 °C. Section 2.2 compares the teaching rule with real and standard-atmosphere profiles.

What the question is asking

You are asked to do two things. First, fill in the first two columns of handout Table 1: the heights go up in steps of 5.6 km, and at each step the percentage of the atmosphere above you is half of the value at the step below. Second, draw a graph of the result, with the percentage of the atmosphere above on the horizontal (x) axis and height on the vertical (y) axis. The arithmetic rule is developed immediately below; its mechanical origin is derived in Section 2.2 within Question 2.

Method, step by step

1. **Heights.** Start at sea level (0 km) and keep adding 5.6 km: 0, 5.6, 11.2, 16.8, … handout Table 1 has room for ten rows, so the last height is $9 \times 5.6 = 50.4$ km.
2. **Percentages.** Start at 100 % and halve the previous value for every new row: 100, 50, 25, 12.5, … Equivalently, in row n (counting sea level as $n = 0$) the value is $100 \times \left(\frac{1}{2}\right)^n$.
3. **Axes.** Draw the x-axis from 0 to 100 % and the y-axis from 0 to 60 km. Choose round-number scales that use most of the grid, for example 20 % per large square across and 10 km per large square up. Label each axis with the quantity *and* its unit.
4. **Points.** Plot the ten points as small, neat crosses or dots.
5. **Curve.** Draw *one smooth curve* through the points, freehand, without a ruler. It should bend gradually, with no sharp corners.
6. **Leave room at the top** of the graph for the second x-axis that you will add in Question 2.

Worked solution

Heights. Each height is the previous one plus 5.6 km, or $5.6 \times n$ for step n:

$$0,\quad 5.6,\quad 5.6 \times 2 = 11.2,\quad 5.6 \times 3 = 16.8,\quad 22.4,\quad 28.0,\quad 33.6,\quad 39.2,\quad 44.8,\quad 50.4 \text{ km}. \tag{1.25}$$

Percentages. Each value is half of the one before:

$$100 \xrightarrow{\times\frac{1}{2}} 50 \xrightarrow{\times\frac{1}{2}} 25 \xrightarrow{\times\frac{1}{2}} 12.5 \xrightarrow{\times\frac{1}{2}} 6.25 \xrightarrow{\times\frac{1}{2}} 3.125 \tag{1.26}$$

$$\xrightarrow{\times\frac{1}{2}} 1.5625 \xrightarrow{\times\frac{1}{2}} 0.78125 \xrightarrow{\times\frac{1}{2}} 0.390625 \xrightarrow{\times\frac{1}{2}} 0.1953125 \tag{1.27}$$

Rounded to three significant figures these are the values shown in red in handout Table 1 above. Table 1.2 sets out the same working in the general form $100 \times (\frac{1}{2})^n$, and Figure 1.1 shows the completed graph.

Table 1.2: Working for handout Table 1. Step n is the number of 5.6 km halvings completed.

Step n	Height $5.6n$ (km)	$(\frac{1}{2})^n$	% above	% below
0	0.0	1	100	0
1	5.6	0.5	50	50
2	11.2	0.25	25	75
3	16.8	0.125	12.5	87.5
4	22.4	0.0625	6.25	93.75
5	28.0	0.03125	3.125	96.9
6	33.6	0.015625	1.56	98.4
7	39.2	0.0078125	0.781	99.2
8	44.8	0.00390625	0.391	99.6
9	50.4	0.001953125	0.195	99.8

What the graph shows. Starting from 100 % at sea level, the curve sweeps up and to the left: the percentage above falls quickly at first and then ever more slowly. Above about 30 km the curve runs almost parallel to the height axis, creeping towards 0 % but never reaching it. The '% below' column makes the same point the other way round: half of the atmosphere lies below 5.6 km, three quarters below 11.2 km, and 99.8 % below 50.4 km. The atmosphere is overwhelmingly concentrated close to the ground.

Common pitfalls

- **Subtracting instead of halving.** Taking 50 % away at each step gives 100, 50, 0, which would mean the atmosphere ends at 11.2 km. The rule halves *whatever is left*, so the sequence is 100, 50, 25, 12.5, …
- **Swapping the axes.** The handout asks for the percentage on the x-axis and height on the y-axis.
- **Joining the dots with straight lines.** The rule describes a smooth, continuously curving relationship. A dot-to-dot line with corners gives wrong readings in Question 3.
- **Worrying that the top points sit almost on the axis.** Values of 0.4 % or 0.2 % are genuinely tiny on a scale running to 100 %. That is the point of the graph, not a mistake.
- **Axis labels without units.** Every axis needs a quantity and a unit, for example 'Height above sea level (km)'.

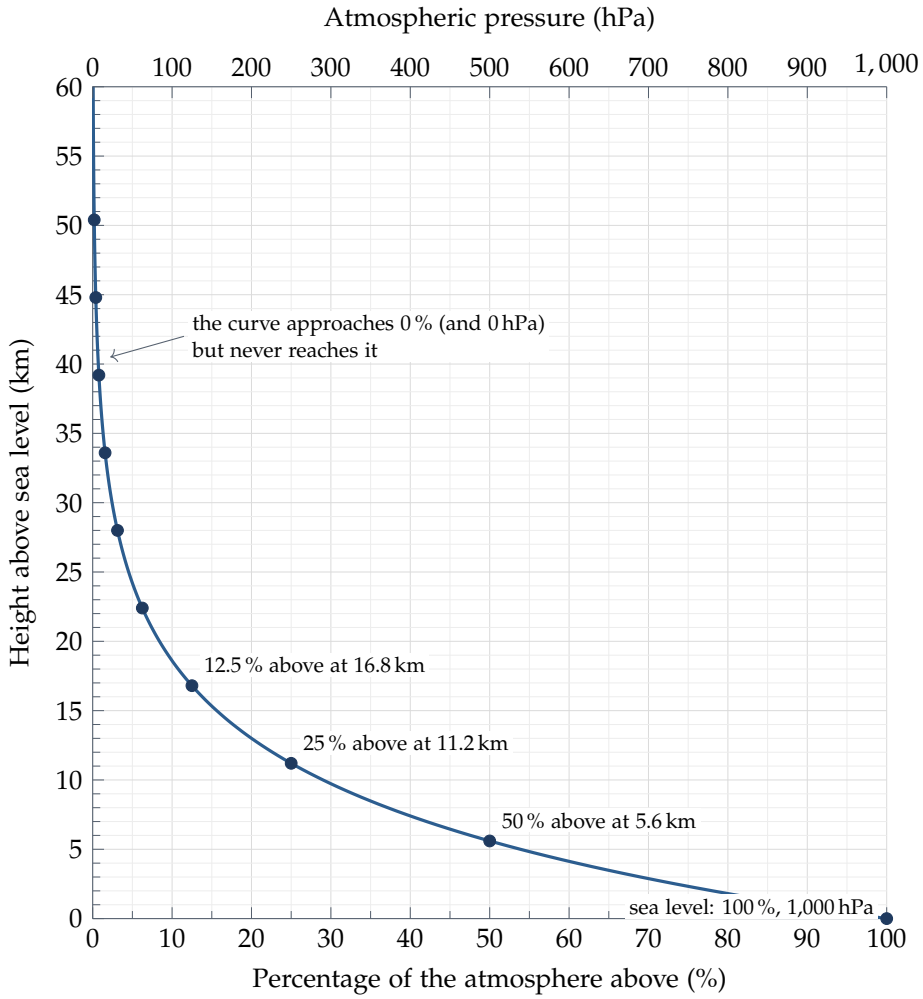


Figure 1.1: Completed graph for Questions 1 and 2. The dots are the ten values of handout Table 1 and the curve is the halving rule, $100 \times (\frac{1}{2})^{z/5.6}$. The top axis (Question 2) shows the corresponding pressure: because pressure is proportional to the amount of air above, one curve serves for both axes.

2 Question 2: adding a pressure axis

Question 2

There is a direct relationship between height and atmospheric pressure. As the density of the atmosphere drops by 50 % when we go up 5.6 km, the pressure also drops by 50 %. At sea level, the pressure is approximately 1,000 hPa. Add the pressure for each of your heights in the 3rd column of handout Table 1 and add a second x-axis to your graph to show atmospheric pressure.

What Question 2 means in ordinary words

Question 1 expressed the amount of atmosphere above each height as a percentage. Question 2 asks you to express the *same whole-column information* as pressure. If only half as much atmospheric mass is above a level, its weight per square metre is half as large, so the pressure is approximately half as large. Thus 100 % above corresponds to about 1,000 hPa, 50 % to 500 hPa, 25 % to 250 hPa, and so on. The rest of this section explains why that works rather than asking you to accept it as a trick.

2.1 Why pressure falls with height, from definitions to calculus

Question 2 looks simple—halve 1,000 hPa whenever the percentage above halves—but that proportionality contains most of the physics of the vertical atmosphere. We therefore build it from first principles: force, weight and pressure; finite changes; the derivative; integration; hydrostatic balance; compressibility and the ideal gas law; and finally the exponential barometric law. Every symbol is defined when it first enters, and every dimensional cancellation is shown.

Force, weight, pressure and algebraic rearrangement

Newton's second law relates net force, mass and acceleration:

$$\boxed{F = ma}. \tag{2.1}$$

Near the Earth's surface, gravity gives a freely falling object an acceleration of magnitude g. The gravitational force on a mass m—its **weight**—is therefore

$$\boxed{W = mg}. \tag{2.2}$$

The SI unit of force is the newton:

$$1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}. \tag{2.3}$$

Pressure answers: *how much force is acting on each unit of area?* It is defined as force per unit area:

$$\boxed{p = \frac{F}{A}}. \tag{2.4}$$

Multiplying both sides of Equation (2.4) by A gives the equivalent pressure-force relation

$$\boxed{F = pA}. \tag{2.5}$$

This is why the pressure force on the bottom of an atmospheric slab is written $p(z)A$: $p(z)$ is the pressure there, and multiplying by the face area converts pressure into force.

The pascal is defined directly from this relationship:

$$1\,\mathrm{Pa} = 1\,\mathrm{N/m^2}. \tag{2.6}$$

Thus

$$(\mathrm{Pa})(\mathrm{m^2}) = \left(\frac{\mathrm{N}}{\mathrm{m^2}}\right)(\mathrm{m^2}) = \mathrm{N}, \tag{2.7}$$

as required for a force.

Worked example: deriving the mass of the atmosphere above one square metre

At the ground, imagine a horizontal patch of area A. Above that patch is a tall column of air extending to the top of the atmosphere. If the atmosphere is approximately hydrostatic, the pressure force associated with that patch balances the gravitational weight of all the air in the column above it:

$$p_0 A = mg. \tag{2.8}$$

We want m, so divide both sides by g:

$$\boxed{m = \frac{p_0 A}{g}}. \tag{2.9}$$

For $p_0 = 1{,}000\,\mathrm{hPa} = 100{,}000\,\mathrm{Pa}$ and $A = 1\,\mathrm{m^2}$,

$$\begin{aligned} m &= \frac{(100{,}000\,\mathrm{Pa})(1\,\mathrm{m^2})}{9.81\,\mathrm{m/s^2}} \\ &= \frac{100{,}000\,\mathrm{N}}{9.81\,\mathrm{m/s^2}} \\ &= \frac{100{,}000\,\mathrm{kg\,m/s^2}}{9.81\,\mathrm{m/s^2}} \\ &\approx 10{,}200\,\mathrm{kg}. \end{aligned} \tag{2.10}$$

The cancellation of $\mathrm{m/s^2}$ leaves kilograms. This is a useful habit: units are an independent check on the algebra.

What does the capital delta Δ mean?

The Greek capital letter delta, Δ, means a **finite change in a quantity**. The rule is always

$$\boxed{\Delta(\text{quantity}) = (\text{final value}) - (\text{initial value}).} \tag{2.11}$$

If we move from height z to height $z + \Delta z$, then

$$\Delta z = (z + \Delta z) - z = \Delta z. \tag{2.12}$$

That looks circular only because Δz was chosen as the name of the height increment in the first place.

For pressure, Equation (2.11) becomes

$$\boxed{\Delta p = p_{\text{final}} - p_{\text{initial}}.} \tag{2.13}$$

Moving upward from z to $z + \Delta z$ gives the specific finite-change relation

$$\boxed{\Delta p = p(z + \Delta z) - p(z)}. \tag{2.14}$$

If pressure decreases upward, then $p(z + \Delta z) < p(z)$ and therefore $\Delta p < 0$. A negative *change* does not mean pressure itself is negative; it means the final pressure is smaller than the initial pressure.

Average rate of change: $\Delta y / \Delta x$

A rate of change compares how much an output changes with how much its input changes. For a function $y = f(x)$, move from input x to input $x + h$. The input change is

$$(x + h) - x = h. \tag{2.15}$$

The output changes from $f(x)$ to $f(x + h)$, so the output change is

$$f(x + h) - f(x). \tag{2.16}$$

The average rate of change is therefore

$$\boxed{\frac{f(x + h) - f(x)}{h}}. \tag{2.17}$$

The numerator and denominator are different quantities: h is the *input change*; $f(x + h) - f(x)$ is the *resulting output change*. On a graph these are the horizontal *run* and vertical *rise*.

For atmospheric pressure, substitute Equation (2.14) into the generic rate-of-change structure in Equation (2.17):

$$\boxed{\frac{\Delta p}{\Delta z} = \frac{p(z + \Delta z) - p(z)}{\Delta z}}. \tag{2.18}$$

Equation (2.18) is the average pressure change per metre across a finite layer.

Why the derivative is defined by a limit

A finite difference such as $\Delta p / \Delta z$ gives an *average* gradient across a layer. Atmospheric density, temperature and pressure vary continuously, so we often want the gradient at one exact height. That is the problem solved by the **derivative**.

Start with an ordinary function $f(x)$. Between two distinct points x and $x + h$, the slope of the secant line is

$$\frac{f(x + h) - f(x)}{h}. \tag{2.19}$$

Now make the horizontal separation h smaller:

$$h = 1, \quad 0.1, \quad 0.01, \quad 0.001, \ldots \tag{2.20}$$

The second point approaches the first, and the secant slope approaches the slope of the tangent at x. We define that limiting slope to be the derivative:

$$\boxed{f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}}. \tag{2.21}$$

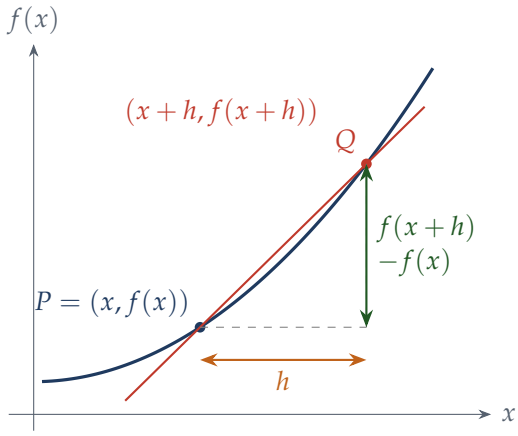


Figure 2.1: The derivative is the limiting slope obtained by bringing the second point Q arbitrarily close to the first point P. The horizontal change h and vertical change $f(x+h) - f(x)$ are not the same thing.

The notation $h \to 0$ means 'h approaches zero'. It does *not* mean that we substitute $h = 0$ into the fraction. Doing that would produce $0/0$, which is undefined. A limit asks instead: *what value does the fraction approach for non-zero values of h that can be made arbitrarily small?*

Worked derivative from first principles: $f(x) = x^2$

Use the definition rather than a memorised differentiation rule:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\ &= \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h\to 0} \frac{h(2x+h)}{h}. \end{aligned} \tag{2.22}$$

For every non-zero h we may cancel h:

$$f'(x) = \lim_{h\to 0}(2x+h) = 2x. \tag{2.23}$$

Thus

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x}(x^2) = 2x.} \tag{2.24}$$

At $x = 2$ the instantaneous slope is $2(2) = 4$.

For pressure as a function of height, make the substitutions

$$f \longrightarrow p, \qquad x \longrightarrow z, \qquad h \longrightarrow \Delta z. \tag{2.25}$$

Equation (2.21) then becomes

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}z} = \lim_{\Delta z\to 0} \frac{p(z+\Delta z) - p(z)}{\Delta z}.} \tag{2.26}$$

This is not yet the hydrostatic equation. It is simply the *definition* of the derivative of pressure with respect to height. Hydrostatic physics will tell us what this derivative equals.

What does the lowercase d mean?

The symbols $\mathrm{d}p$ and $\mathrm{d}z$ are called **differentials**. In an introductory interpretation, they represent infinitesimal changes associated with pressure and height. The notation

$$\frac{\mathrm{d}p}{\mathrm{d}z} \tag{2.27}$$

means the derivative of p with respect to z—the limiting local rate of change defined in Equation (2.26).

It is useful, but slightly informal, to say that $\mathrm{d}z$ is a 'tiny height change'. Formally, the derivative is defined by the limit of finite differences; we do not need to imagine a mysterious smallest possible number. Once derivatives have been defined, differential notation becomes extremely useful because equations such as

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g \tag{2.28}$$

can be written as

$$\mathrm{d}p = -\rho g \, \mathrm{d}z, \tag{2.29}$$

which is the natural form to integrate.

Integration: adding infinitely many small contributions

Differentiation asks, 'what is the local rate of change?' Integration asks the inverse kind of question: 'what total change results when all of those local changes are accumulated?'

Suppose a quantity $q(z)$ varies with height and we want to add its contribution across a layer. Divide the layer into many thin slices of width Δz. A first approximation to the total is a sum

$$q(z_1)\Delta z + q(z_2)\Delta z + q(z_3)\Delta z + \cdots . \tag{2.30}$$

As the slices become thinner and their number increases without bound, this limiting sum becomes a **definite integral**:

$$\boxed{\int_{z_1}^{z_2} q(z)\,\mathrm{d}z.} \tag{2.31}$$

The elongated symbol $\int$ grew historically from the idea of a sum. The $\mathrm{d}z$ identifies the variable over which the accumulation is being performed, and the limits z_1 and z_2 identify the start and end of the interval.

The **Fundamental Theorem of Calculus** connects differentiation and integration. If $F'(z) = q(z)$, then

$$\boxed{\int_a^b q(z)\,\mathrm{d}z = F(b) - F(a).} \tag{2.32}$$

In words: integrating a rate of change reconstructs the accumulated change. This is why, once hydrostatic balance gives us $\mathrm{d}p/\mathrm{d}z$, we can integrate it to recover how pressure itself varies with height.

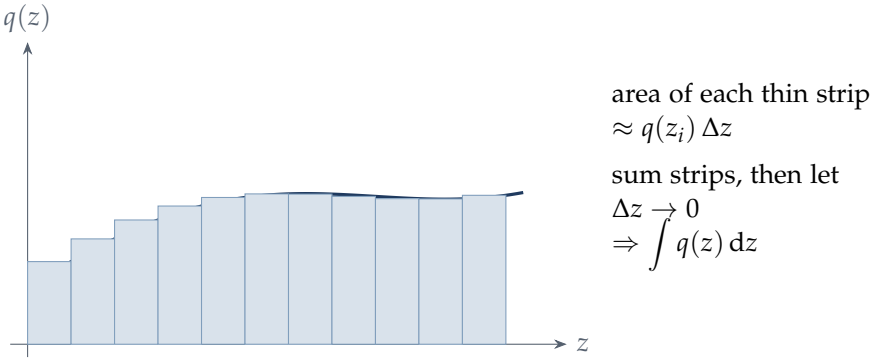


Figure 2.2: An integral as accumulation. A finite sum of thin rectangles approximates the area under a curve; the definite integral is the limit as the rectangles become arbitrarily thin. The same mathematics accumulates atmospheric mass or pressure changes with height.

Exponentials and logarithms: why they appear in atmospheric pressure

An **exponential** function has the variable in the exponent, for example

$$y = e^{kx}. \tag{2.33}$$

The special number

$$e \approx 2.71828 \tag{2.34}$$

is the natural base for calculus because

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x} e^{kx} = k e^{kx}.} \tag{2.35}$$

The derivative is therefore proportional to the function itself. This is why differential equations of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = ky \tag{2.36}$$

have exponential solutions. If $k < 0$, the quantity decays exponentially.

A **logarithm** reverses an exponential. The natural logarithm ln is the inverse of e^x:

$$\ln(e^x) = x, \qquad e^{\ln x} = x. \tag{2.37}$$

Two rules used later are

$$\ln\left(\frac{a}{b}\right) = \ln a - \ln b, \qquad \int \frac{1}{x}\,\mathrm{d}x = \ln|x| + C. \tag{2.38}$$

When pressure appears inside a logarithm, the physically meaningful expression is a dimensionless ratio such as $\ln(p/p_0)$, not the logarithm of a dimensional pressure by itself.

2.2 Why pressure and density fall with height

Pressure is the weight of the air above you

Imagine a vertical column of atmosphere standing on a horizontal patch of ground of area A. Gravity pulls every parcel in that column downward. Under ordinary hydrostatic conditions the ground and the surrounding atmosphere support that weight, so the pressure at the base is directly related to the mass of air above.

Start from the pressure definition in Equation (2.4). Its rearranged pressure-force form, Equation (2.5), gives the upward force associated with the surface pressure. The column's weight is given by Equation (2.2). Equating those two forces in vertical balance gives

$$p_0 A = mg. \tag{2.39}$$

Now isolate the column mass m. Divide both sides of Equation (2.39) by g:

$$\boxed{m = \frac{p_0 A}{g}.} \tag{2.40}$$

For the rounded sea-level pressure used in this practical,

$$p_0 = 1{,}000\,\text{hPa} = 100{,}000\,\text{Pa} = 100{,}000\,\text{N/m}^2, \tag{2.41}$$

and for $A = 1\,\text{m}^2$,

$$\begin{aligned} m &= \frac{p_0 A}{g} \\ &= \frac{(100{,}000\,\text{N/m}^2)(1\,\text{m}^2)}{9.81\,\text{m/s}^2} \\ &= \frac{100{,}000\,\text{N}}{9.81\,\text{m/s}^2} \\ &= \frac{100{,}000\,\text{kg m/s}^2}{9.81\,\text{m/s}^2} \\ &\approx 10{,}200\,\text{kg}. \end{aligned} \tag{2.42}$$

Thus the atmosphere above each square metre near sea level has a mass of roughly

$$\boxed{1.0 \times 10^4\,\text{kg} \approx \text{ten metric tonnes.}} \tag{2.43}$$

The familiar phrase 'pressure is the weight of the air above you' is therefore literal, provided we understand it as *weight per unit horizontal area in a hydrostatic atmosphere*.

Why are we not crushed by ten tonnes of air?

The ten-tonne figure is the mass in a tall column above $1\,\text{m}^2$; it is not a single downward force applied only to the top of a person. Fluid pressure acts on surfaces in all directions. The fluids and gases inside the body are also at nearly the surrounding pressure, so most external pressure is balanced internally. We mainly feel *pressure differences*, such as across the eardrum during a rapid ascent or descent.

Now climb a mountain. Some of the atmospheric column that used to be above you is now below you. The mass remaining above is smaller, so its weight per unit area—the pressure—is smaller.

What does it mean to say that a layer 'supports the weight of the air above'?

'Support' is only shorthand; air is not a rigid shelf. At any horizontal level, the pressure there represents the downward weight of the entire air column above that level per unit area. For a *thin slab*, the pressure just below the slab must be slightly larger than the pressure just above it. That small difference in pressure force is what balances the slab's own weight. In

symbols,

$$p_{\text{bottom}}A - p_{\text{top}}A = mg, \tag{2.44}$$

so

$$\boxed{p_{\text{bottom}} - p_{\text{top}} = \frac{mg}{A}.} \tag{2.45}$$

Stack many such slabs on top of one another and all those small pressure differences add up. The pressure at the bottom then equals the weight per unit area of all the slabs above. This is the precise meaning behind the phrase 'pressure decreases upward because lower layers have more air above them'.

Key idea

At height z, hydrostatic pressure is the weight per unit area of all the atmosphere above that level. If gravitational acceleration is treated as constant,

$$\boxed{\frac{p(z)}{p_0} \approx \frac{m_{\text{above}}(z)}{m_{\text{above}}(0)}}. \tag{2.46}$$

Therefore

$$\boxed{\text{percentage of atmosphere above height } z = \frac{p(z)}{p_0} \times 100\,\%.} \tag{2.47}$$

If half the atmospheric mass remains above you, the pressure is half the sea-level value. This is the logic behind the paired '% above' and 'pressure' columns in the practical.

The simple whole-column argument is useful, but it does not yet tell us *how rapidly* pressure changes from one height to the next. For that we analyse a thin slab of atmosphere and derive the hydrostatic differential equation.

Hydrostatic balance: deriving $\mathrm{d}p/\mathrm{d}z = -\rho g$ from first principles

The word **hydrostatic** literally refers to a fluid in mechanical balance. In atmospheric science it means that, to an excellent approximation over ordinary weather scales, the vertical pressure-gradient force balances gravity. Air can still move horizontally and can still have small vertical motions; hydrostatic balance says that the huge background vertical forces are very nearly in balance.

We will derive the equation slowly. Every symbol used below has already been introduced in the mathematical build-up within Question 2.

Step 1: choose a coordinate direction

Take vertical height z to increase upward. That sign convention matters:

$$\boxed{\text{upward} = +z, \qquad \text{downward} = -z.} \tag{2.48}$$

A force pointing upward will therefore be written as positive; a force pointing downward will be written as negative.

Step 2: imagine a thin slab of air

Let the slab have

$$\text{horizontal area} = A, \qquad \text{thickness} = \Delta z. \tag{2.49}$$

Its base is at height z, so its top is at

$$z + \Delta z. \tag{2.50}$$

Its volume is the prism relation already established in Equation (1.3). Combining it with the density definition, Equation (1.6), gives the slab mass in Equation (1.8). Using the weight relation Equation (2.2), the slab's downward weight is

$$W = (\rho A \Delta z) g = \rho g A \Delta z. \tag{2.51}$$

Step 3: convert the pressures on the two faces into forces

By the pressure definition Equation (2.4), force on a face is given by Equation (2.5). At the bottom face the pressure force is upward; at the top face it is downward. Together with the downward weight from Equation (2.51), the three signed forces are

$$F_{\text{bottom}} = +p(z)A, \qquad F_{\text{top}} = -p(z+\Delta z)A, \qquad F_g = -\rho g A \Delta z. \tag{2.52}$$

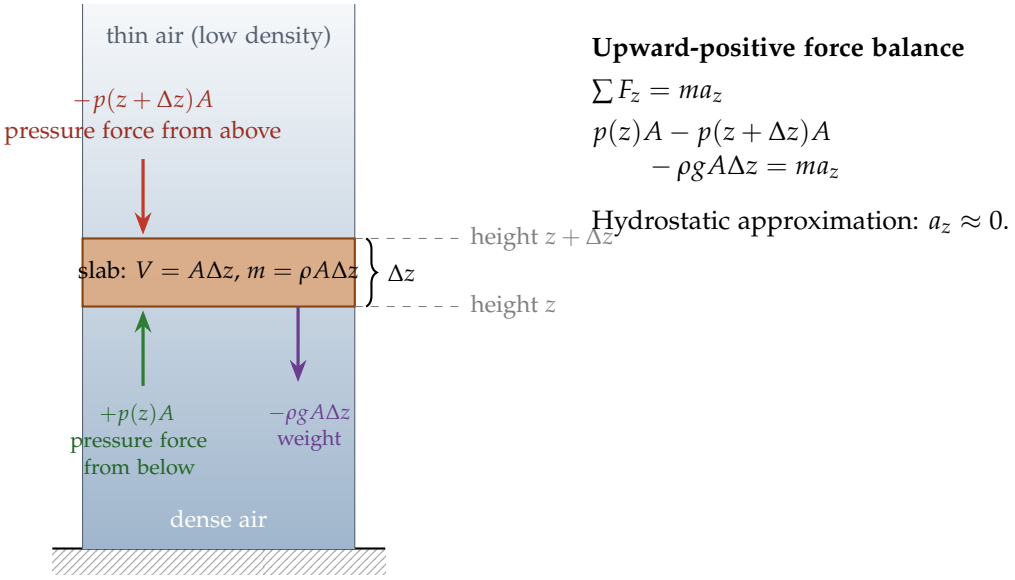


Figure 2.3: Forces on a thin slab of air. The lower pressure force is upward; the upper pressure force and gravity are downward. In hydrostatic balance the net vertical force is approximately zero.

Step 4: apply Newton's second law

Newton's second law was introduced in Equation (2.1). In the vertical direction it is $\sum F_z = ma_z$. Substituting the three forces from Equation (2.52):

$$p(z)A - p(z+\Delta z)A - \rho g A \Delta z = ma_z. \tag{2.53}$$

Under hydrostatic conditions the mean vertical acceleration is negligible compared with gravity, so set

$$a_z \approx 0. \tag{2.54}$$

Then

$$p(z)A - p(z+\Delta z)A - \rho g A \Delta z = 0. \tag{2.55}$$

This is already the physical statement of hydrostatic balance in finite-layer form.

Step 5: do the algebra one operation at a time

Every term in Equation (2.55) contains the same horizontal area A. Divide the entire equation by A:

$$p(z) - p(z+\Delta z) - \rho g \Delta z = 0. \tag{2.56}$$

Move the pressure terms so that 'top minus bottom' appears on the left:

$$\begin{aligned} p(z) - p(z+\Delta z) &= \rho g \Delta z, \\ -\big[p(z+\Delta z) - p(z)\big] &= \rho g \Delta z, \\ p(z+\Delta z) - p(z) &= -\rho g \Delta z. \end{aligned} \tag{2.57}$$

But the finite pressure change was defined in Equation (2.14). Therefore

$$\boxed{\Delta p = -\rho g \Delta z.} \tag{2.58}$$

This finite-layer equation says: *the pressure change across a thin layer equals minus the weight of that layer per unit horizontal area.*

Step 6: turn the finite difference into a local derivative

Divide Equation (2.58) by Δz:

$$\frac{\Delta p}{\Delta z} = -\rho g. \tag{2.59}$$

Using the explicit expression for the pressure change,

$$\frac{p(z+\Delta z) - p(z)}{\Delta z} = -\rho g. \tag{2.60}$$

The left side is the *average* pressure gradient across the finite slab. To obtain the gradient at one exact height, let the slab become arbitrarily thin:

$$\lim_{\Delta z \to 0} \frac{p(z+\Delta z) - p(z)}{\Delta z} = -\rho(z) g. \tag{2.61}$$

The derivative definition specialised to pressure, Equation (2.26), tells us that the entire limit on the left is $\mathrm{d}p/\mathrm{d}z$. Hence

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g.} \tag{2.62}$$

This is the **hydrostatic equation**.

The density is written more precisely as $\rho(z)$ because, as the slab shrinks onto the height z, the representative density of the slab becomes the *local* density at that height. In compact atmospheric notation we normally omit the explicit (z) and simply write ρ.

Physical meaning of the hydrostatic equation

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g} \tag{2.63}$$

means that the local vertical pressure gradient is exactly what is needed to support the local air against gravity.

- $\mathrm{d}p/\mathrm{d}z < 0$: pressure decreases upward.
- Larger ρ: more mass in each metre-thick layer, so pressure must fall more rapidly with height.
- Smaller ρ: less mass in each metre-thick layer, so pressure falls more slowly with height.

Step 7: check the units

The left side has units

$$\left[\frac{\mathrm{d}p}{\mathrm{d}z}\right] = \frac{\mathrm{Pa}}{\mathrm{m}}. \tag{2.64}$$

The right side has

$$\begin{aligned}[\rho g] &= \frac{\mathrm{kg}}{\mathrm{m}^3}\frac{\mathrm{m}}{\mathrm{s}^2} \\ &= \frac{\mathrm{kg}}{\mathrm{m}^2\,\mathrm{s}^2}.\end{aligned} \tag{2.65}$$

Since

$$1\,\mathrm{Pa} = \frac{1\,\mathrm{N}}{1\,\mathrm{m}^2} = \frac{\mathrm{kg\,m/s}^2}{\mathrm{m}^2} = \frac{\mathrm{kg}}{\mathrm{m\,s}^2}, \tag{2.66}$$

we have

$$\frac{\mathrm{Pa}}{\mathrm{m}} = \frac{\mathrm{kg}}{\mathrm{m}^2\,\mathrm{s}^2}, \tag{2.67}$$

so both sides match.

Step 8: put numbers on the gradient

Near sea level, take

$$\rho \approx 1.2\,\mathrm{kg/m}^3. \tag{2.68}$$

Then

$$\frac{\mathrm{d}p}{\mathrm{d}z} \approx -(1.2)(9.81) \approx -11.8\,\mathrm{Pa/m}. \tag{2.69}$$

Because 100 Pa=1 hPa, a 1 hPa decrease requires approximately

$$\frac{100}{11.8} \approx 8.5\,\mathrm{m}. \tag{2.70}$$

At 10 km, where density is only about $0.41\,\mathrm{kg/m}^3$,

$$\frac{\mathrm{d}p}{\mathrm{d}z} \approx -(0.41)(9.81) \approx -4.0\,\mathrm{Pa/m}, \tag{2.71}$$

so a 1 hPa drop takes roughly

$$\frac{100}{4.0} \approx 25\,\mathrm{m}. \tag{2.72}$$

This is the mathematical reason pressure falls more slowly in absolute hPa per kilometre high in the atmosphere.

Where this equation comes from

What kind of equation is this? Equation (2.62) is a **first-order differential equation**: it relates a function, pressure $p(z)$, to its first derivative with respect to height. By itself it is not yet enough to give a numerical pressure at every height, because the right-hand side contains another unknown function, density $\rho(z)$. The ideal gas law in Section 2.2 supplies the missing relationship between ρ, p and T.

What is exact and what is approximate? Newton's second law is exact. The step $a_z \approx 0$ is the hydrostatic approximation. It is extraordinarily accurate for the large-scale atmosphere because vertical accelerations are usually tiny compared with g. It can become poor inside strong convective updraughts, explosive eruptions and other rapidly accelerating vertical flows.

What each symbol means.

Symbol	Unit	Meaning
z	m	Starting height of the slab.
Δz	m	Finite thickness of the slab; the top is at $z + \Delta z$.
A	m^2	Horizontal face area. It cancels, so the result does not depend on the imagined column width.
$p(z)$	Pa or hPa	Pressure at the bottom face.
$p(z + \Delta z)$	Pa or hPa	Pressure at the top face.
Δp	Pa or hPa	Finite pressure change: top minus bottom.
ρ	$\mathrm{kg/m}^3$	Air density in the slab; in the differential limit, local density $\rho(z)$.
g	$\mathrm{m/s}^2$	Magnitude of gravitational acceleration.
$\mathrm{d}p/\mathrm{d}z$	Pa/m	Local vertical pressure gradient, i.e. the derivative of pressure with respect to height.

From the local hydrostatic equation back to the whole air column

The earlier result that pressure measures the weight of all air above can also be recovered by integration. Start with

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g. \tag{2.73}$$

Write it in differential form:

$$\mathrm{d}p = -\rho g \, \mathrm{d}z. \tag{2.74}$$

Integrate from our chosen height z upward to the top of the atmosphere, where pressure tends toward zero:

$$\int_{p(z)}^{0} \mathrm{d}p = -g \int_{z}^{\infty} \rho(z') \, \mathrm{d}z'. \tag{2.75}$$

The prime on z' simply marks it as the dummy height variable used inside the integral. Evaluating the left side gives

$$-p(z) = -g \int_{z}^{\infty} \rho(z') \, \mathrm{d}z', \tag{2.76}$$

so

$$\boxed{p(z) = g \int_{z}^{\infty} \rho(z') \, \mathrm{d}z'.} \tag{2.77}$$

The integral

$$\int_z^\infty \rho \, \mathrm{d}z \tag{2.78}$$

has units $\mathrm{kg/m^3} \times \mathrm{m} = \mathrm{kg/m^2}$ and is therefore the **mass of atmosphere above unit horizontal area**. If that column mass per unit area is m/A, then

$$p(z) = g\frac{m}{A}, \tag{2.79}$$

which rearranges to

$$\boxed{m = \frac{p(z)A}{g}}. \tag{2.80}$$

So the simple whole-column argument and the differential hydrostatic equation are two scales of the same physics: the integral adds the weight contributions of all the infinitesimal layers above us.

Air is compressible: the ideal gas law

Water is almost incompressible: its density is practically the same at the surface of the sea and at the bottom of a deep lake. As a result, water pressure increases in equal steps as you dive deeper (about one extra atmosphere for every 10 m of depth). Air behaves very differently, because air is **compressible**: squeeze it and it becomes denser. The link between pressure, density and temperature of a gas is the **ideal gas law**:

$$p = \rho \, R_\mathrm{d} \, T \qquad \Longleftrightarrow \qquad \rho = \frac{p}{R_\mathrm{d} \, T}, \tag{2.81}$$

where $R_\mathrm{d} = 287\,\mathrm{J/(kg\,K)}$ is the gas constant for dry air and T is in kelvin. For the standard sea-level conditions of 1,013.25 hPa and 15 °C:

$$\rho = \frac{101{,}325\,\mathrm{Pa}}{287 \times 288.15\,\mathrm{K}} = 1.225\,\mathrm{kg/m^3}. \tag{2.82}$$

So a typical classroom measuring 10 m × 8 m × 3 m contains about 290 kg of air.

Where this equation comes from

Where it comes from. Unlike the hydrostatic equation, the gas law is not derived from a single mechanical principle; it is the combination of three separate relationships discovered experimentally between the seventeenth and nineteenth centuries, each varying one property of a gas while holding the others fixed:

- **Boyle's law** (1662): at constant temperature, $p \propto 1/V$ — squeeze a gas into half the volume and its pressure doubles.
- **Charles's law** (1780s): at constant pressure, $V \propto T$ — heat a gas and it expands in direct proportion to its absolute temperature.
- **Avogadro's law** (1811): at the same pressure and temperature, equal volumes of *any* gas contain equal numbers of molecules, so $V \propto n$, the amount of substance.

Combining all three gives the **combined** (or 'universal') **gas law**

$$pV = nR^*T. \tag{2.83}$$

Here n is the number of moles of gas and $R^* = 8.314\,\mathrm{J/(mol\,K)}$ is the **universal gas**

constant, the same for every gas. (Kinetic theory later showed $R^* = N_A k_B$, the product of the Avogadro constant $N_A = 6.022 \times 10^{23}$ mol^{-1} and the Boltzmann constant $k_B = 1.380{,}649 \times 10^{-23}$ J/K.)

To turn Equation (2.83) into the form used in Equation (2.81), write the number of moles as $n = m/M$, where m is the mass of gas present and M is its **molar mass** (the mass of one mole). Substituting and dividing by volume V, and using the density definition in Equation (1.6):

$$pV = \frac{m}{M} R^* T \quad \Longrightarrow \quad p = \frac{m}{V} \cdot \frac{R^*}{M} T \quad \Longrightarrow \quad p = \rho \underbrace{\left(\frac{R^*}{M} \right)}_{= R_\mathrm{d}} T. \tag{2.84}$$

The combination R^*/M is called the **specific gas constant**; for dry air it is given the symbol R_d. Because dry air is itself a mixture (Table 4.1), M is the *mean* molar mass of that mixture, weighted by how common each gas is: $M_\mathrm{d} = 0.7808 \times 28.01 + 0.2095 \times 32.00 + 0.0093 \times 39.95 + \cdots \approx 28.97\,\mathrm{g/mol} = 0.028{,}97\,\mathrm{kg/mol}$. So

$$R_\mathrm{d} = \frac{R^*}{M_\mathrm{d}} = \frac{8.314}{0.02897} \approx 287\,\mathrm{J/(kg\,K)}, \tag{2.85}$$

which is exactly the value used throughout this practical. In short: R_d is not a fundamental constant of nature in its own right — it is what the one universal gas constant R^* becomes once you have converted from 'per mole' to 'per kilogram', specifically for the particular mixture of gases that makes up dry air.

Why T must be in kelvin. The gas law comes from kinetic theory, in which T measures the average kinetic energy of the gas molecules' random motion. That energy is genuinely zero only at **absolute zero**, so the temperature scale in the equation must have its zero at that physical point — the kelvin scale. The Celsius scale instead puts its zero at the freezing point of water, an arbitrary (if convenient) choice, so a Celsius temperature cannot be substituted directly. The number 273.15 that connects them, $T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$, is simply the size of that offset: by international agreement, the freezing point of water (0 °C) sits 273.15 kelvin above absolute zero.

What each symbol means.

Symbol	Unit (here)	Meaning
p	hPa (or Pa)	Pressure of the air.
ρ	kg/m^3	Density of the air.
R_d	J/(kg K)	Specific gas constant for *dry* air, $= R^*/M_\mathrm{d}$.
T	K	Absolute (kelvin) temperature — never °C in this formula.
R^*	J/(mol K)	Universal gas constant (same value for every gas).
M_d	kg/mol	Mean molar mass of dry air.

Where the numbers come from. $R_\mathrm{d} = 287\,\mathrm{J/(kg\,K)}$ is *calculated*, as shown above, from the universal gas constant R^* (itself built from the two fixed SI constants N_A and k_B) and the measured composition of dry air from Table 4.1 (Section 4.1), which fixes M_d. It is not a separately measured or looked-up number; every atmospheric-science textbook quotes

$R_d \approx 287\,\mathrm{J/(kg\,K)}$ because they are all doing this same calculation with the same (very well known) atmospheric composition. The constant 273.15 is a defined conversion factor, fixed by international agreement on where the Celsius and kelvin scales sit relative to one another, not something measured in a laboratory.

Equation (2.81) says that, at a given temperature, the density of air is proportional to its pressure. Now put the two equations together and follow the chain of reasoning:

1. Near the ground there is a lot of air above, so the pressure is high.
2. High pressure squashes the air, so it is dense (gas law).
3. Dense air means the pressure falls quickly as you go up (hydrostatic equation).
4. Higher up there is less air above, the pressure is lower, the air is less squashed and less dense, so the pressure falls more slowly.

The rate at which the pressure falls is always proportional to how much pressure is left. Quantities that behave like this decay **exponentially**, which is the subject of the next section.

A helpful picture is a tall stack of foam mattresses: the ones at the bottom are squashed flat by the weight of those above, while the ones near the top are hardly compressed at all. Figure 2.4 shows the same effect in the atmosphere: layers that each contain the same mass of air get thicker and thicker with height.

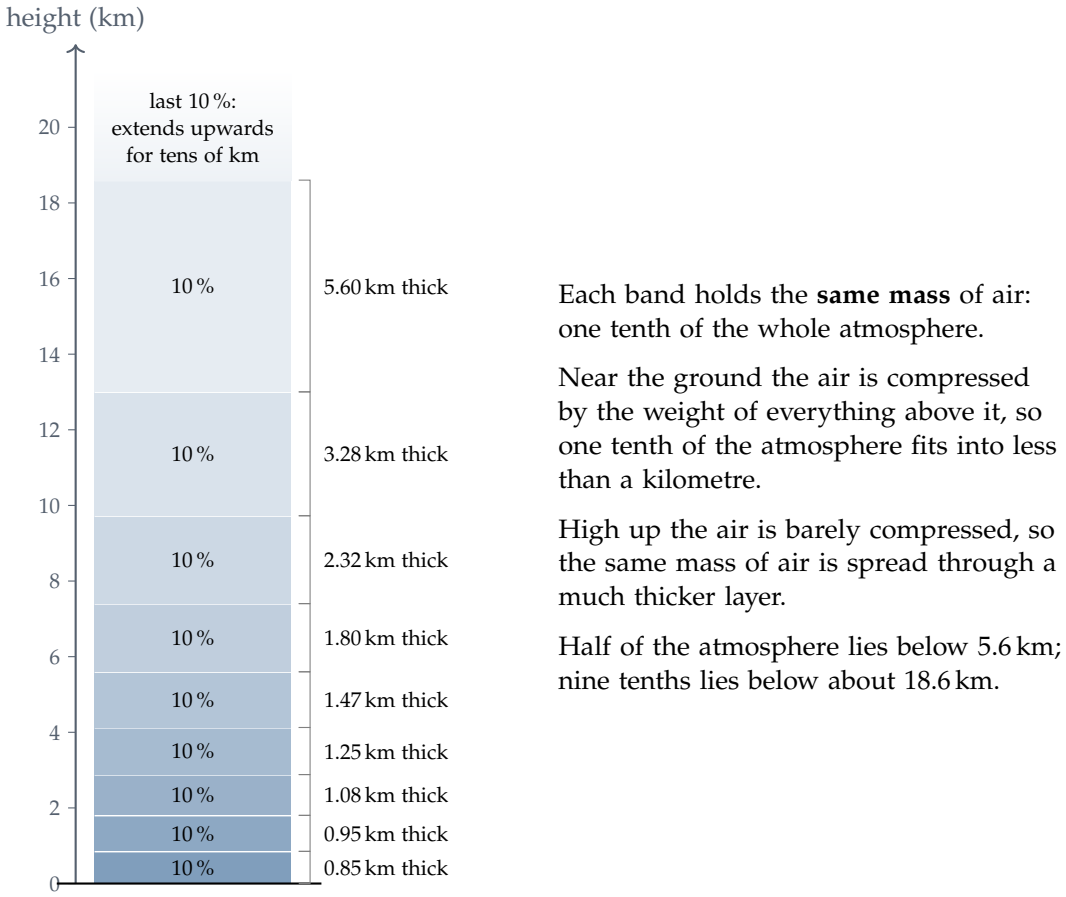


Figure 2.4: The atmosphere divided into ten bands of equal mass, calculated with the 5.6 km halving rule of Section 1.2. The bands become thicker with height because air is compressible.

Check your understanding

2.1 Roughly what mass of air lies above one square metre of the ground at sea level?

2.2 In your own words, why is the air near the ground denser than the air a few kilometres up?

2.3 Near sea level, roughly how far must you climb for the pressure to fall by 1 hPa? Why must you climb further for the same drop at 10 km?

2.4 Using the gas law, show that air at 500 hPa and $-21\,^{\circ}\mathrm{C}$ has a density of about $0.69\,\mathrm{kg/m^3}$. Why is this slightly more than half the sea-level density, even though the pressure is exactly half?

Where does the 5.6 km come from? A complete derivation

Going further: from hydrostatic balance to an exponential atmosphere

This derivation is an excellent example of how algebra, differentiation and integration work together in atmospheric science. We begin with two equations already derived: hydrostatic balance, Equation (2.62), and the ideal gas law, Equation (2.81).

Step 1: eliminate density. Substitute the gas-law expression for ρ into the hydrostatic equation:

$$\begin{aligned}\frac{\mathrm{d}p}{\mathrm{d}z} &= -\left(\frac{p}{R_\mathrm{d}T}\right)g \\ &= -\frac{g}{R_\mathrm{d}T}\,p.\end{aligned} \tag{2.86}$$

So

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{g}{R_\mathrm{d}T}p.} \tag{2.87}$$

The derivative of pressure is proportional to pressure itself. That mathematical structure is the hallmark of exponential decay.

Step 2: state the temperature assumption. The real atmosphere is not isothermal, but to obtain the simple halving model we temporarily assume

$$T = \text{constant}. \tag{2.88}$$

Then g, R_d and T can all be treated as constants. Define the **scale height**

$$\boxed{H = \frac{R_\mathrm{d}T}{g}.} \tag{2.89}$$

Why does $H = R_d T/g$ let us replace $g/(R_d T)$ by $1/H$?

This is only taking the reciprocal of both sides. From

$$H = \frac{R_d T}{g}, \tag{2.90}$$

we have

$$\frac{1}{H} = \frac{1}{R_d T/g} = \frac{g}{R_d T}. \tag{2.91}$$

Therefore

$$-\frac{g}{R_d T} p = -\frac{1}{H} p = -\frac{p}{H}. \tag{2.92}$$

No new physical law has been introduced. The symbol H simply packages a recurring group of constants into one length scale.

The unit check for Equation (2.89) is:

$$\frac{\mathrm{J/(kg\,K)\,K}}{\mathrm{m/s^2}} = \frac{\mathrm{m^2/s^2}}{\mathrm{m/s^2}} = \mathrm{m}, \tag{2.93}$$

because $1\,\mathrm{J/kg} = 1\,\mathrm{m^2/s^2}$. Equation (2.87) becomes

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{1}{H} p. \tag{2.94}$$

Step 3: separate the variables. We want all factors involving p with $\mathrm{d}p$, and all factors involving z with $\mathrm{d}z$. Starting from

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{p}{H}, \tag{2.95}$$

multiply by $\mathrm{d}z$:

$$\mathrm{d}p = -\frac{p}{H}\,\mathrm{d}z. \tag{2.96}$$

Now divide by p:

$$\boxed{\frac{\mathrm{d}p}{p} = -\frac{1}{H}\,\mathrm{d}z.} \tag{2.97}$$

This operation is called **separation of variables**.

Step 4: integrate between known limits. At the ground let

$$z = 0, \qquad p = p_0. \tag{2.98}$$

At a general height z, pressure is $p(z)$. Integrate both sides of Equation (2.97) between those two states:

$$\int_{p_0}^{p(z)} \frac{1}{p'}\,\mathrm{d}p' = -\frac{1}{H}\int_0^z \mathrm{d}z'. \tag{2.99}$$

The primes on p' and z' simply mark dummy integration variables. They prevent us from confusing the running variable inside the integral with the final endpoint.

The antiderivative of $1/p'$ is $\ln p'$, while the antiderivative of 1 with respect to z' is z'. Therefore

$$\begin{aligned} \left[\ln p'\right]_{p_0}^{p(z)} &= -\frac{1}{H}\left[z'\right]_0^z, \\ \ln p(z) - \ln p_0 &= -\frac{z}{H}. \end{aligned} \tag{2.100}$$

Use the logarithm rule $\ln a - \ln b = \ln(a/b)$:

$$\boxed{\ln\left(\frac{p(z)}{p_0}\right) = -\frac{z}{H}.} \tag{2.101}$$

The ratio $p(z)/p_0$ is dimensionless, so taking its logarithm is physically meaningful.

Step 5: undo the logarithm. Because the exponential and natural logarithm are inverse functions, exponentiate both sides:

$$\begin{aligned} \exp\left[\ln\left(\frac{p(z)}{p_0}\right)\right] &= \exp\left(-\frac{z}{H}\right), \\ \frac{p(z)}{p_0} &= e^{-z/H}. \end{aligned} \tag{2.102}$$

Multiply by p_0:

$$\boxed{p(z) = p_0 e^{-z/H}.} \tag{2.103}$$

This is the isothermal barometric law.

Step 6: interpret the scale height. If $z = H$,

$$\frac{p(H)}{p_0} = e^{-1} = \frac{1}{e} \approx 0.368. \tag{2.104}$$

So one scale height is the height over which pressure falls to about 36.8 % of its starting value.

Step 7: find the half-height. Set $p(z) = p_0/2$:

$$\begin{aligned} \frac{p_0}{2} &= p_0 e^{-z/H}, \\ \frac{1}{2} &= e^{-z/H}. \end{aligned} \tag{2.105}$$

Take natural logs:

$$\ln\left(\frac{1}{2}\right) = -\frac{z}{H}. \tag{2.106}$$

Since $\ln(1/2) = -\ln 2$,

$$-\ln 2 = -\frac{z}{H}, \tag{2.107}$$

so

$$\boxed{z_{1/2} = H \ln 2 \approx 0.693H.} \tag{2.108}$$

Step 8: choose the scale height that corresponds to a 5.6 km half-height. If

$$z_{1/2} = 5.6\,\text{km}, \tag{2.109}$$

then

$$H = \frac{5.6}{\ln 2} = 8.08\,\text{km}. \tag{2.110}$$

Use $H = R_d T/g$ to infer the uniform temperature represented by this teaching model:

$$\begin{aligned} T &= \frac{gH}{R_d} \\ &= \frac{(9.81)(8080)}{287} \\ &\approx 276\,\text{K} \\ &\approx 3\,^\circ\text{C}. \end{aligned} \tag{2.111}$$

So the 5.6 km halving rule is equivalent to an isothermal atmosphere at roughly 3 °C. It is a useful approximation, not a universal constant of nature.

How can the effective model temperature be $+3\,^\circ$C when the standard atmosphere is about $-21\,^\circ$C at 5.6 km?

The two temperatures answer different questions.

The $-21\,^\circ$C value is a local temperature at one height. Using the standard tropospheric lapse rate from Question 5,

$$T(5.6\,\text{km}) = 15 - (6.5)(5.6) = -21.4\,^\circ\text{C}. \tag{2.112}$$

This asks: 'what is the temperature of the standard atmosphere *at* 5.6 km?'

The $+3\,^\circ$C value is an effective constant temperature for an imaginary isothermal column. We forced a model in which the temperature is the same at *every* height to reproduce exactly the handout's rule $p(5.6\,\text{km}) = p_0/2$. Solving that model backwards gave $T \approx 276\,\text{K} \approx 3\,^\circ\text{C}$. It asks: 'what single constant temperature would make this simple exponential model have a half-height of exactly 5.6 km?'

In the real atmosphere T is a function of height, $T = T(z)$, so the governing equation is

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{pg}{R_d T(z)}, \tag{2.113}$$

and integration gives

$$\boxed{\ln\left(\frac{p(z)}{p_0}\right) = -\int_0^z \frac{g}{R_d T(z')}\,\mathrm{d}z'.} \tag{2.114}$$

So pressure at 5.6 km depends on the temperature structure of the *whole column below*, not just the endpoint temperature there. With the standard-atmosphere lapse-rate profile, the pressure at 5.6 km is about $0.492p_0$, and the exact half-pressure level is about 5.48 km. That is why the handout's 5.6 km rule is a very useful rounded approximation even though the local temperature there is nowhere near 3 °C.

Why the real atmosphere departs from the rule. In the real troposphere T decreases strongly with height. Since

$$H = \frac{R_d T}{g}, \tag{2.115}$$

colder air has a smaller scale height: pressure decreases more rapidly with geometric height than the uniform-3 °C model predicts. That is why the rule is very good low down but increasingly overestimates pressure high in the atmosphere.

How well does the rule match real measurements?

We can test the rule against the two weather-balloon soundings used later in the practical (handout Table 3a and Table 3b in Section 6). Rearranging Equation (1.11) gives the height at which the rule predicts each pressure, $z = 5.6 \log_2(1000/p)$. Table 2.1 compares these predictions with the measured heights of the same pressure levels and with the International Standard Atmosphere.

Table 2.1: Heights (in km) at which selected pressure levels were found over Hillsborough and St John's on 21 March 1990, compared with the 5.6 km halving rule and with the International Standard Atmosphere.

Pressure level	Halving rule	Hillsborough	St John's	Standard atmosphere
500 hPa	5.60	5.41	5.60	5.57
400 hPa	7.40	7.02	7.25	7.19
250 hPa	11.20	10.16	10.45	10.36
100 hPa	18.60	16.13	16.25	16.18
30 hPa	28.33	23.69	23.83	23.85
20 hPa	31.61	26.22	26.40	26.48

The agreement at 500 hPa is remarkable: the rule puts the 500 hPa level at exactly 5.6 km, and over St John's it was measured at 5,604 m. Higher up, the rule places each pressure level too high, by about 1 km at 250 hPa and by about 5 km at 20 hPa, because the air up there is much colder than the 3 °C that the rule effectively assumes. The rule is a teaching tool, not a law of nature, but it captures the essential behaviour: pressure and density fall off exponentially with height.

Check your understanding

2.1 What percentage of the atmosphere is above you at 16.8 km? What is the pressure there according to the rule?

2.2 What fraction of the whole atmosphere lies *between* 5.6 km and 11.2 km?

2.3 Use Equation (1.18) to find the height at which the pressure has fallen to 100 hPa.

2.4 Why can the amount of atmosphere above you never quite reach zero, according to the rule?

What the question is asking

Pressure is the weight of the air above you (Section 2.2). So if half of the atmosphere is above you, the pressure is half of its sea-level value; if a quarter is above you, it is a quarter of the sea-level value, and so on. The pressure column is therefore just the percentage column scaled so that 100 % corresponds to 1,000 hPa.

Method, step by step

1. **Pressure column.** Multiply each percentage by 10 (because 100 % ↔ 1,000 hPa), or equivalently halve 1,000 hPa repeatedly: 1000, 500, 250, ...
2. **Second axis.** Along the *top* edge of your graph, draw a second horizontal scale. Place 0 hPa directly above 0 %, 500 hPa above 50 % and 1,000 hPa above 100 %, with the other values in between (100 hPa above every 10 %). Label it 'Atmospheric pressure (hPa)'.
3. **Do not draw a second curve.** The existing curve already works for both axes, because the two scales are exactly proportional.

Worked solution

Multiplying each percentage in handout Table 1 by 10 (or, equivalently, halving 1,000 hPa repeatedly):

Height (km)	0	5.6	11.2	16.8	22.4	28.0	33.6	39.2	44.8	50.4
% above	100	50	25	12.5	6.25	3.125	1.56	0.781	0.391	0.195
Pressure (hPa)	1000	500	250	125	62.5	31.25	15.6	7.81	3.91	1.95

These are the red values in the third column of handout Table 1 (Section 1). The second axis is shown along the top of Figure 1.1: each division of 10 % on the bottom axis corresponds to 100 hPa on the top axis.

In general, the pressure at height z according to the rule is

$$p(z) = 1{,}000\,\text{hPa} \times \left(\tfrac{1}{2}\right)^{z/5.6}. \tag{2.116}$$

Going further: does density really halve as well?

The handout says that the density and the pressure both drop by 50 % for every 5.6 km. For pressure this is exactly what the rule means, because pressure measures the amount of air above you. Density, however, also depends on temperature: by the gas law, $\rho = p/(R_d T)$. The air at 5.6 km is typically about 36 °C colder than at sea level, and cold air is denser, so the density there is about 56 % of its sea-level value rather than exactly 50 %. The halving rule treats the atmosphere as if it had one uniform temperature (about 3 °C, see Section 1.2), in which case pressure and density would halve together exactly.

Common pitfalls

- **Drawing the pressure axis in reverse.** High pressure goes with a high percentage of the atmosphere above, so 1,000 hPa belongs above 100 %, on the right.
- **Plotting a second, separate curve.** It would lie exactly on top of the first one. One curve and two labelled scales is all that is needed.
- **Using** 1,013.25 hPa**.** This is the true standard sea-level pressure, but the handout uses 1,000 hPa. Using 1013 would not be wrong physically, but your answers would then differ from everyone else's by about one percent.

3 Question 3: reading the graph

Question 3

Using the graph you just drew, estimate the percentage of atmosphere above and the total pressure for the following heights:

Location (height)	% above	Pressure (hPa)
Portstewart Strand (0 m)	100	1000
Top of Mount Everest (8,848 m)	33.4	334
Cruising altitude of a jumbo jet (12 km)	22.6	226
Machu Picchu, Peru (2,430 m)	74.0	740
World record hot-air balloon (21,027 m)	7.4	74

What Question 3 means in ordinary words

For each named place or height, start on the vertical height axis, move horizontally until you reach the curve, and then read two descriptions of that same point: the **percentage of the atmosphere's total mass still above that height** from the bottom axis, and the **atmospheric pressure** from the top axis. You are not calculating a new kind of quantity here; you are using the graph from Questions 1–2 as a lookup chart.

3.1 Graph reading, interpolation and exact checking

This question turns the curve from Questions 1–2 into a measuring instrument. A hand-drawn graph gives an estimate; the exponential formula gives a numerical check. The two should agree within the precision of the graph. The crucial practical skills are unit conversion, following the *curve* rather than a straight chord, and keeping percentage and pressure on their correct axes.

What the question is asking

You are using your graph as a 'look-up' tool: for any height, the curve tells you what percentage of the atmosphere is above, and the top axis tells you the pressure. Because the values come from reading a hand-drawn graph, everyone's answers will differ slightly. You can check your readings with the formula from Section 1.2.

Method, step by step

1. **Convert every height to kilometres**, because your graph's height axis is in km: 8,848 m = 8.848 km, 2,430 m = 2.43 km, 21,027 m = 21.027 km.
2. Find the height on the vertical axis and draw a light pencil line *horizontally* across until it meets the curve.
3. From that point, draw a line *vertically* down to the bottom axis and read off the percentage of the atmosphere above.
4. From the same point, go vertically *up* to the top axis and read off the pressure (or simply multiply the percentage by 10).
5. **Check** with the rule: percentage above $= 100 \times 0.5^{z/5.6}$ (with z in km).

Worked solution

Figure 3.1 shows the graph readings. The exact values from the rule are calculated below; graph readings within about two percentage points (about 20 hPa) of these are perfectly acceptable.

Location	z (km)	$n = z/5.6$	0.5^n	% above	Pressure (hPa)
Portstewart Strand	0	0	1	100	1000
Mount Everest	8.848	1.580	0.3345	33.4	334
Jumbo jet	12.0	2.143	0.2264	22.6	226
Machu Picchu	2.43	0.4339	0.7402	74.0	740
Hot-air balloon record	21.027	3.755	0.0741	7.4	74

For example, for Everest: $n = 8.848/5.6 = 1.580$ halvings, and $0.5^{1.580} = 0.3345$, so 33.4 % of the atmosphere is above the summit and the pressure is $1000 \times 0.3345 = 334$ hPa.

Interpretation. Standing on the summit of Everest, two thirds of the atmosphere is below you. A jumbo jet cruises above more than three quarters of the atmosphere, and the record-breaking hot-air balloon climbed above more than nine tenths of it. Even at Machu Picchu, about a quarter of the atmosphere is below you.

Going further: how realistic are these numbers?

The halving rule is a simplification, so it is worth comparing its predictions with the International Standard Atmosphere (which starts from 1,013.25 hPa at sea level and has a realistic temperature profile) and, where possible, with measurements.

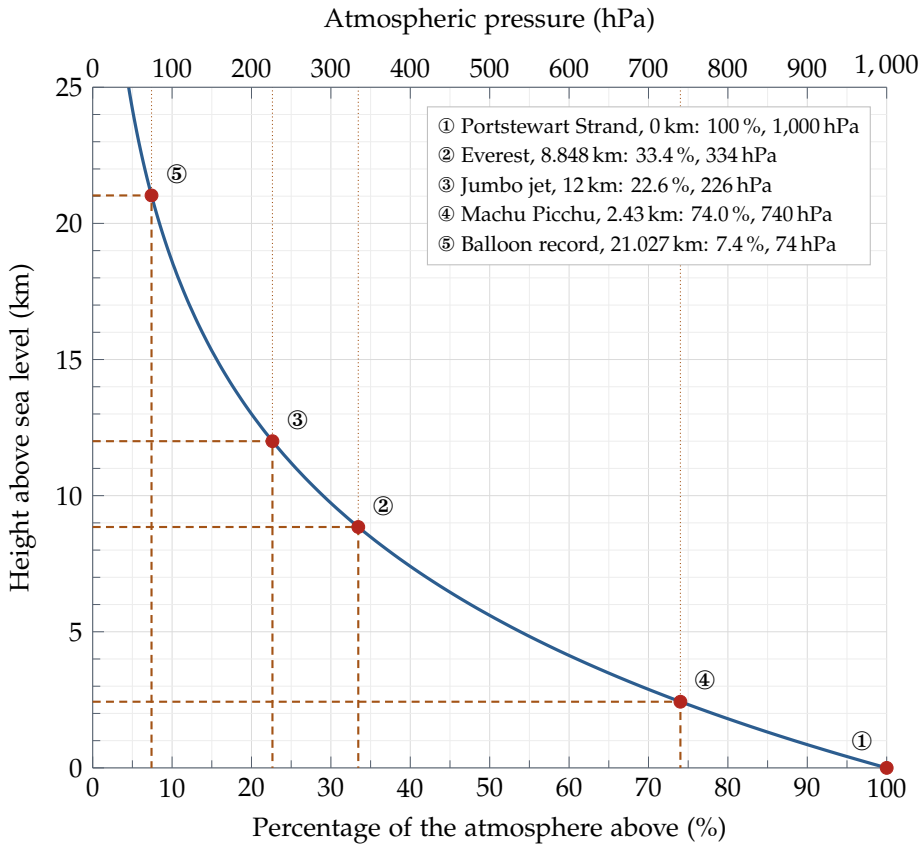


Figure 3.1: Reading the graph for Question 3 (the lowest 25 km of Figure 1.1, enlarged). Dashed lines: across from the height to the curve, then down to read the percentage. Dotted lines: up to the top axis to read the pressure.

Location	Rule (hPa)	Standard atmosphere (hPa)	Comment
Machu Picchu	740	753	Excellent agreement: 74.0 % of the sea-level value by the rule, 74.4 % in the standard atmosphere.
Mount Everest	334	314	The pressure measured on the summit in 1981 was about 337 hPa, higher than the standard atmosphere because Everest lies at 28° N, where the air column is warmer and therefore taller than in mid-latitudes.
Jumbo jet (12 km)	226	193	The rule is too high: above about 10 km the air is much colder than the 3 °C the rule effectively assumes, so the pressure falls faster.
Balloon record (21 km)	74	47	The rule overestimates the pressure by more than half, for the same reason.

The balloon record in the question was set by Vijaypat Singhania over Mumbai, India, on 26 November 2005. He flew in a sealed, pressurised capsule, and Question 4 shows why he had to.

Common pitfalls

- **Forgetting to convert metres to kilometres.** 8,848 m is 8.848 km, not 8,848 km. In the formula, 8848/5.6 would give a nonsensical 1580 halvings.
- **Reading between the table values with a ruler.** If you draw a straight line between the plotted points rather than following the curve, you will overestimate the percentage. For Machu Picchu a straight line between 0 km (100 %) and 5.6 km (50 %) gives 78.3 % instead of 74.0 %; for Everest it gives 35.5 % instead of 33.4 %. The curve bends, so follow it.
- **Reading the wrong axis.** Percentages are on the bottom axis, pressures on the top axis.

4 Question 4: oxygen at altitude

Question 4

At height, physical activity is more taxing and, at even higher elevations, jets have pressurised cabins. This is because the partial pressure of gases within atmosphere drops with altitude and although the percentage of each gas remains the same, there are fewer gas molecules present. At sea level 78 % of the atmosphere is nitrogen and 21 % is oxygen. Or in other words, the partial pressure of nitrogen at sea level is 780 hPa and the partial pressure of oxygen at sea level is 210 hPa. However, the partial pressure of each gas drops at the same rate as the total atmospheric pressure. Keeping this in mind, use your answers to question 3 to first calculate the partial pressure of oxygen at different elevations. Then use your answer to calculate the percentage of oxygen available compared to the atmosphere at sea-level.

Location (height)	**Partial pressure of O_2 (hPa)**	**% oxygen**
Portstewart Strand (0 m)	210	21
Top of Mount Everest (8,848 m)	70	7.0
Cruising altitude of a jumbo jet (12 km)	47.5	4.7
Machu Picchu, Peru (2,430 m)	155	15.5
World record hot-air balloon (21,027 m)	15.5	1.6

What Question 4 means in ordinary words

The air at altitude is still about 21 % oxygen *by molecular proportion* (for dry air). What changes is the total number of air molecules packed into each breath because total pressure is lower. Therefore the oxygen molecules contribute a smaller *partial pressure*. The question first asks for that oxygen partial pressure in hPa. It then asks you to express that pressure using the worksheet's 'effective oxygen percentage'. Do not confuse this effective percentage with the actual composition of the air: the actual dry-air oxygen fraction remains about 21 %.

4.1 What '21% oxygen' actually means

The familiar composition figures refer to *dry air*. Question 4 is therefore a good place to distinguish dry air from moist air, mole fraction from partial pressure, and oxygen concentration from oxygen availability. The handout deliberately uses the clean dry-air approximation $p_{O_2} \approx 0.21p$; we derive it and then show the small humidity correction so that the approximation is explicit rather than mysterious.

What the atmosphere is made of

The atmosphere is a mixture of gases held to the Earth by gravity. Table 4.1 lists the main gases in dry air (air with the water vapour removed). The percentages are *by volume*, which for gases is the same as the percentage of molecules: '21 % oxygen' means that 21 out of every 100 molecules of air are oxygen molecules.

Table 4.1: Composition of dry air near the ground. The handout rounds nitrogen to 78 % and oxygen to 21 %, and these notes do the same.

Gas	Formula	Percentage by volume
Nitrogen	N_2	78.08
Oxygen	O_2	20.95
Argon	Ar	0.93
Carbon dioxide	CO_2	about 0.04 (over 420 parts per million, and rising)
Neon, helium, methane, krypton, hydrogen and others	various	less than 0.003 in total
Water vapour (not part of 'dry' air)	H_2O	highly variable, from almost 0 up to about 4

Dry air versus the air we actually breathe

The familiar figures 78 % nitrogen and 21 % oxygen refer to **dry air**: the mixture left after water vapour is excluded. Real atmospheric air is usually **moist air**, because it contains a variable amount of water vapour.

If water vapour occupies a mole fraction x_v, the remaining fraction $1 - x_v$ is dry air. Oxygen then occupies

$$x_{O_2,\,\text{moist}} = 0.2095(1 - x_v). \tag{4.1}$$

of the *whole moist mixture*. For example, if water vapour makes up 2 % of the molecules,

$$x_{O_2,\,\text{moist}} = 0.2095(0.98) = 0.205 \approx 20.5\,\%. \tag{4.2}$$

The nitrogen-to-oxygen ratio within the dry component has not changed; water vapour has simply taken up part of the total mixture and therefore diluted the dry-air fractions slightly.

The same idea appears in partial pressures. If the water-vapour partial pressure is e, then the dry-air pressure is

$$p_d = p - e, \tag{4.3}$$

and the oxygen partial pressure in moist ambient air is more precisely

$$\boxed{p_{O_2} = 0.2095\,(p - e).} \tag{4.4}$$

The practical deliberately neglects this small refinement in Equation (4.4) and uses the dry-air approximation Equation (4.6), which is entirely appropriate for learning the main altitude effect. The important point is to know what approximation has been made.

A crucial fact for this practical is that, in the lowest 80 to 100 km of the atmosphere, the proportions of the *dry-air* constituents are essentially the same at every height. The air is constantly stirred by winds and turbulence, so the gases never get the chance to separate out by weight. The air on the summit of Everest is still 21 % oxygen. What changes with height is not the *recipe* of the air but the *amount* of air: there are fewer molecules of everything in each lungful. We will use this idea in Question 4.

4.2 Mixtures of gases: partial pressure

Dalton's law of partial pressures

In a mixture of gases, all gas molecules strike surfaces and contribute to the measured total pressure. We can conceptually separate that total into contributions from each gas. The pressure contribution associated with one gas is its **partial pressure**. For an ideal mixture, that contribution is proportional to that gas's share of the molecules. This relationship is **Dalton's law**:

$$p_{\text{gas}} = x_{\text{gas}}\, p_{\text{total}}, \qquad p_{\text{total}} = p_{\text{N}_2} + p_{\text{O}_2} + p_{\text{Ar}} + \dots \tag{4.5}$$

where x_{gas} is the fraction of molecules that belong to that gas (0.78 for nitrogen, 0.21 for oxygen). At sea level, with a total pressure of 1,000 hPa, the partial pressure of nitrogen is $0.78 \times 1000 = 780$ hPa and that of oxygen is $0.21 \times 1000 = 210$ hPa, exactly as stated in the handout.

Because the proportions of the gases do not change with height (Section 4.1), every partial pressure falls *in step* with the total pressure. If the total pressure halves, the oxygen partial pressure halves too:

$$p_{\text{O}_2}(z) = 0.21 \times p(z). \tag{4.6}$$

Where this equation comes from

Where it comes from. Dalton's law, published by the chemist John Dalton in 1801–02, follows directly from the ideal gas law (Equation (2.81)) applied separately to each gas in a mixture. For an ideal-gas mixture, intermolecular potential forces are neglected and molecular collisions are treated as perfectly elastic; each species therefore contributes independently to the pressure as if it alone occupied the container at the same temperature. Writing the ideal gas law in 'moles' form for gas i in a fixed volume V at temperature T: $p_i V = n_i R^* T$. Summing this over every gas present and using V, T the same for all of them:

$$\sum_i p_i V = \left(\sum_i n_i\right) R^* T = n_{\text{total}} R^* T = p_{\text{total}} V \quad \Longrightarrow \quad \sum_i p_i = p_{\text{total}}. \tag{4.7}$$

Dividing the equation for gas i by this total equation gives $p_i / p_{\text{total}} = n_i / n_{\text{total}} = x_i$, the **mole fraction** (which, by Avogadro's law, is the same as the fraction of molecules, and the same as the percentage by volume from Table 4.1). Rearranged, this is exactly Equation (4.5). So Dalton's law is really just 'the ideal gas law, applied once per gas, then added up.'

What each symbol means.

Symbol	Unit (here)	Meaning
p_{gas} (e.g. p_{O_2})	hPa	**Partial pressure**: the pressure that one particular gas in the mixture would exert on its own, at the same temperature, if it alone filled the container.
x_{gas}	dimensionless (or %)	**Mole fraction**: the proportion of all the gas molecules present that belong to this one gas — identical to the '% by volume' of Table 4.1.
p_{total}	hPa	The total (measured) air pressure — the sum of every gas's partial pressure.
n_i, n_{total}	mol	Number of moles of gas i, and of all gases combined.

Where the numbers come from. The mole fractions $x_{N_2} = 0.78$ and $x_{O_2} = 0.21$ are not derived from any equation; they are *measured* quantities, established by direct chemical analysis of air samples. The two-gas composition of air was first worked out in the 1770s by Antoine Lavoisier, Joseph Priestley and Henry Cavendish (using combustion and absorption experiments to show that only part of air supports burning and breathing); today the proportions are confirmed to high precision by techniques such as gas chromatography and mass spectrometry, and are essentially the same everywhere on Earth below about 100 km because turbulence keeps the atmosphere thoroughly mixed (Section 4.1). The 1,000 hPa sea-level pressure used to turn x_{gas} into an actual partial pressure in hPa is the same rounded reference value introduced in Section 2.2.

Why partial pressure matters for breathing

Oxygen gets from the air in your lungs into your blood by *diffusion*, and the rate of diffusion depends on the *partial pressure* of oxygen, not on its percentage. At altitude the air is still 21 % oxygen, but each breath contains fewer oxygen molecules, the oxygen partial pressure is lower, and less oxygen reaches the blood. The resulting shortage of oxygen in the body is called **hypoxia**. Its effects include breathlessness, headache, fatigue and impaired judgement, and it is the cause of altitude sickness. Many people start to notice the effects above about 2,500 m.

Definition: effective (equivalent) oxygen percentage

The **effective oxygen percentage** at a height z is the percentage of oxygen that air at sea level would need to contain in order to have the same oxygen partial pressure as the air at height z:

$$\boxed{\text{effective } O_2 \text{ percentage} = \frac{p_{O_2}(z)}{p_0} \times 100\,\% = 21\,\% \times \frac{p(z)}{p_0}.} \tag{4.8}$$

On Slieve Donard (900 hPa), $p_{O_2} = 0.21 \times 900 = 189$ hPa, so the effective oxygen percentage is $189/1000 \times 100 = 18.9$ %: breathing there is like breathing sea-level air containing only 18.9 % oxygen instead of 21 %. Altitude-training tents and hypoxic chambers are described in exactly these terms.

Where this equation comes from

Where it comes from. This is not a separate physical law — it is Dalton's law (Equation (4.5)) combined with a definition invented for human convenience: rather than quoting an oxygen partial pressure in hPa, which most people have no intuition for, physiologists re-express it as 'the sea-level oxygen percentage that would feel the same'. Substituting $p_{O_2}(z) = 0.21\,p(z)$ (Equation (4.6)) into the definition and cancelling gives the second form, $21\,\% \times p(z)/p_0$, showing that the effective oxygen percentage is always exactly 21 % of whatever fraction of the atmosphere is above you.

Symbols and where the numbers come from. $p_{O_2}(z)$ [hPa] and $p_0 = 1{,}000$ hPa are exactly as in Dalton's law above; 21 % is the same measured oxygen mole fraction from Table 4.1. Nothing new is measured or assumed for this formula — every number in it has already been introduced.

This is also why airliners have **pressurised cabins**. At a cruising altitude of 10 to 12 km the outside pressure is only about a quarter of its sea-level value, far too low to breathe safely. Aviation regulations require the cabin to be kept at a pressure no lower than that found at about 2,400 m (8,000 ft), roughly 750 hPa, which, as we will see in Question 4, is almost exactly the altitude of Machu Picchu.

Going further: the water vapour in your lungs

Air in the lungs is saturated with water vapour at body temperature, and this water vapour contributes its own partial pressure of about 63 hPa. Only the remainder is available to the other gases, so the oxygen partial pressure in the lungs is roughly $0.21 \times (p - 63\,\text{hPa})$. At sea level this makes little difference ($0.21 \times 937 \approx 197$ hPa), but at very low pressures it matters a great deal. Barometric pressure measured on the summit of Everest during the 1981 American Medical Research Expedition was about 337 hPa, giving $0.21 \times (337 - 63) \approx 58$ hPa, less than a third of the sea-level value. This is why most climbers use bottled oxygen near the summit, and why the zone above about 8,000 m is known to mountaineers as the 'death zone'.

Check your understanding

4.1 On the summit of a mountain the total pressure is 600 hPa. What percentage of the air molecules there are oxygen?

4.2 At the same summit, what is the partial pressure of oxygen, and what is the effective oxygen percentage?

4.3 Why does it become harder to breathe at altitude, even though the air is still 21 % oxygen?

What the question is asking

There are two steps.

1. **The partial pressure of oxygen.** Oxygen makes up 21 % of the molecules at every height, so its partial pressure is always 21 % of the total pressure you found in Question 3 (Dalton's law, Section 4.2).

2. **The 'percentage of oxygen available compared to the atmosphere at sea level'.** This is the *effective oxygen percentage* defined in Section 4.2: the oxygen partial pressure expressed as a percentage of the total sea-level pressure of 1,000 hPa. It answers the question 'breathing this air is like breathing sea-level air containing what percentage of oxygen?'

Method, step by step

1. Take the total pressure p at each location from Question 3.
2. Partial pressure of oxygen: $p_{O_2} = 0.21 \times p$.
3. Effective oxygen percentage: $\dfrac{p_{O_2}}{1{,}000\,\text{hPa}} \times 100$, which is simply p_{O_2} divided by 10.
4. Shortcut and check: the effective oxygen percentage is also 21 times the fraction of the atmosphere above you. Everest, with 33.4 % of the atmosphere above, gives $21 \times 0.334 = 7.0\,\%$.

Worked solution

Location	p (hPa)	$p_{O_2} = 0.21\,p$ (hPa)	Effective O_2 (%)	Share of sea-level O_2
Portstewart Strand	1000	$0.21 \times 1000 = 210$	$210/10 = 21.0$	100 %
Mount Everest	334	$0.21 \times 334 = 70.1$	$70.1/10 = 7.0$	33 %
Jumbo jet	226	$0.21 \times 226 = 47.5$	$47.5/10 = 4.7$	23 %
Machu Picchu	740	$0.21 \times 740 = 155$	$155/10 = 15.5$	74 %
Balloon record	74	$0.21 \times 74 = 15.5$	$15.5/10 = 1.6$	7 %

(For the jet, $0.21 \times 226 = 47.46$, which gives 4.7 %; if you round to 47.5 hPa first you may get 4.8 %. Either is fine.)

What these numbers mean. At Machu Picchu each breath delivers only about three quarters of the oxygen it would at sea level; it is like breathing sea-level air containing 15.5 % oxygen, and many visitors feel breathless when walking uphill. On the summit of Everest the effective oxygen percentage is only about 7 %, a third of the sea-level value; most climbers use bottled oxygen, and the first ascent without it was not made until 1978. At an airliner's cruising altitude the effective figure is below 5 %, and anyone exposed to it would lose useful consciousness within seconds, which is why the cabin is pressurised and why oxygen masks drop if the pressurisation fails. At the height of the balloon record (1.6 %) an unprotected person could not survive at all.

Going further: why the balloonist needed a pressurised capsule

Low oxygen is not the only danger at great heights. At a pressure of about 63 hPa, reached at roughly 18 to 19 km in the real atmosphere, water boils at normal body temperature (37 °C), so the moisture in the lungs, mouth and eyes would begin to boil away. This height is known as the *Armstrong limit*. The balloon record height of 21 km lies above it, so the flight was only possible inside a sealed, pressurised capsule. Airliner cabins are pressurised to the equivalent of no more than about 2,400 m, practically the altitude of Machu Picchu, so the cabin air on a long-haul flight has an effective oxygen percentage of roughly 15 %, the same as in the table above.

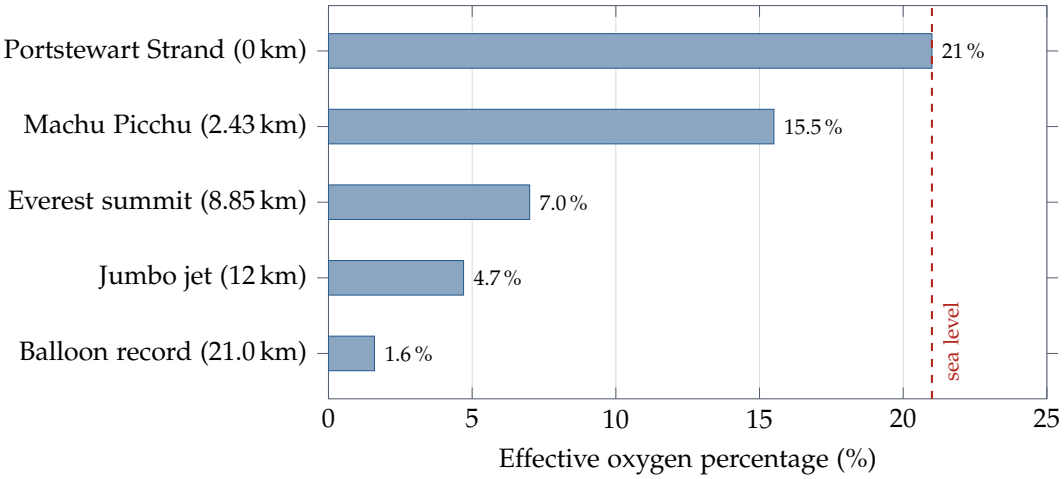


Figure 4.1: Effective oxygen percentage at the five locations of Question 4: the percentage of oxygen that sea-level air would need to contain to supply the same amount of oxygen per breath.

Two ways of describing the oxygen 'available'

The wording 'percentage of oxygen available compared to the atmosphere at sea level' can be read in two ways, and it helps to see that they are consistent.

- **Effective oxygen percentage** (the answer expected on the handout): the oxygen partial pressure as a percentage of the total sea-level pressure, $p_{O_2}/1{,}000\,\text{hPa} \times 100$. At sea level this is 21 %.
- **Share of the sea-level oxygen**: the oxygen partial pressure as a percentage of the *sea-level oxygen partial pressure*, $p_{O_2}/210\,\text{hPa} \times 100$. At sea level this is 100 %, and at any height it equals the percentage of the atmosphere above you from Question 3.

The first number is always 0.21 times the second. What does *not* change is the composition: at every one of these heights the air is still 21 % oxygen by volume.

Common pitfalls

- **Saying that the air at altitude contains less than 21 % oxygen.** It does not; the *proportion* is unchanged. There are simply fewer molecules of every gas in each breath.
- **Taking 21 % of the percentage above instead of the pressure.** $0.21 \times 33.4 = 7.0$ happens to give the effective percentage directly, but it is not a partial pressure; make sure you label which number is which.
- **Mixing up hPa and %.** Partial pressures are in hPa; the effective oxygen figures are percentages.

5 Question 5: the standard atmosphere

Question 5

Not only do the pressure and density change with altitude, the temperature does as well. We will have a closer look at what happens in the troposphere, the layer in which most weather phenomena occur. In this layer, temperature decreases by 6.5 °C/km in the lowest 11 km; this is called the *average* or *normal lapse rate*. Based on this average lapse rate, complete the drop in temperature for 'standard atmosphere' in handout Table 2.

Height (m)	Sea level	1000	2000	4000	6000	8000	10000	11000	14000	16000
Temperature (°C)	15	8.5	2	−11	−24	−37	−50	−56.5	−56.5	−56.5

Table 2 of the handout: temperature drop in the troposphere for the 'standard atmosphere'.

What Question 5 means in ordinary words

The practical tells you that, in the standard atmosphere below 11 km, temperature falls by 6.5 °C for every 1 km you climb. That number is the **lapse rate**. Starting from 15 °C at sea level, subtract 6.5 °C for each kilometre of height. The question is therefore repeated subtraction with careful attention to units and negative temperatures.

5.1 Temperature structure and the meaning of a lapse rate

The word *lapse rate* is a rate-of-change statement. We first locate the troposphere within the thermal layers of the atmosphere, then define the finite and local temperature gradients, explain the sign convention, and only then plug the standard-atmosphere value into the table.

5.2 How temperature changes with height: the layers of the atmosphere

Pressure and density fall steadily with height, but temperature does not. It falls, then stays roughly constant, then rises, then falls again and finally rises steeply. These changes in the direction of the temperature trend are used to divide the atmosphere into layers (Figure 5.1). The boundary at the top of each layer is called a 'pause': the tropopause, the stratopause and the mesopause.

The troposphere

The **troposphere** is the lowest layer, extending from the ground to about 8 km over the poles, about 11 km in mid-latitudes and 16 to 18 km over the tropics. It contains roughly three quarters of the mass of the atmosphere and almost all of its water vapour, clouds and weather. Its name comes from the Greek *tropos*, 'turning', because the air in it is constantly overturning and mixing.

Temperature in the troposphere decreases with height, on average by about 6.5 °C/km. The reason is that *the troposphere is heated from below*. Most of the sunlight that reaches the Earth passes through the air and is absorbed by the ground and the oceans. The warm surface then heats the air in contact with it; that heat is carried upwards by rising currents of air (convection), by turbulence and by the release of latent heat when water vapour condenses into cloud. Air that

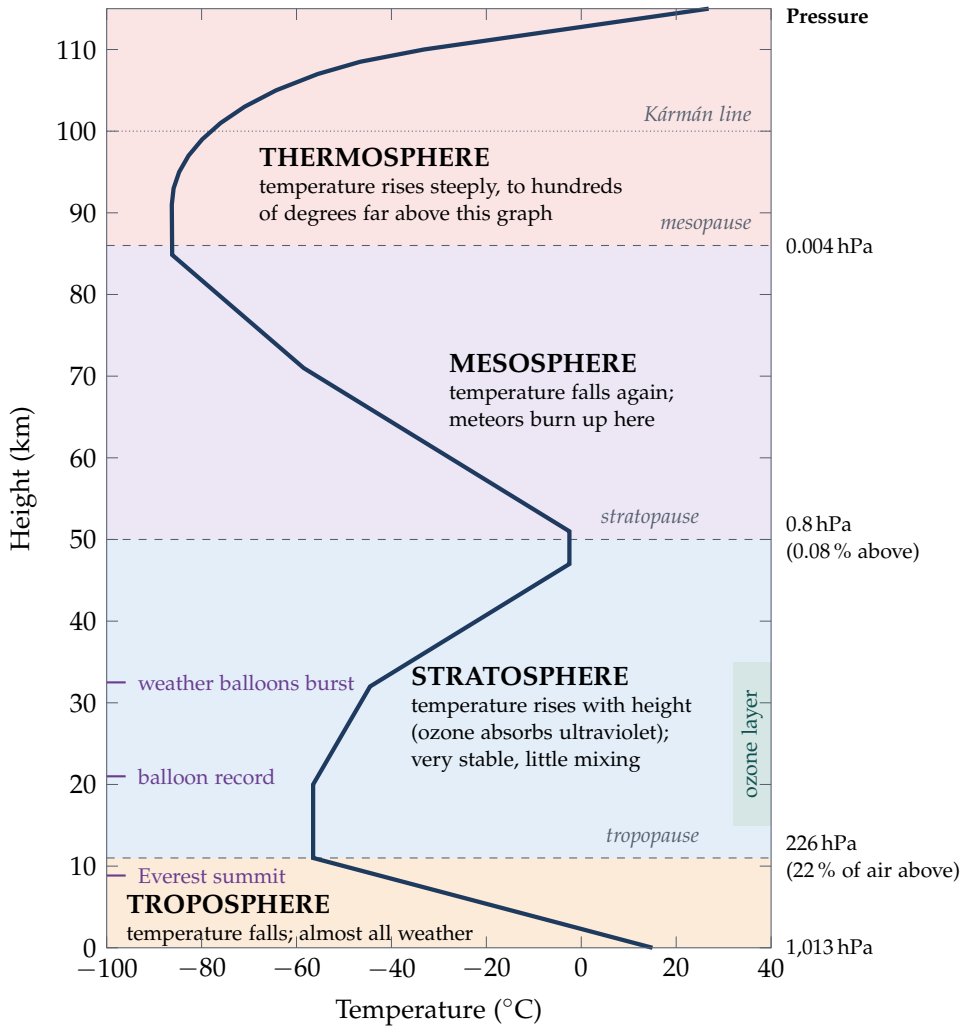


Figure 5.1: The layers of the atmosphere. The curve is the temperature profile of the International Standard Atmosphere (the U.S. Standard Atmosphere, 1976, above 86 km), an idealised profile representing average mid-latitude conditions. The labels on the right give the pressure at the boundaries between layers. The heights of the boundaries vary with latitude and season.

rises also expands, because the pressure around it is lower, and expanding air cools (Section 7.1). The further you are from the heated surface, the colder the air tends to be.

The stratosphere

Above the tropopause lies the **stratosphere**, which extends to about 50 km. In the standard atmosphere the temperature stays constant at −56.5 °C from 11 to 20 km and then increases with height, reaching about 0 °C at the **stratopause**. This warming is caused by **ozone** (O_3), which absorbs ultraviolet radiation from the Sun. Most ozone lies between about 15 and 35 km, but the heating *per kilogram of air* is greatest near the top of the stratosphere, where the air is thin and the ultraviolet sunlight is most intense.

Warm air lying on top of colder air is very stable (Section 7.1), so there is little vertical mixing in the stratosphere and the air tends to lie in horizontal layers, or *strata*. This is why long-haul aircraft like to cruise in the lowest part of the stratosphere, where the air is usually smooth, and why particles injected into the stratosphere by large volcanic eruptions can remain there for a year or more.

The mesosphere and the thermosphere

In the **mesosphere** (about 50 to 85 km) the temperature falls with height once more, because there is little ozone heating and the air loses heat by radiating it to space. The **mesopause**, at about 85 to 90 km, is the coldest part of the atmosphere, at around −90 °C. Most meteors burn up in the mesosphere.

In the **thermosphere**, above the mesopause, the temperature rises steeply because the very thin gas absorbs the Sun's most energetic (extreme ultraviolet and X-ray) radiation. Temperatures reach several hundred degrees Celsius, and more than 1,000 °C when the Sun is active. Yet the air is so thin that it contains very little heat; an astronaut would not be burned by it. The aurora and the orbit of the International Space Station (about 400 km) are in the thermosphere.

Table 5.1: Summary of the layers of the atmosphere (approximate heights for mid-latitudes).

Layer	Height	Temperature trend	Main reason and notable features
Troposphere	0 to 11 km	Falls, about 6.5 °C/km	Heated from below by the surface; convection and mixing; contains almost all water vapour and weather.
Stratosphere	11 to 50 km	Constant, then rises	Ozone absorbs solar ultraviolet; very stable; contains the ozone layer.
Mesosphere	50 to 85 km	Falls	Little ozone heating; cooling by radiation to space; coldest air at the mesopause.
Thermosphere	Above about 85 km	Rises steeply	Absorption of extreme ultraviolet and X-rays; extremely thin air; aurora.

5.3 Lapse rates, stability and inversions

The lapse rate

Definition: lapse rate

The **lapse rate**, Γ (the Greek capital letter gamma), answers the question: *how many degrees colder does the air become for each kilometre we go upward?* It is defined as the rate at which temperature *decreases* with height:

$$\boxed{\Gamma = -\frac{\Delta T}{\Delta z} = \frac{T_{\text{lower}} - T_{\text{upper}}}{z_{\text{upper}} - z_{\text{lower}}}.} \tag{5.1}$$

It is usually given in °C/km. A *positive* lapse rate means that temperature falls as you go up, which is the normal situation in the troposphere. A lapse rate of zero means the temperature does not change with height (an *isothermal* layer), and a *negative* lapse rate means temperature rises with height (an *inversion*).

Finite lapse rate versus local temperature gradient

The practical mostly uses a lapse rate averaged across a finite layer,

$$\Gamma_{\text{layer}} = -\frac{\Delta T}{\Delta z} = -\frac{T(z + \Delta z) - T(z)}{\Delta z}. \tag{5.2}$$

If we shrink the layer thickness toward zero, the finite difference becomes the derivative:

$$\boxed{\Gamma(z) = -\frac{\mathrm{d}T}{\mathrm{d}z}.} \tag{5.3}$$

Thus $\mathrm{d}T/\mathrm{d}z$ is the local vertical temperature gradient. In the ordinary troposphere $\mathrm{d}T/\mathrm{d}z < 0$ because temperature decreases upward, so meteorologists place a minus sign in the definition of Γ to make the ordinary lapse rate positive.

Example: if $\mathrm{d}T/\mathrm{d}z = -6.5\,\mathrm{K/km}$, then

$$\Gamma = -(-6.5) = 6.5\,^{\circ}\mathrm{C/km}. \tag{5.4}$$

In an inversion, $\mathrm{d}T/\mathrm{d}z > 0$, so $\Gamma < 0$.

The recipe in Equation (5.1) is always the same: *temperature drop divided by height gain*. The only place where it is easy to go wrong is with negative temperatures, so it helps to write the subtraction out in full.

Where this equation comes from

Where it comes from. Unlike the hydrostatic equation or the gas law, the lapse rate is not derived from a deeper physical law at all — it is simply a *definition*, in exactly the same way that 'speed' is defined as distance divided by time. It is a convenient, standard way of describing something we can measure (how temperature varies with height) without yet explaining *why* it varies that way. The minus sign in Equation (5.1) is a deliberate choice of convention: temperature normally *falls* with height in the troposphere, and meteorologists wanted that everyday, normal case to correspond to a positive number.

What each symbol means.

Symbol	Unit (here)	Meaning
Γ	°C/km	Lapse rate: how many degrees the temperature falls for every kilometre climbed.
T_{lower}, T_{upper}	°C	Temperature at the bottom and top of the layer.
z_{lower}, z_{upper}	km	Height of the bottom and top of the layer.
$\Delta T = T_{\text{upper}} - T_{\text{lower}}$	°C	Change in temperature across the layer (negative when it gets colder going up).
$\Delta z = z_{\text{upper}} - z_{\text{lower}}$	km	Change in height across the layer (always positive here, since 'upper' is higher).

Where the numbers come from. There is no universal constant in this equation — every lapse rate you calculate in this practical (6.6 °C/km over Hillsborough in Question 9, for instance) is worked out fresh from two temperatures and two heights that were *measured* by a specific radiosonde on a specific day (Section 6.2). The one number that recurs, 6.5 °C/km, is different: it is not measured on one day but is the long-term, whole-globe *average* environmental lapse rate, adopted by international agreement as part of the standard atmosphere (Section 5 explains this further). It is close to, but distinctly smaller than, the dry adiabatic lapse rate derived just below — the reason for that gap is explored in Question 13.

Worked example: the lapse rate of the standard atmosphere

In the standard atmosphere the temperature is 15 °C at sea level and −56.5 °C at 11 km. The lapse rate is

$$\Gamma = \frac{T_{\text{lower}} - T_{\text{upper}}}{z_{\text{upper}} - z_{\text{lower}}} = \frac{15 - (-56.5)}{11 - 0} = \frac{71.5}{11} = 6.5\,^{\circ}\text{C/km}. \tag{5.5}$$

Notice that subtracting a negative number is the same as adding: $15 - (-56.5) = 15 + 56.5 = 71.5$.

What the question is asking

The 'standard atmosphere' is an agreed, idealised description of the average state of the atmosphere, used for example to calibrate aircraft altimeters. It starts at 15 °C at sea level and cools by 6.5 °C/km up to 11 km, where it reaches the tropopause. You are asked to fill in the missing temperatures using that lapse rate. The values already given for 11,000, 14,000 and 16,000 m show what happens above the tropopause: the temperature stays constant.

Where this equation comes from

Where it comes from. The working formula is a rearrangement of the finite lapse-rate definition in Equation (5.1). Fix the lower level at the surface ($z_{\text{lower}} = 0$, $T_{\text{lower}} = 15\,^{\circ}\text{C}$) and let the upper level be the general height z:

$$\Gamma = \frac{15 - T(z)}{z - 0} \quad \Longrightarrow \quad \boxed{T(z) = 15 - \Gamma z}. \tag{5.6}$$

For the standard lapse rate $\Gamma = 6.5\,^{\circ}\mathrm{C/km}$, Equation (5.6) becomes

$$\boxed{T(z) = 15 - 6.5z}, \qquad z \text{ in km.} \tag{5.7}$$

So this equation contains no new physics beyond Section 5.3 — it is that same definition, simply run in the opposite direction (given Γ and a starting temperature, predict T at any height, rather than given two temperatures, calculate Γ).

Where the numbers come from. Unlike g or R_d, none of the three numbers here — 15 °C, 6.5 °C/km, and the sea-level pressure 1,013.25 hPa used elsewhere in the standard atmosphere — is derived mathematically from anything more fundamental. They are all part of the **International Standard Atmosphere (ISA)**, a single, agreed reference profile adopted by the International Civil Aviation Organization (building on earlier national standards from the 1920s) so that instrument manufacturers, pilots and scientists everywhere use exactly the same reference numbers. The values were chosen, by international committee, to be reasonably representative of the real global-average atmosphere: 15 °C (288.15 K) approximates the mean sea-level temperature of the Earth, 1,013.25 hPa the mean sea-level pressure, and 6.5 °C/km the mean tropospheric lapse rate found by averaging huge numbers of real radiosonde and aircraft observations. None of the three follows from theory alone — they are conventions, fixed once and then used by everyone for consistency, in exactly the way the 1,000 hPa used elsewhere in this practical is a rounded convention rather than a physically derived number.

Method, step by step

1. Convert each height to **kilometres**, because the lapse rate is given per kilometre.
2. Work out how much cooler it is than at sea level: $6.5 \times z$ (with z in km).
3. Subtract that from the sea-level temperature using Equation (5.7).
4. Check your arithmetic with the values already given: at 1,000 m you should get 8.5, and at 11,000 m you should get -56.5.

Worked solution

Using Equation (5.7):

$$\begin{aligned}
2{,}000\,\mathrm{m}: &\quad T = 15 - 6.5 \times 2 = 15 - 13 = 2\,^{\circ}\mathrm{C} \\
4{,}000\,\mathrm{m}: &\quad T = 15 - 6.5 \times 4 = 15 - 26 = -11\,^{\circ}\mathrm{C} \\
6{,}000\,\mathrm{m}: &\quad T = 15 - 6.5 \times 6 = 15 - 39 = -24\,^{\circ}\mathrm{C} \\
8{,}000\,\mathrm{m}: &\quad T = 15 - 6.5 \times 8 = 15 - 52 = -37\,^{\circ}\mathrm{C} \\
10{,}000\,\mathrm{m}: &\quad T = 15 - 6.5 \times 10 = 15 - 65 = -50\,^{\circ}\mathrm{C} \\
\text{check at } 11{,}000\,\mathrm{m}: &\quad T = 15 - 6.5 \times 11 = 15 - 71.5 = -56.5\,^{\circ}\mathrm{C}\ \checkmark
\end{aligned} \tag{5.8}$$

A quick way to see the pattern: every 2,000 m the temperature falls by $2 \times 6.5 = 13\,^{\circ}\mathrm{C}$, so the sequence from 2,000 m to 10,000 m is 2, −11, −24, −37, −50.

Above 11,000 m the standard atmosphere is isothermal: the temperature stays at −56.5 °C all the way up to 20 km. This is the lower stratosphere (Section 5.2), and 11,000 m is the tropopause of the standard atmosphere.

Common pitfalls

- **Multiplying by the height in metres.** $6.5 \times 2000 = 13000$ is clearly wrong. Either convert to km first, or use the lapse rate per metre, $0.006{,}5\,^{\circ}\mathrm{C/m}$.
- **Sign errors below zero.** $15 - 26 = -11$, not $+11$. Once the temperatures are negative, each one should be *more* negative than the one below it.
- **Carrying on cooling above** 11 km**.** Continuing at $6.5\,^{\circ}\mathrm{C/km}$ would give $-76\,^{\circ}\mathrm{C}$ at 14 km. The table itself shows that the cooling stops at the tropopause.

Check your understanding

Q5.1 What is the temperature of the standard atmosphere at 5.6 km?

Q5.2 At what height does the standard atmosphere reach $0\,^{\circ}\mathrm{C}$? (This height is called the *freezing level*.)

6 Question 6: plotting the radiosonde profiles

Question 6

handout Table 3a and Table 3b give the pressure, height and temperatures from a radiosonde measurement taken on the 21st March 1990 for Hillsborough (Northern Ireland) and St John's, Newfoundland (Canada). Plot the temperature values versus height for these stations and for the standard atmosphere (handout Table 2) on the graph on the final page (use different colours for the different plots).

Table 3a. Hillsborough (21/03/1990)

Pressure (hPa)	Height (m)	Temperature (°C)
1001	38	11.05
853	1330	−1.25
850	1358	−1.25
835	1503	−1.05
735	2504	−7.45
573	4405	−15.25
549	4725	−18.05
500	5414	−22.45
460	6025	−25.45
400	7018	−32.85
282	9383	−51.05
250	10164	−49.05
216	11116	−44.65
150	13524	−49.65
120	14969	−53.85
100	16126	−56.45
70	18379	−58.65
50	20482	−58.45
30	23691	−60.05
20	26218	−56.25

Table 3b. St John's (21/03/1990)

Pressure (hPa)	Height (m)	Temperature (°C)
1005	140	−1.65
938	687	−2.05
700	3006	−2.85
612	4062	−6.65
500	5604	−15.85
458	6263	−19.85
400	7248	−27.05
307	9103	−41.05
250	10446	−53.45
212	11482	−62.05
200	11843	−58.05
191	12141	−54.65
100	16254	−56.25
88.9	17004	−55.85
73.6	18190	−56.65
50.5	20566	−54.85
30	23834	−56.25
20	26404	−54.05
18.1	27047	−53.65
10	30828	−54.05

What Question 6 means in ordinary words

Each row of the radiosonde tables is one measurement made at a particular height as a weather balloon rose. To turn the list into a vertical picture of the atmosphere, put **temperature on the horizontal axis** and **height on the vertical axis**. Plot every measured pair (T, z) for Hillsborough and St John's, then add the standard-atmosphere values. Joining each profile in height order lets you see where temperature falls, stays nearly constant or rises.

Questions 6–14 are one continuous investigation

Do not treat the final questions as separate exercises. They form one chain:

1. **Q6 - Build the evidence:** turn the tables into vertical temperature profiles.
2. **Q7 - Find reversals:** identify layers where temperature rises with height (inversions).
3. **Q8 - Find the boundary:** locate the first major level where normal tropospheric cooling stops (the tropopause).
4. **Q9 - Quantify the slope:** use Hillsborough's surface and tropopause values to calculate an average lapse rate.
5. **Q10 - Add a tropical comparison:** plot Florida on the same axes and locate its tropopause by the same rule.
6. **Q11 - Compare the four profiles:** infer that tropospheric thickness varies geographically, then explain why warm tropical columns are deeper.
7. **Q12 - Return to pressure:** compare two almost equal height intervals and distinguish absolute from fractional pressure change.
8. **Q13 - Correct the buoyancy slogan:** use pressure, density and adiabatic warming to explain why cold upper air does not simply fall to the ground.
9. **Q14 - Connect atmosphere and ocean:** compare the two surface temperatures and explain the maritime/ocean-current contrast and its seasonality.

The graph you construct in Question 6 is therefore not just an answer to one question: it is the evidence base for Question 7, Question 8, Question 9, Question 10 and Question 11.

Q6 plotting recipe: exactly what value goes where

For this graph every plotted point has the form $(x, y) = (T, z)$.

1. **Horizontal x-axis = temperature, T, in °C.** Colder/more negative values are further left; warmer/more positive values are further right.
2. **Vertical y-axis = height, z, in metres.** Larger heights are higher on the page.
3. **Do not plot pressure in Q6.** Pressure is the first column of Tables 3a and 3b, but the question asks for *temperature versus height*, so use only the second and third columns for coordinates.
4. **Example - Hillsborough.** The row 500 hPa, 5414 m, −22.45 °C becomes the point

$$\boxed{(x, y) = (-22.45,\ 5414).} \tag{6.1}$$

   The 500 hPa value identifies the pressure level but is not a coordinate on this graph.
5. **Example - St John's.** The row 250 hPa, 10446 m, −53.45 °C becomes

$$\boxed{(x, y) = (-53.45,\ 10446).} \tag{6.2}$$

6. **Example - standard atmosphere.** Table 2 says −11 °C at 4,000 m, so plot $(-11, 4000)$.
7. **Plot in ascending height order.** Start with the lowest point for one dataset and work upward row by row. This prevents you from accidentally joining unrelated levels.
8. **Join neighbouring points in the same dataset with straight segments.** Do not connect a Hillsborough point to a St John's point. Do not fit one smooth curve through all datasets.
9. **Give each profile a distinct colour/symbol and make a legend.** Keep the same identity when Florida is added in Question 10.

If you are drawing your own axes rather than using the handout's final-page grid, a practical common scale is about −80 °C to 40 °C horizontally and 0 to 35,000 m vertically. That leaves room for Florida and all radiosonde points.

6.1 Where the sounding data come from and how to plot them

A radiosonde table is not just a list of arbitrary numbers: it is a vertical sequence of measurements made as a balloon ascends. Understanding standard pressure levels, significant levels and plotting conventions explains why the heights are irregularly spaced and why measured points should be joined in height order.

6.2 Measuring the atmosphere in the vertical: radiosondes

How a radiosonde works

The data you will plot in this practical were measured by **radiosondes**. A radiosonde is a small instrument package carried upward through the atmosphere by a hydrogen- or helium-filled balloon (Figure 6.2). As it rises, at about 5 m/s, it measures pressure, temperature and humidity and transmits the readings to a receiving station on the ground by radio, every second or two. Modern radiosondes also carry a GPS receiver, and the way the sonde drifts with the wind gives the wind speed and direction at every level. The balloon expands as the pressure around it falls,

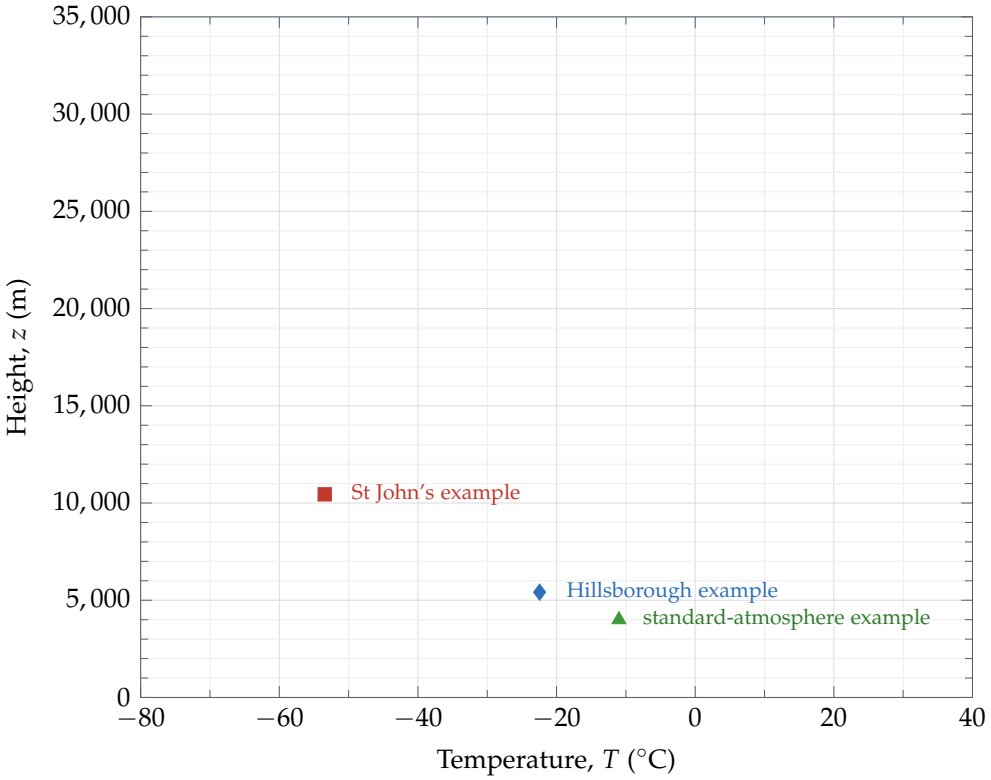


Figure 6.1: How one row becomes one plotted point in Question 6. The horizontal coordinate comes from temperature; the vertical coordinate comes from height. Pressure is deliberately not plotted on these axes.

until, after an hour and a half to two hours, at a height of about 30 to 35 km, it bursts; the sonde then falls back to Earth on a small parachute.

Radiosondes are launched simultaneously, at 00 and 12 UTC (midnight and midday Greenwich time), from hundreds of stations around the world, giving forecasters a three-dimensional snapshot of the atmosphere twice a day. A set of measurements from one ascent is called a **sounding** or an **upper-air profile**.

Reading a sounding

A sounding is usually reported as a list of levels, each with a pressure, a height and a temperature, exactly like handout Table 3a and Table 3b of the practical. Two kinds of level appear in such lists.

- **Standard (mandatory) pressure levels**, such as 850 hPa, 500, 250 and 100, which are reported by every station so that the soundings can be compared and mapped.
- **Significant levels**, chosen wherever the profile changes direction or slope, for example at the base and top of an inversion or at the tropopause. They are selected so that joining the listed points with *straight lines* reproduces the measured profile closely.

The levels listed in handout Table 3a and Table 3b are a mixture of the two (for example, 850 hPa and 500 are standard levels, while 853 hPa, 735 and 282 over Hillsborough are significant levels). This has three practical consequences for you: the points are unevenly spaced in height; you should join them with straight lines rather than a smooth curve; and the tropopause and the edges of any inversions will themselves appear as listed points.

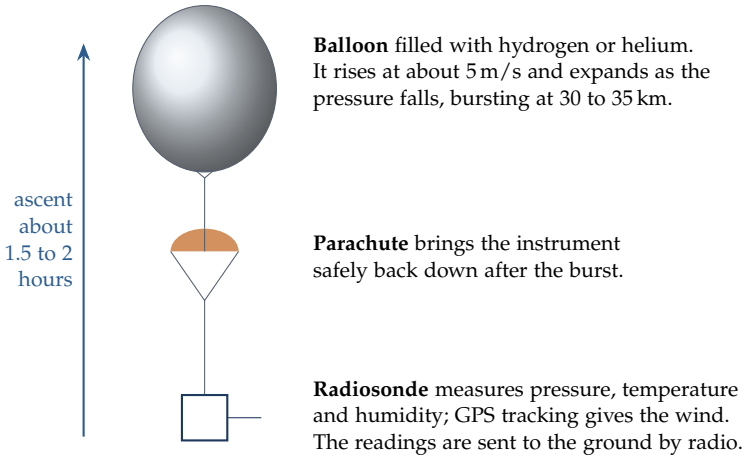


Figure 6.2: A radiosonde flight (not to scale; in reality the instrument hangs tens of metres below the balloon).

Plotting conventions

Atmospheric scientists normally plot height on the vertical axis and temperature on the horizontal axis, because that makes the graph look like the atmosphere itself, with 'up' being up. This is the layout used throughout this practical. (Professional forecasters in the UK plot soundings on a special chart called a *tephigram*, which uses a logarithmic pressure scale instead of height, but the idea is the same.)

Check your understanding

6.1 In which layer of the atmosphere do weather balloons usually burst?

6.2 Why is the upper stratosphere warmer than the lower stratosphere?

6.3 Convert $-56.5\,^{\circ}\mathrm{C}$ to kelvin.

6.4 The temperature falls from $20\,^{\circ}\mathrm{C}$ at the ground to $-10\,^{\circ}\mathrm{C}$ at 5 km. What is the lapse rate? Is this air more or less stable than the standard atmosphere?

6.5 Give two reasons why inversions matter for people living beneath them.

What the question is asking

Each row of handout Table 3a and Table 3b is one level measured by the radiosonde as it rose (Section 6.2). Plotting temperature against height turns the list of numbers into a picture of the vertical structure of the atmosphere over each station on that day. The standard atmosphere from handout Table 2 is plotted as a reference, so that you can see how the real atmosphere differs from the average. You will add a fourth curve (Florida) in Question 10, so plan your axes to leave room for it.

Method, step by step

1. **Axes.** Put temperature on the horizontal axis and height on the vertical axis, so that 'up' on the page is up in the atmosphere. The data (including Florida in Question 10) run from about −76 °C to 27 °C and from 0 to about 31,000 m. A temperature axis from −80 °C to 40 °C (for example 20 °C per large square) and a height axis from 0 to 35,000 m (for example 5,000 m per large square) will fit everything. Temperatures increase to the right, so the negative values are on the left.
2. **Colours and symbols.** Choose one colour per profile and keep to it, for example blue for Hillsborough, red for St John's, green for the standard atmosphere and purple for Florida. Using a different symbol for each profile as well (dot, square, triangle, cross) helps if the graph is photocopied in black and white.
3. **Plot the points.** For each row, find the height on the vertical axis and the temperature on the horizontal axis, and mark the point. Remember that the heights in handout Table 2 and handout Table 3a and Table 3b are in *metres*.
4. **Join the points of each profile** in order of height with straight lines (Section 6.2 explains why straight lines, rather than a smooth curve, are the right choice for radiosonde data).
5. **Add a key** (legend) identifying each profile, and a title.

Worked solution

Figure 6.4 shows the three profiles. The completed graph for the whole practical, including the marks for Questions 7 and 8 and the Florida profile of Question 10, is Figure 10.1.

What to notice.

- **Below about** 9,000 m all three profiles lean to the left: temperature falls with height at broadly similar rates (compare the slopes), but each starts from a different surface temperature: 15 °C in the standard atmosphere, 11.05 °C at Hillsborough and −1.65 °C at St John's.
- **Each profile has a 'corner'** between about 9,000 and 12,000 m, where the temperature stops falling. This is the tropopause (Question 8).
- **Above the corner** the temperatures stay roughly constant, between about −54 °C and −62 °C: the lower stratosphere, close to the standard atmosphere's −56.5 °C.
- **St John's is almost vertical in the lowest** 3,000 m: the temperature changes by only 1.2 °C between 140 and 3,006 m. Such a layer is extremely stable.
- **Above about** 1,700 m**, St John's is warmer than Hillsborough**, by 6 to 8 °C, even though it is 12.7 °C colder at the surface. We return to this in Question 14.

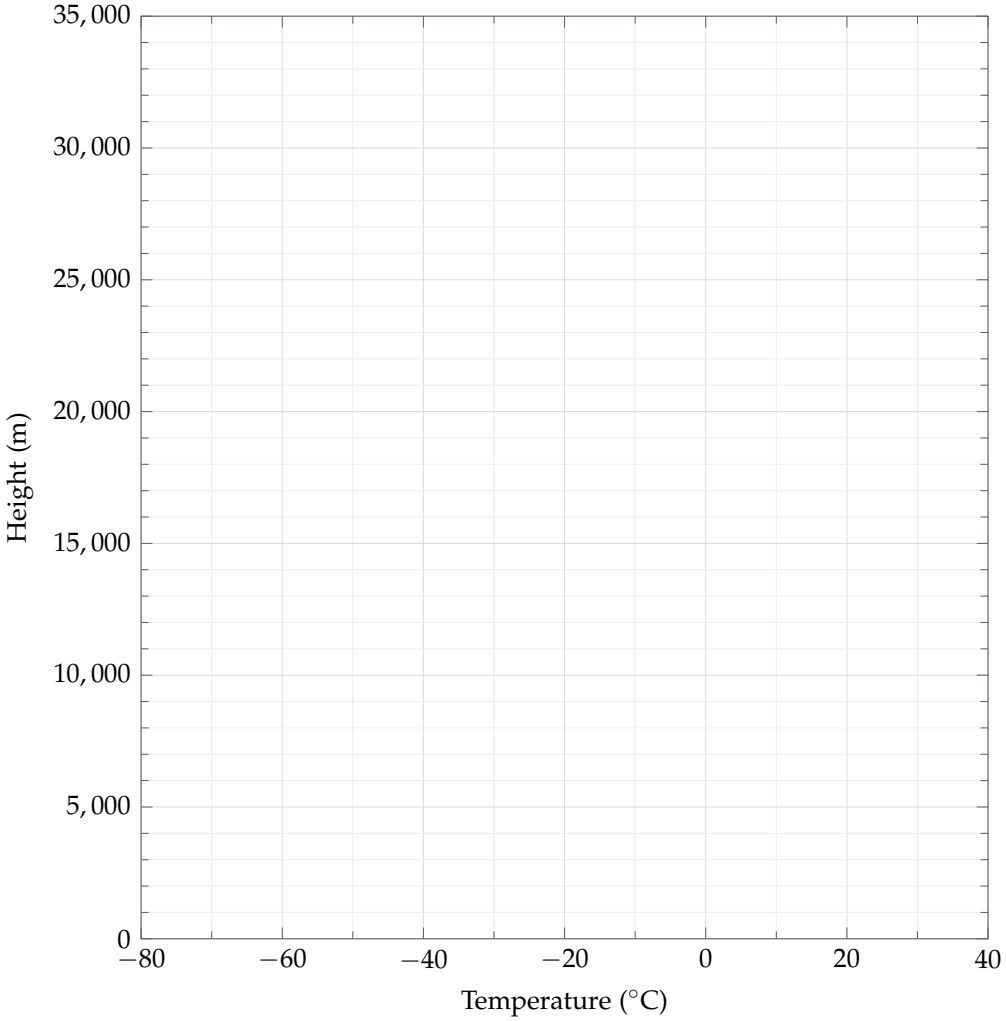


Figure 6.3: Blank plotting template for Questions 6–10. Plot Hillsborough, St John's and the standard atmosphere first; then add Florida in Question 10. Use Figure 6.1 if you are unsure which table columns become x and y.

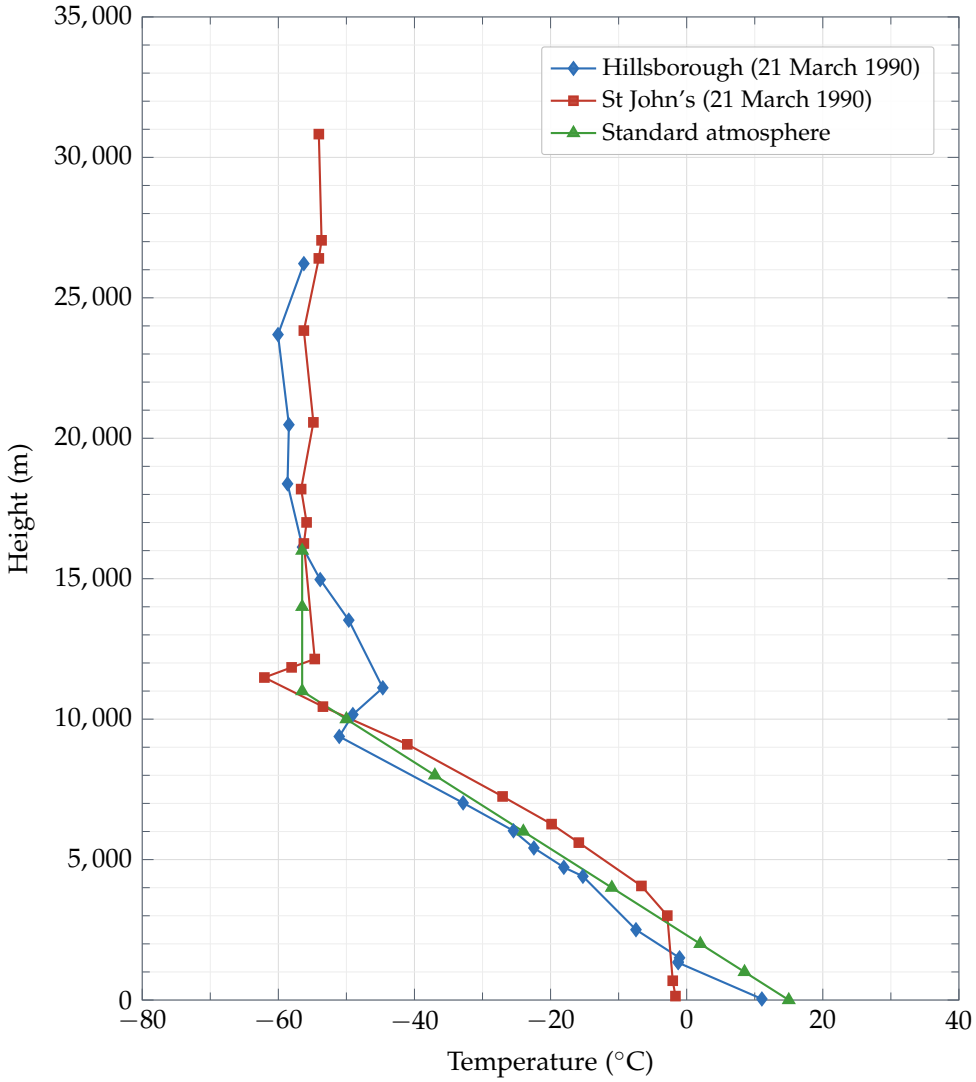


Figure 6.4: Completed plot for Question 6: temperature against height for Hillsborough and St John's on 21 March 1990 (handout Table 3a and Table 3b), with the standard atmosphere (handout Table 2) for comparison. Points are joined with straight lines.

Common pitfalls

- **Misreading large heights.** 10,164 m is ten thousand one hundred and sixty-four metres; it is easy to plot it at 1,016 m by mistake. Check that each profile rises steadily up the page.
- **A temperature axis that runs the wrong way.** Colder (more negative) temperatures belong on the left, warmer on the right.
- **Uneven scales.** Each large square must represent the same number of degrees (or metres) all the way along the axis.
- **Mixing up the columns.** In handout Table 3a and Table 3b the first column is pressure, not height. Plot height (second column) against temperature (third column).
- **No key.** Without a legend, a graph with four coloured lines cannot be read by anyone else.

How to read the finished profile before answering Q7–Q11

Once the points are connected, read the graph from the bottom upward. Because temperature is on the horizontal axis and height on the vertical axis:

- line moving **up and left**: temperature is decreasing with height - the usual tropospheric behaviour;
- line nearly **vertical**: temperature changes very little with height - nearly isothermal;
- line moving **up and right**: temperature is increasing with height - an inversion (Question 7);
- the **first major corner** where persistent leftward cooling stops: candidate tropopause (Question 8);
- the vertical distance from the surface to that corner: tropospheric thickness, compared among profiles in Question 11.

The graph shows the pattern; the tables give the exact numbers. Use the graph to identify *where to look*, then return to the relevant table row for the precise height and temperature.

7 Question 7: inversion layers

Question 7

You will see that in some situations the temperature increases with height in the troposphere. Such layers are called *inversion layers*. Circle these on the graph.

What Question 7 means in ordinary words

Normally, your temperature profile moves left toward colder values as you go upward. An **inversion** is the opposite: over some height interval, the air gets warmer as you climb. On a temperature-versus-height graph, that part of the line bends or leans to the **right** as height increases. Question 7 asks you to find and circle those height intervals, not merely layers where cooling is slow.

7.1 Why inversions matter: adiabatic motion and stability

An inversion is identified geometrically from the sounding, but its importance comes from parcel physics. To understand why an inversion is such a strong lid on vertical mixing, we derive the dry adiabatic lapse rate from the first law of thermodynamics, distinguish environmental from parcel lapse rates, and then use those ideas to define stability.

Environmental and adiabatic lapse rates

It is important to distinguish between two different kinds of lapse rate.

The **environmental lapse rate** (ELR) describes the *actual* temperature of the air at different heights at a given place and time, as measured by a weather balloon. It changes from day to day and from layer to layer. Averaged over the globe and over time, the environmental lapse rate in the troposphere is about 6.5 °C/km; this is the 'average' or 'normal' lapse rate used in the standard atmosphere.

The **adiabatic lapse rates** describe what happens to a *parcel* of air (a balloon-sized blob, imagined to be wrapped in an invisible skin) when it is lifted or lowered without exchanging heat with its surroundings. A rising parcel moves into lower pressure, so it expands; in expanding it pushes the surrounding air out of the way, uses up some of its own internal energy and cools. A sinking parcel is compressed and warms. You have met both effects if you have felt a bicycle pump warm up as you compress the air in it, or felt the cold spray from an aerosol can as the gas expands.

- An unsaturated ('dry') parcel cools at the **dry adiabatic lapse rate** of about 9.8 °C/km.
- A saturated parcel, inside a cloud, cools more slowly, at the **saturated adiabatic lapse rate**, typically 4 to 7 °C/km, because condensation releases latent heat that partly offsets the cooling.

Going further: deriving the dry adiabatic lapse rate from the first law

The result

$$\boxed{\Gamma_d \approx 9.8\,^{\circ}\mathrm{C/km}} \tag{7.1}$$

is not an arbitrary number. It follows from energy conservation. The derivation looks compact in textbooks, so here every step is unpacked.

1. What is an air parcel? An air parcel is an imagined small blob of air that moves while remaining identifiable. It is not enclosed by a rigid wall. As pressure outside changes, the parcel can expand or contract. We assume it stays close enough to pressure equilibrium with its surroundings that its internal pressure is essentially the environmental pressure at the same height.

2. What does adiabatic mean? A process is **adiabatic** when no heat crosses the parcel boundary. In differential notation,

$$\boxed{\delta q = 0.} \tag{7.2}$$

Thermodynamics often writes heat as δq rather than $\mathrm{d}q$ because heat is energy transferred during a process, not a state variable possessed by the parcel. For this introductory derivation the important statement is simply: no heat is supplied from outside and none is removed to outside.

3. First law per unit mass. For one kilogram of dry air, the first law can be written

$$\delta q = c_v \, \mathrm{d}T + p \, \mathrm{d}\alpha, \tag{7.3}$$

where

$$\alpha = \frac{1}{\rho} \tag{7.4}$$

is **specific volume**, the volume occupied by one kilogram of air. Its units are m^3/kg. The term $c_v \mathrm{d}T$ is the change in internal energy per kilogram for an ideal gas; the term $p\mathrm{d}\alpha$ is expansion work per kilogram.

Why does $p\mathrm{d}\alpha$ represent work? Mechanical work is force times distance. A pressure p acting over area A gives force pA. If a parcel boundary moves outward a small distance $\mathrm{d}x$, the work is

$$\mathrm{d}W = (pA)\mathrm{d}x = p(A\mathrm{d}x) = p \, \mathrm{d}V. \tag{7.5}$$

For one kilogram, V is the specific volume α, so the work per kilogram is $p\mathrm{d}\alpha$.

For an adiabatic parcel, set $\delta q = 0$:

$$0 = c_v \mathrm{d}T + p\mathrm{d}\alpha. \tag{7.6}$$

4. Rewrite the gas law using specific volume. Since $\rho = 1/\alpha$, the dry-air gas law

$$p = \rho R_\mathrm{d} T \tag{7.7}$$

becomes

$$p\alpha = R_\mathrm{d} T. \tag{7.8}$$

5. Differentiate the gas law. Both p and α can change as the parcel moves, so differentiate the product $p\alpha$ using the product rule:

$$\mathrm{d}(p\alpha) = p\mathrm{d}\alpha + \alpha \mathrm{d}p. \tag{7.9}$$

The right side of Equation (7.8) gives

$$\mathrm{d}(R_\mathrm{d}T) = R_\mathrm{d}\mathrm{d}T \tag{7.10}$$

because R_d is constant. Hence

$$p\mathrm{d}\alpha + \alpha\mathrm{d}p = R_\mathrm{d}\mathrm{d}T. \tag{7.11}$$

Solve for $p\mathrm{d}\alpha$:

$$p\mathrm{d}\alpha = R_\mathrm{d}\mathrm{d}T - \alpha\mathrm{d}p. \tag{7.12}$$

6. Substitute into the first law. Put Equation (7.12) into Equation (7.6):

$$\begin{aligned} 0 &= c_v\mathrm{d}T + \big(R_\mathrm{d}\mathrm{d}T - \alpha\mathrm{d}p\big) \\ &= (c_v + R_\mathrm{d})\mathrm{d}T - \alpha\mathrm{d}p. \end{aligned} \tag{7.13}$$

Move the pressure term to the other side:

$$(c_v + R_\mathrm{d})\mathrm{d}T = \alpha\mathrm{d}p. \tag{7.14}$$

For an ideal gas, Mayer's relation is

$$\boxed{c_p = c_v + R_\mathrm{d}.} \tag{7.15}$$

Using Equation (7.15),

$$\boxed{c_p\mathrm{d}T = \alpha\mathrm{d}p.} \tag{7.16}$$

7. Connect pressure change to height change. A parcel displaced slowly through a hydrostatic environment experiences the environmental pressure change

$$\mathrm{d}p = -\rho g\mathrm{d}z. \tag{7.17}$$

Use $\alpha = 1/\rho$ in Equation (7.16):

$$\begin{aligned} c_p\mathrm{d}T &= \frac{1}{\rho}\big(-\rho g\mathrm{d}z\big) \\ &= -g\mathrm{d}z. \end{aligned} \tag{7.18}$$

The density cancels:

$$c_p\mathrm{d}T = -g\mathrm{d}z. \tag{7.19}$$

Divide by $c_p\mathrm{d}z$:

$$\boxed{\frac{\mathrm{d}T}{\mathrm{d}z} = -\frac{g}{c_p}.} \tag{7.20}$$

Because meteorological lapse rate is defined as $\Gamma = -\mathrm{d}T/\mathrm{d}z$,

$$\boxed{\Gamma_d = \frac{g}{c_p}.} \tag{7.21}$$

8. Insert the numbers. For dry air near ordinary atmospheric temperatures,

$$c_p \approx 1{,}005\,\mathrm{J/(kg\,K)}. \tag{7.22}$$

Therefore

$$\Gamma_d = \frac{9.81\,\mathrm{m/s^2}}{1{,}005\,\mathrm{J/(kg\,K)}}. \tag{7.23}$$

Since $1\,\mathrm{J} = 1\,\mathrm{kg\,m^2/s^2}$,

$$\frac{\mathrm{m/s^2}}{\mathrm{m^2/(s^2\,K)}} = \mathrm{K/m}. \tag{7.24}$$

Numerically,

$$\begin{aligned} \Gamma_d &= 0.009{,}76\,\mathrm{K/m} \\ &= 9.76\,\mathrm{K/km} \\ &= 9.76\,^\circ\mathrm{C/km} \approx 9.8\,^\circ\mathrm{C/km}. \end{aligned} \tag{7.25}$$

A change of one kelvin has the same size as a change of one degree Celsius, which is why the numerical lapse rate is the same in K/km and °C/km.

9. Physical interpretation. As an unsaturated parcel rises, environmental pressure decreases. The parcel expands and does work on its surroundings. With no heat supplied from outside, that work must come from the parcel's internal energy, so its temperature falls. The dry adiabatic lapse rate is the rate at which that cooling occurs in the idealised dry, reversible limit.

10. Why saturated air cools more slowly. If the parcel is saturated, some water vapour condenses as the parcel cools. Condensation releases latent heat into the parcel. That energy partly offsets the cooling caused by expansion, so the saturated adiabatic lapse rate is smaller than the dry value. Unlike Γ_d, it is not one universal constant; it varies with temperature, pressure and moisture content, commonly lying around 4–7 °C/km in the troposphere.

Stability in brief

Imagine nudging a parcel of air upwards. It cools at the adiabatic rate. If the surrounding air cools with height *more slowly* than that, the parcel soon finds itself colder, and therefore denser, than its surroundings, and it sinks back to where it started: the air is **stable**. If the surrounding air cools with height *faster* than the parcel does, the parcel stays warmer and lighter than its surroundings and keeps rising: the air is **unstable**, which leads to convection, cumulus clouds and showers.

So the smaller the environmental lapse rate, the more stable the air. An isothermal layer is very stable, and an inversion, in which the temperature actually increases with height, is the most stable arrangement of all.

Temperature inversions

Definition: temperature inversion

A **temperature inversion** is a layer of air in which the temperature *increases* with height, the reverse (inversion) of the usual tropospheric pattern. Its lapse rate is negative.

Inversions form in several ways:

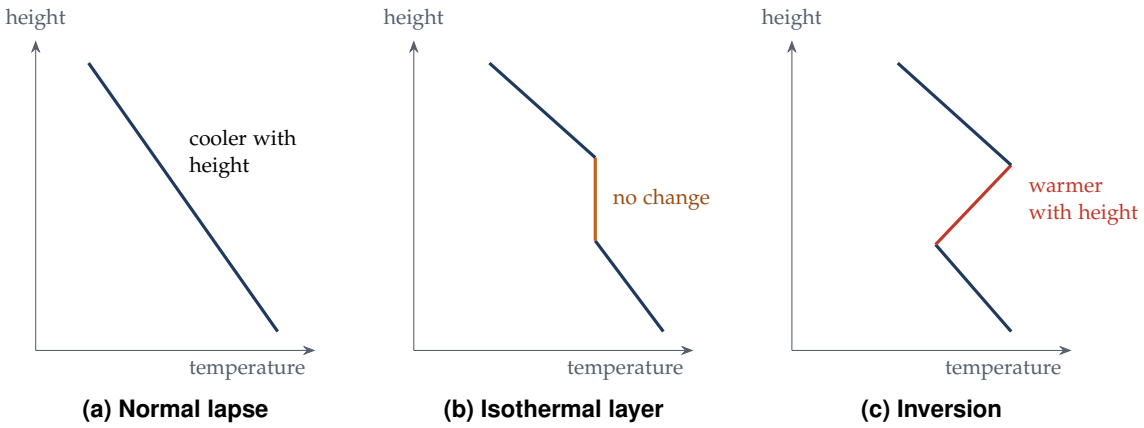


Figure 7.1: Three kinds of temperature profile. (a) The normal tropospheric pattern, with temperature falling with height. (b) An isothermal layer, in which temperature does not change with height. (c) An inversion, in which temperature rises with height. Layers (b) and (c) are very stable and act as a lid on vertical motion.

- **Radiation (nocturnal) inversions.** On clear, calm nights the ground loses heat by radiation and cools the air next to it, leaving warmer air above. These are responsible for frost hollows and valley fog.
- **Subsidence inversions.** In areas of high pressure, air sinks slowly over a large area and warms by compression. A warm layer forms, typically one to two kilometres above the ground, capping the air below.
- **Frontal inversions.** At weather fronts, warm air rides up and over colder, denser air, so the temperature rises as you go from the cold air into the warm air above it.
- **Advection inversions.** Warm air blowing over a cold surface, such as cold sea, is cooled from below.
- **The stratospheric inversion.** On the largest scale, the whole of the upper stratosphere is an inversion, caused by ozone heating. A sharp temperature increase is often found just above the tropopause.

Because they are so stable, inversions act as *lids*. They stop air from rising through them, so they trap smoke, pollution, haze and fog beneath them, flatten the tops of clouds and suppress showers. Severe pollution episodes in cities almost always occur under an inversion.

What the question is asking

An inversion (Section 7.1) is any layer in which the temperature *rises* with height. On your graph, a normal tropospheric profile leans to the left as it goes up (colder with height). An inversion is a stretch where the line leans to the *right* instead (warmer with height). You are asked to find these stretches and circle them.

Method, step by step

1. Look along each profile on your graph for any stretch where the line moves to the right as it goes up.
2. Confirm it in handout Table 3a and Table 3b: an inversion is present wherever a temperature is *higher* (less negative) than the temperature in the row *below* it. For example, at Hillsborough the temperature goes from −1.25 at 1,358 m to −1.05 at 1,503 m: a rise of 0.2 °C.
3. Circle each inversion you find, and note its height range.

Worked solution

Scanning handout Table 3a and Table 3b row by row for temperature increases gives the layers listed below. The two prominent inversions, which everyone should circle, are the ones marked with a star.

Profile	Layer (m)	Temperature change	Where it is and what it means
Hillsborough	1330–1358	−1.25 → −1.25 (no change)	An isothermal layer: very stable.
Hillsborough	1358–1503	−1.25 → −1.05 (0.2 °C)	A shallow inversion in the lower troposphere, capping the well-mixed layer below it. Too small to see at the scale of the full graph (Figure 7.2a).
Hillsborough⋆	9383–11116	−51.05 → −44.65 (6.4 °C)	A strong inversion starting at the tropopause (Figure 7.2b).
St John's⋆	11482–12141	−62.05 → −54.65 (7.4 °C)	A strong inversion starting at the tropopause (Figure 7.2b).
Florida (Question 10)	16000–18000	−75.6 → −73.3 (2.3 °C)	Warming above the tropopause, in the stratosphere.

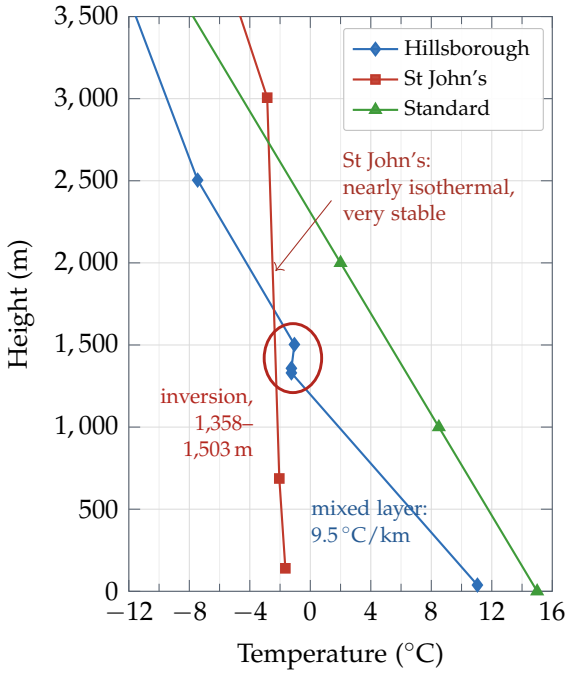
Above about 12,000 m both soundings also show several small ups and downs of one or two degrees. These are small wiggles in an almost isothermal lower stratosphere and are not normally circled.

A point worth understanding. Both prominent inversions begin exactly at the tropopause. Strictly speaking, then, they mark the *top* of the troposphere and the base of the stratosphere, rather than lying inside the troposphere. They are what makes the tropopause so easy to spot: the temperature does not merely stop falling, it jumps up by several degrees. A sharp warming like this just above the tropopause is common in middle latitudes. The only inversion that lies genuinely *within* the troposphere in these data is the small one over Hillsborough at about 1,400 m, which you can only find by reading the table.

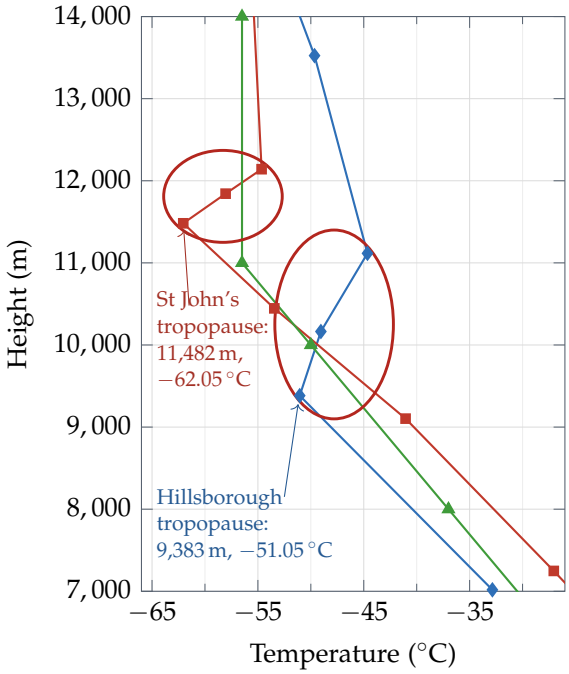
The layer beneath the Hillsborough inversion. Between the ground (38 m) and 1,330 m, the temperature at Hillsborough fell from 11.05 °C to −1.25 °C, a lapse rate of

$$\frac{11.05-(-1.25)}{1.330-0.038}=\frac{12.30}{1.292}=9.5\,^{\circ}\mathrm{C/km}, \tag{7.26}$$

very close to the dry adiabatic lapse rate of 9.8 °C/km (Section 7.1). A lapse rate this close to the dry adiabatic rate is the signature of a well-mixed layer, in which air warmed by the ground has been churned up and down by convection. The inversion at about 1,400 m acts as the lid on that mixing. Over St John's, by contrast, the lowest 3,000 m are nearly isothermal (lapse rates of only 0.3 to 0.7 °C/km): a very stable layer of cold air with no convection at all.



(a) The lowest 3,500 m.



(b) The tropopause region, 7,000–14,000 m.

Figure 7.2: Close-up views of the profiles for Questions 7 and 8. (a) The small inversion at about 1,400 m over Hillsborough, sitting on top of a well-mixed layer, and the nearly isothermal lowest layer over St John's. (b) The strong inversions (circled) that begin at the tropopause over Hillsborough and St John's. The standard atmosphere, in green, has a sharp corner but no inversion.

Common pitfalls

- **Looking only at the graph.** The Hillsborough inversion at 1,400 m is only 0.2 °C: invisible at the scale of the full graph but obvious in the table.
- **Circling layers where the temperature merely falls slowly.** A small lapse rate (such as St John's lowest 3,000 m) makes a layer very stable, but it is only an inversion if the temperature actually *rises*.
- **Circling the whole stratosphere.** The small wiggles above about 12,000 m are not what the question is after.

8 Question 8: the tropopause

Question 8

The tropopause is defined as the top boundary of the troposphere where the temperature ceases to decrease with height. Mark this boundary on the graph and note the temperature and the height for each below:

	Height (m)	Temperature (°C)
Standard atmosphere	11000	−56.5
Hillsborough	9383	−51.05
St John's	11482	−62.05

What Question 8 means in ordinary words

Start at the ground and follow each temperature profile upward. The **tropopause** is the first major level where the usual tropospheric cooling with height stops: above it, temperature becomes nearly constant or begins to rise. The question asks you to identify that turning level for each profile and report both its height and temperature.

8.1 The tropopause as a physical and operational boundary

The handout uses the intuitive definition: the level where tropospheric cooling with height ceases. We use that rule for the practical, then compare it with the formal World Meteorological Organization lapse-rate definition and explain why real tropopauses vary with latitude, season and weather.

8.2 The tropopause

The handout defines the **tropopause** as 'the top boundary of the troposphere where the temperature ceases to decrease with height'. On a plotted temperature profile it is the height at which the curve, having leaned steadily towards colder temperatures, stops doing so and turns vertical or bends back towards warmer temperatures.

Going further: the official definition

Real profiles are wiggly, so meteorologists use a precise rule agreed by the World Meteorological Organization in 1957: the tropopause is *the lowest level at which the lapse rate decreases to* 2 °C/km *or less, provided that the average lapse rate between this level and all higher levels within* 2 km *does not exceed* 2 °C/km. The second condition prevents a thin isothermal or inversion layer inside the troposphere from being mistaken for the tropopause. For the profiles in this practical, the handout's simpler definition and the official one give the same answers.

The tropopause is not at the same height everywhere (Table 8.1). It is highest over the tropics and lowest over the poles, it is generally higher in summer than in winter, and it moves up and

down with the weather: it is lower in cold air, for example beneath depressions, and higher in warm air beneath anticyclones. Question 11 explores the reasons.

Table 8.1: Typical height and temperature of the tropopause.

Region	Typical height	Typical temperature
Tropics	16 to 18 km	−75 °C to −80 °C
Mid-latitudes	9 to 13 km	−50 °C to −60 °C
Polar regions	8 to 9 km	−45 °C to −60 °C

A surprising consequence, which you will see for yourself in Question 10, is that the *coldest* tropopause temperatures on Earth are found above the *warmest* surfaces: the tropical troposphere is so deep that the air keeps cooling for many more kilometres before the tropopause is reached.

The tropopause also acts as a lid on the weather. The rising air in a thunderstorm can punch up to the tropopause, but the stable stratosphere above stops it rising much further, so the air spreads out sideways and forms the flat, anvil-shaped top of a cumulonimbus cloud.

Method, step by step

1. Follow each profile upwards from the ground. As long as the line keeps leaning to the left, you are in the troposphere.
2. The tropopause is the *first* level at which the temperature stops decreasing: above it the temperature stays the same or increases. On the graph it is the corner where the line turns vertical or bends back to the right.
3. Read the exact height and temperature of that corner from the table rather than from the graph; because the tropopause is a significant level, it appears as a row in the table.
4. Mark it on your graph, for example with a short horizontal line in the profile's colour.

Worked solution

- **Standard atmosphere:** the temperature falls steadily to −56.5 °C at 11,000 m and then stays constant. Tropopause: 11,000 m, −56.5 °C.
- **Hillsborough:** the temperature falls to −51.05 °C at 9,383 m; the next reading, at 10,164 m, is warmer (−49.05 °C). Tropopause: 9,383 m, −51.05 °C.
- **St John's:** the temperature falls to −62.05 °C at 11,482 m; the next reading, at 11,843 m, is warmer (−58.05 °C). Tropopause: 11,482 m, −62.05 °C.

The three tropopauses are marked on the completed graph, Figure 10.1, and shown close up in Figure 7.2b. In each case the tropopause is also the coldest point of the profile below 20,000 m.

Interpretation. On 21 March 1990 the tropopause was about 2 km lower over Hillsborough than over St John's, and lower than in the standard atmosphere. Hillsborough is further north, and the air column over it was colder above about 1,700 m (Question 6); colder columns are shallower, so the tropopause is lower (see Question 11).

Going further: checking against the official definition

The World Meteorological Organization definition (Section 8.2) asks for the lowest level at which the lapse rate falls to 2 °C/km or less and stays at or below that value, on average, for the next 2 km. At Hillsborough, the lapse rate just below 9,383 m is 7.7 °C/km; just above it the temperature *rises* (a negative lapse rate), and it keeps rising up to 11,116 m, well over 1.7 km higher. So 9,383 m satisfies the official definition too. The same is true of 11,482 m at St John's, where the lapse rate changes abruptly from 8.3 °C/km below to strongly negative above.

Common pitfalls

- **Choosing the coldest point anywhere in the profile.** Over Hillsborough the lowest temperature of all (−60.05 °C) is at 23,691 m, high in the stratosphere. The tropopause is the *first* point at which the cooling stops, going up from the ground.
- **Choosing the top of the inversion instead of its base.** The tropopause is where the temperature *stops falling*, which is the bottom of the warm layer (9,383 m at Hillsborough), not the top (11,116 m).

9 Question 9: the lapse rate over Hillsborough

Question 9

Based on your graph and handout Table 3a, calculate the lapse rate for the part of the atmosphere below the tropopause for Hillsborough (have another look at question 5 to work out how to calculate the lapse rate):

Lapse rate below the tropopause at Hillsborough: 6.6 °C/km

What Question 9 means in ordinary words

Use only two endpoints: Hillsborough's near-surface observation and its tropopause. Find how many degrees the temperature drops between those endpoints, find how many kilometres of height separate them, then divide **temperature drop by height gain**. The result is the *average* lapse rate of the whole troposphere below the tropopause; it does not mean every smaller layer had exactly that lapse rate.

9.1 Average lapse rate versus local lapse rate

Here the required value is an *average* over the whole Hillsborough troposphere. That means a finite temperature difference divided by a finite height difference, not the derivative at one particular level. The calculation is therefore the same rise/run logic used earlier for average gradients, with the meteorological sign convention chosen so ordinary cooling with height gives a positive lapse rate.

Method, step by step

1. Take the bottom of the layer from the first row of handout Table 3a (the surface) and the top of the layer from Question 8 (the tropopause).
2. Temperature drop: $T_{\text{surface}} - T_{\text{tropopause}}$. Be careful with the minus signs.
3. Height gain: $z_{\text{tropopause}} - z_{\text{surface}}$, converted to kilometres.
4. Lapse rate = temperature drop ÷ height gain, in °C/km.

Worked solution

Surface: 38 m, 11.05 °C. Tropopause: 9,383 m, −51.05 °C.

$$\begin{aligned} \text{temperature drop} &= 11.05 - (-51.05) = 11.05 + 51.05 = 62.10\,^{\circ}\text{C} \\ \text{height gain} &= 9383 - 38 = 9{,}345\,\text{m} = 9.345\,\text{km} \\ \Gamma &= \frac{62.10}{9.345} = 6.65\,^{\circ}\text{C/km} \approx 6.6\,^{\circ}\text{C/km} \end{aligned} \tag{9.1}$$

(If you divide by the height in metres, you get $62.10/9345 = 0.006{,}65\,^{\circ}\text{C/m}$; multiply by 1000 to convert to °C/km.)

Interpretation. Averaged over the whole troposphere, the air over Hillsborough cooled with height at 6.6 °C/km, almost exactly the 6.5 °C/km of the standard atmosphere. This

average hides a lot of variation, however: the lapse rate was about 9.5 °C/km in the lowest 1,300 m, negative in the small inversion above that, and between about 4 and 9 °C/km in the layers further up (see the next box).

Going further: layer by layer, and the other profiles

Applying the same calculation to each pair of neighbouring rows in handout Table 3a gives the lapse rate of each layer:

Layer (m)	Lapse rate (°C/km)	Layer (m)	Lapse rate (°C/km)
38–1330	9.5	4725–5414	6.4
1330–1358	0 (isothermal)	5414–6025	4.9
1358–1503	−1.4 (inversion)	6025–7018	7.5
1503–2504	6.4	7018–9383	7.7
2504–4405	4.1	9383–10164	−2.6 (inversion)
4405–4725	8.8	10164–11116	−4.6 (inversion)

The same surface-to-tropopause calculation for the other profiles gives:

- St John's: $\dfrac{-1.65-(-62.05)}{11.482-0.140}=\dfrac{60.40}{11.342}=5.3\,^{\circ}\mathrm{C/km}$, lowered by the nearly isothermal bottom 3 km;
- Florida (Question 10): $\dfrac{27.2-(-75.6)}{16.0-0}=\dfrac{102.8}{16.0}=6.4\,^{\circ}\mathrm{C/km}$;
- standard atmosphere: 6.5 °C/km, by definition.

Common pitfalls

- **The double negative.** $11.05-(-51.05)=62.10$, not -40.0.
- **Getting a negative lapse rate.** If you calculate $(T_{\text{surface}}-T_{\text{tropopause}})/(z_{\text{surface}}-z_{\text{tropopause}})$, the denominator is negative ($38-9383=-9345$) and you get -6.6. The minus sign simply records that temperature *falls* as height *rises*. By convention a lapse rate is quoted as a positive number when temperature decreases with height, so always divide the temperature *drop* by the height *gain*.
- **Forgetting to convert to kilometres**, which gives 0.006,65 instead of 6.65.

10 Question 10: a tropical profile (Florida)

Question 10

Add the temperature curve for Florida to your graph (handout Table 4). Where does the tropopause occur?

Florida: 16,000 m −75.6 °C

Height (m)	Sea level	2000	4000	6000	8000	10000	12000	14000	16000	18000
Temperature (°C)	27.2	15.2	6.1	−6.7	−17.9	−32.6	−50.5	−64.5	−75.6	−73.3

Table 4 of the handout: temperature drop in the troposphere for Florida.

What Question 10 means in ordinary words

Plot the Florida temperature measurements on the same axes as the other profiles. Then use the same tropopause test from Question 8: move upward until the temperature stops decreasing. Florida is included so that you can compare a deep, warm low-latitude troposphere with the shallower mid-latitude profiles.

10.1 Why a warm tropical surface can have a very cold tropopause

Florida begins far warmer than the other profiles, yet its tropopause is colder because the troposphere is much deeper. The profile therefore provides a useful warning: surface temperature and tropopause temperature do not vary in the same direction. What matters is how many kilometres the air continues to cool before stable stratospheric conditions are reached.

Method, step by step

1. Reuse the *same axes* as Question 6. Do not make a separate Florida graph, because the purpose is comparison.
2. Each Table 4 pair is again $(x, y) = (T, z)$. For example, 10,000 m and −32.6 °C becomes $(-32.6, 10000)$; 16,000 m and −75.6 °C becomes $(-75.6, 16000)$.
3. Plot all ten Florida points in a new colour/symbol (for example purple) and join them in ascending height order with straight lines.
4. Apply the same test as in Question 8: going up from the ground, find the first level above which the temperature no longer decreases.
5. Once you locate that turn on the graph, read the exact height and temperature from Table 4 rather than estimating the decimals from the graph.

Worked solution

The Florida temperatures fall at every level up to 16,000 m (−75.6 °C) and then rise to −73.3 °C at 18,000 m. The tropopause is therefore at 16,000 m, −75.6 °C.

Because handout Table 4 gives temperatures only every 2,000 m, the true tropopause could lie anywhere between about 15,000 m and 17,000 m; 16,000 m is the best estimate these data allow. A height of 16 to 17 km is typical of the tropical tropopause.

What stands out. Florida has by far the warmest surface (27.2 °C) but the *coldest* tropopause (−75.6 °C) of all four profiles, because its troposphere is so much deeper: the air keeps cooling, at a very ordinary lapse rate of about 6.4 °C/km, for 16 km instead of 9 to 11 km. The completed graph for the whole practical is shown in Figure 10.1.

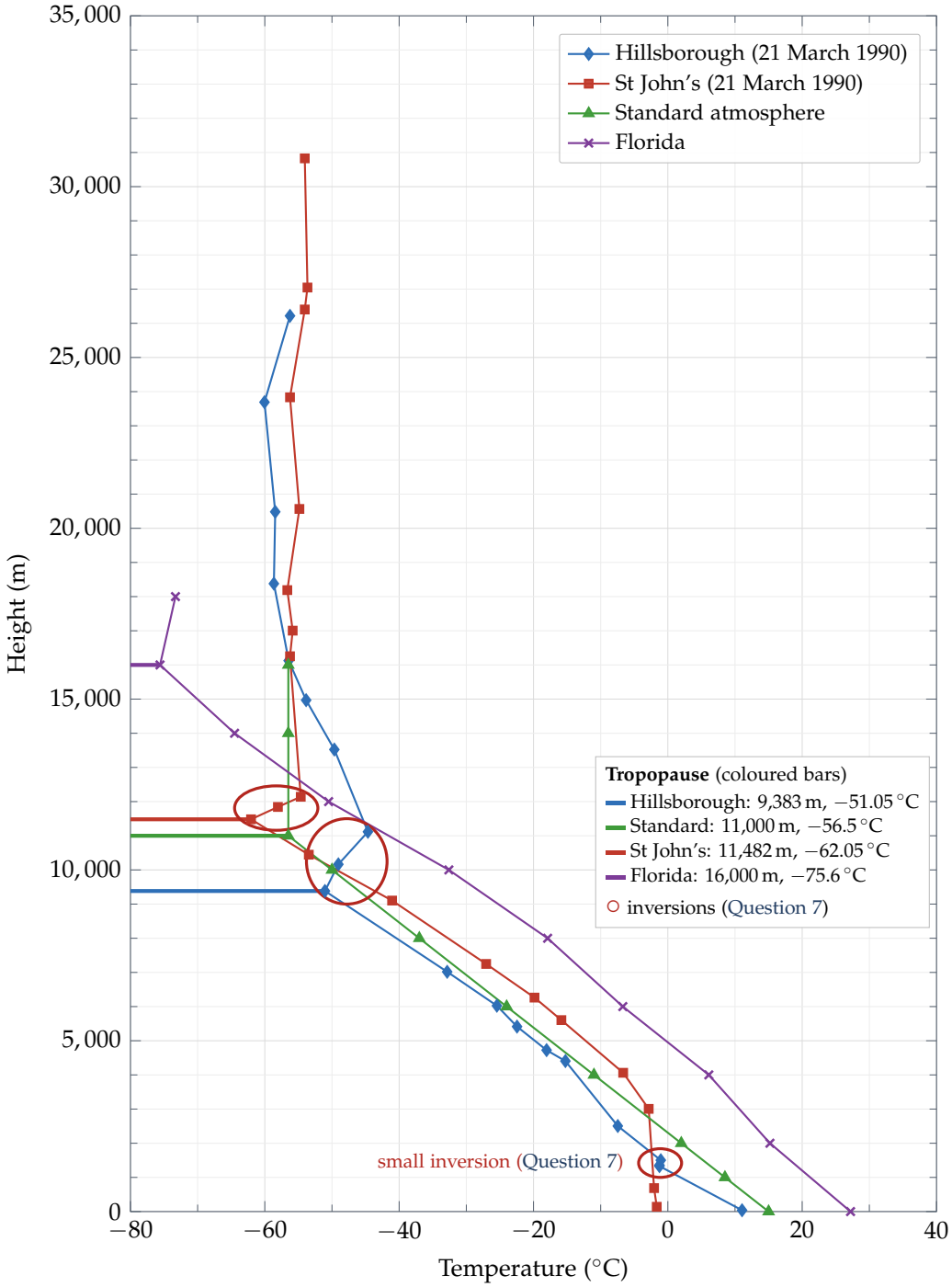


Figure 10.1: The completed graph for Questions 6 to 10: temperature profiles for Hillsborough and St John's (21 March 1990), the standard atmosphere and Florida. Coloured bars on the left mark the tropopause of each profile (Questions 8 and 10); red ellipses mark the inversion layers (Question 7).

11 Question 11: is the troposphere the same thickness everywhere?

Question 11

Examining all your graphs, do you think the troposphere is the same thickness across the globe? Where do you think it is thickest and why is that?

What Question 11 means in ordinary words

Compare the tropopause heights you have just found. If the top of the troposphere occurs at different heights in different places, then the troposphere does not have one fixed global thickness. The question then asks for the physical reason the tropical troposphere is generally deeper: warmer air columns occupy more vertical space, and strong tropical convection mixes air to greater heights.

11.1 From the plotted evidence to the physics

We now use the four tropopause heights as evidence, then explain the pattern with thermal expansion, convection and the hypsometric equation. The hypsometric derivation turns the qualitative statement 'warm air columns are taller' into a quantitative relationship between layer thickness, pressure ratio and mean absolute temperature.

Things to think about

- Compare the tropopause heights you found in Questions 8 and 10. Which profile comes from the lowest latitude, and which from the highest?
- What happens to air when it is heated (Section 2.2)? What does that do to the height of a whole column of air?
- Where on Earth is the surface heated most strongly, and what does strong surface heating do to the air above it?

Model answer

No. The tropopause was at 16,000 m over Florida, about 11,000 m to 11,500 m in the standard atmosphere and at St John's, and only 9,383 m over Hillsborough, so the troposphere is deepest at low latitudes and becomes shallower towards the poles. It is thickest over the equator and the tropics (about 16 to 18 km) and thinnest over the poles (about 8 km). This is because the tropical surface is the warmest: warm air expands, so a warm column of air is taller than a cold one, and the strong surface heating drives vigorous convection (towering thunderstorm clouds) that carries warm, moist air and mixes the troposphere up to much greater heights before the stable stratosphere stops it.

Explanation

The evidence. The four profiles are arranged from low to high latitude in Figure 11.1. The deeper the troposphere, the higher and colder the tropopause.

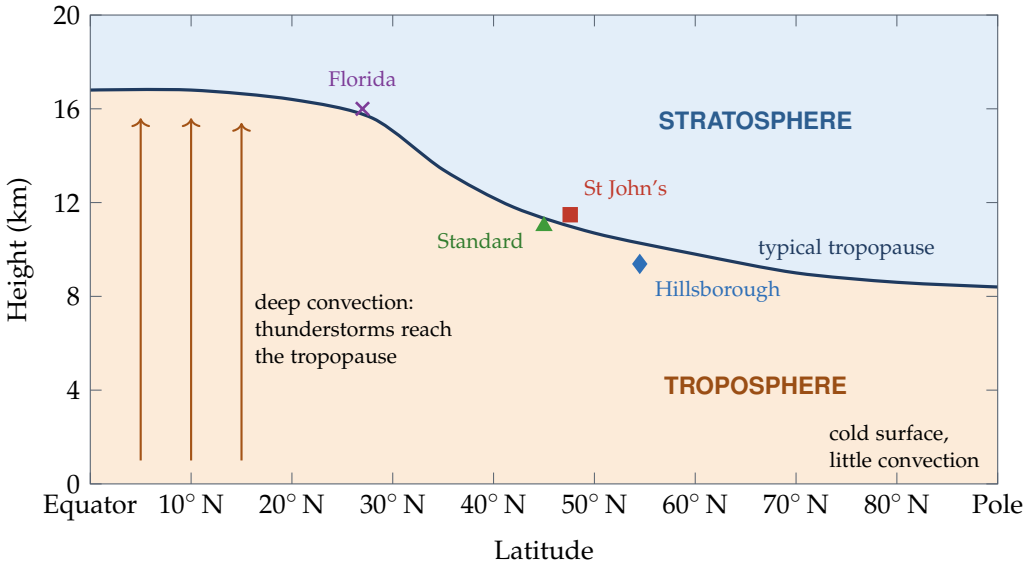


Figure 11.1: Schematic cross-section of the troposphere from the equator to the North Pole. The curve shows the typical height of the tropopause; the symbols show the tropopause heights found in Questions 8 and 10, plotted at the latitude of each profile (the standard atmosphere represents average mid-latitude conditions and is plotted at about 45° N; Florida spans roughly 25° to 31° N). Individual soundings scatter around the typical curve because the tropopause also moves up and down with the weather.

Reason 1: warm air columns are taller. Part A showed that the rate at which pressure falls with height depends on the density of the air, and that warm air is less dense (Section 2.2). In a warm column, therefore, you must climb further to pass through the same mass of air, and every pressure level lies higher up. The whole troposphere is, in effect, inflated by heating. You can see this in the soundings themselves: the 500 hPa level was at 5,604 m over St John's, where the air above 1,700 m was warmer, but only at 5,414 m over Hillsborough, where it was colder.

Reason 2: convection mixes the air to great heights. The tropopause marks the top of the layer that is stirred by rising and sinking air. In the tropics the surface receives the most intense sunshine all year round, and air heated at the surface rises in huge convective clouds. As the water vapour in that air condenses it releases latent heat, which keeps the rising air warmer than its surroundings, so it can rise all the way to 15 to 18 km before the stable stratosphere stops it. Near the poles, the surface is cold, convection is weak, and the layer of well-stirred air is much shallower.

It also changes with season and weather. The tropopause is higher in summer than in winter, and it rises and falls from day to day: it is low in cold air masses and beneath depressions and higher beneath warm anticyclones. The low tropopause over Hillsborough on 21 March 1990 (only 9,383 m) reflects the cold air that lay over Ireland above 1,700 m that day, as well as its latitude.

Going further: deriving the hypsometric equation step by step

The statement 'warm air columns are taller' can be quantified exactly enough for weather work. The result is the **hypsometric equation**. It comes from the same two starting equations used earlier: the hydrostatic equation, Equation (2.62), and the ideal gas law, Equation (2.81).

Step 1: substitute the gas law into hydrostatic balance.

$$\begin{aligned}\frac{\mathrm{d}p}{\mathrm{d}z} &= -\frac{p}{R_\mathrm{d}T}g \\ &= -\frac{pg}{R_\mathrm{d}T}.\end{aligned} \tag{11.1}$$

Step 2: solve for the small height change $\mathrm{d}z$**.** Multiply by $\mathrm{d}z$ and divide by p:

$$\begin{aligned}\mathrm{d}p &= -\frac{pg}{R_\mathrm{d}T}\,\mathrm{d}z, \\ \frac{\mathrm{d}p}{p} &= -\frac{g}{R_\mathrm{d}T}\,\mathrm{d}z.\end{aligned} \tag{11.2}$$

Now rearrange for $\mathrm{d}z$:

$$\boxed{\mathrm{d}z = -\frac{R_\mathrm{d}T}{g}\frac{\mathrm{d}p}{p}.} \tag{11.3}$$

The minus sign is sensible: moving upward means $\mathrm{d}z > 0$ while pressure decreases, so $\mathrm{d}p < 0$.

Step 3: accumulate the height between two pressure surfaces. Let level 1 be the lower level, with pressure p_1 and height z_1, and level 2 the upper level, with $p_2 < p_1$ and height $z_2 > z_1$. Integrate:

$$\int_{z_1}^{z_2}\mathrm{d}z = -\frac{R_\mathrm{d}}{g}\int_{p_1}^{p_2}T\frac{\mathrm{d}p}{p}. \tag{11.4}$$

The exact integral contains the temperature profile $T(p)$. Define an appropriate layer-mean absolute temperature $\overline{T}$ so that it can be taken outside the integral. For the simple treatment here,

$$\int_{z_1}^{z_2}\mathrm{d}z \approx -\frac{R_\mathrm{d}\overline{T}}{g}\int_{p_1}^{p_2}\frac{\mathrm{d}p}{p}. \tag{11.5}$$

Step 4: evaluate both integrals.

$$\int_{z_1}^{z_2}\mathrm{d}z = z_2 - z_1, \tag{11.6}$$

and

$$\int_{p_1}^{p_2}\frac{\mathrm{d}p}{p} = \ln p_2 - \ln p_1 = \ln\left(\frac{p_2}{p_1}\right). \tag{11.7}$$

Therefore

$$z_2 - z_1 = -\frac{R_\mathrm{d}\overline{T}}{g}\ln\left(\frac{p_2}{p_1}\right). \tag{11.8}$$

Using $-\ln(p_2/p_1) = \ln(p_1/p_2)$,

$$\boxed{z_2 - z_1 = \frac{R_\mathrm{d}\overline{T}}{g}\ln\left(\frac{p_1}{p_2}\right).} \tag{11.9}$$

What it says physically. For fixed pressure surfaces p_1 and p_2, the logarithmic pressure ratio is fixed, as are R_d and g. Therefore

$$\boxed{z_2 - z_1 \propto \overline{T}.} \tag{11.10}$$

A warm layer occupies a greater vertical thickness than a cold layer containing the same pressure change. This is the precise mathematical meaning of 'warm columns are taller'.

A numerical comparison. For the layer from 1,000 hPa to 500 hPa,

$$\ln\left(\frac{1000}{500}\right) = \ln 2 \approx 0.693 \tag{11.11}$$

and

$$\frac{R_{\mathrm{d}}}{g} = \frac{287}{9.81} \approx 29.3\,\mathrm{m/K}. \tag{11.12}$$

Hence

$$\Delta z \approx (29.3)(0.693)\overline{T} \approx 20.3\overline{T}\ \mathrm{m}. \tag{11.13}$$

If $\overline{T} = 285\,\mathrm{K}$,

$$\Delta z \approx 20.3(285) = 5{,}790\,\mathrm{m}; \tag{11.14}$$

if $\overline{T} = 250\,\mathrm{K}$,

$$\Delta z \approx 20.3(250) = 5{,}080\,\mathrm{m}. \tag{11.15}$$

The same pair of pressure surfaces is therefore about 710 m farther apart in the warmer column.

Symbol guide.

Symbol	Unit	Meaning
z_1, z_2	m	Geometric heights of the lower and upper pressure surfaces.
p_1, p_2	Pa or hPa	Pressures at the two levels; $p_1 > p_2$. Only their ratio enters the logarithm, so either pressure unit may be used consistently.
$\overline{T}$	K	Mean absolute temperature of the layer.
R_{d}	$\mathrm{J/(kg\,K)}$	Specific gas constant for dry air.
g	$\mathrm{m/s^2}$	Gravitational acceleration.

A common misconception

The troposphere is not deeper at the equator because of the Earth's rotation or its equatorial bulge. Heights here are measured above the local surface, and the difference of 8 to 10 km between tropical and polar tropopauses is due to temperature and convection, as explained above.

12 Question 12: why pressure falls more slowly high up

Question 12

In St John's, how much is the pressure drop between 140 m and 3,006 m? 305 hPa

In St John's, how much is the pressure drop between 23,834 m and 26,404 m? 10 hPa

Seeing that the difference in altitude is almost the same, why is the difference in pressure so dissimilar?

What Question 12 means in ordinary words

The two height intervals are almost the same thickness, but the first loses hundreds of hPa while the second loses only about 10 hPa. The key is to distinguish an **absolute pressure drop** from a **fractional pressure drop**. Near the ground the starting pressure is large, so losing roughly one third of it is hundreds of hPa. High in the atmosphere the starting pressure is tiny, so losing roughly one third is only a few hPa. The hydrostatic reason is that high-altitude air is much less dense, so the same geometric thickness contains far less mass.

12.1 Equal height intervals do not imply equal pressure drops

The two St John's layers have nearly the same geometric thickness, but they contain radically different masses of air. We compare absolute and fractional pressure changes, connect both to the hydrostatic equation, and show why a logarithmic pressure axis reveals the near-exponential structure.

Method, step by step

1. Find the two pairs of heights in handout Table 3b and read off the pressure at each.
2. Subtract: pressure at the lower level minus pressure at the upper level.
3. Compare the two drops, and also the two height differences, then explain using Sections 1.2 and 2.2.

Worked solution

Lower layer: the pressure falls from 1,005 hPa at 140 m to 700 hPa at 3,006 m, a drop of $1005 - 700 = 305$ hPa over a height gain of $3006 - 140 = 2{,}866$ m.

Upper layer: the pressure falls from 30 hPa at 23,834 m to 20 hPa at 26,404 m, a drop of $30 - 20 = 10$ hPa over a height gain of $26404 - 23834 = 2{,}570$ m.

Layer (m)	Height gain (m)	Pressure drop (hPa)	Drop per km (hPa)	Fraction lost
140 to 3006	2866	305	106	$305/1005 = 30\,\%$
23834 to 26404	2570	10	3.9	$10/30 = 33\,\%$

The two layers are of similar thickness (2.9 and 2.6 km), yet the pressure falls about 30 times more in the lower one.

Model answer

Atmospheric pressure decreases exponentially with height: it falls rapidly near the Earth's surface and much more gradually at great heights. The underlying reason is that air is compressible. Pressure is the weight of the air above, and the pressure drop across a layer equals the weight of the air in that layer. Near the surface there is a lot of air pressing down from above, so the air is compressed and dense, and a 2.9 km layer contains a great deal of air. High up there is little air above, so the air is hardly compressed, and a layer of the same thickness contains very little air, so crossing it reduces the pressure only slightly.

Explanation

Absolute versus relative drops. Look at the last column of the table above: each layer removes roughly a *third* of the pressure that was present at its base (30 % and 33 %). This is the defining feature of exponential decay, just as with the halving rule: equal steps in height remove equal *fractions*, not equal *amounts*. At the surface, a third of 1,005 hPa is about 300 hPa; at 24,000 m, a third of 30 hPa is only 10 hPa. Figure 12.1 shows the same thing graphically: on a normal (linear) pressure scale the profile curves sharply, while on a logarithmic scale, where equal distances represent equal *ratios*, it is close to a straight line.

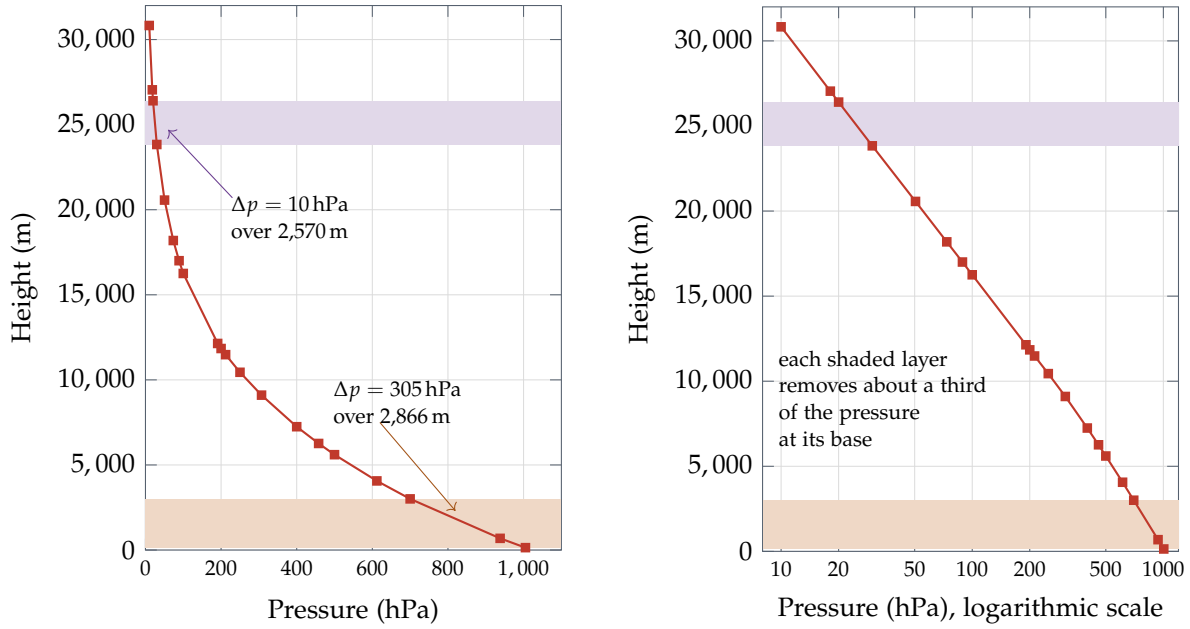
Density. Using the gas law (Section 2.2), the air near the ground over St John's had a density of about 1.29 kg/m^3, while at 24 to 26 km it was only about 0.03 kg/m^3 to 0.05 kg/m^3, some 30 times less. By the hydrostatic equation (Section 2.2), the pressure drop across a layer is proportional to the density of the air in it, which is why the drops differ by a factor of about 30.

Going further: the data obey the hypsometric equation

For an exponential decrease, $p_2 = p_1 e^{-\Delta z/H}$, so each layer's scale height can be measured as $H = \Delta z / \ln(p_1/p_2)$:

$$\text{lower layer: } H = \frac{2866}{\ln(1005/700)} = 7{,}925\,\text{m}, \qquad \text{upper layer: } H = \frac{2570}{\ln(30/20)} = 6{,}338\,\text{m}. \tag{12.1}$$

Theory (Section 1.2) says $H = R_\text{d}\overline{T}/g = 29.3\,\overline{T}$. The average temperature of the lower layer was about $-2.3\,^\circ$C (270.8 K), giving $H = 7{,}930$ m; that of the upper layer was about $-55.2\,^\circ$C (218.0 K), giving $H = 6{,}380$ m. The measured and predicted values agree to within one percent, and the difference between the two layers is explained entirely by their temperatures: the colder upper air is more compressed, so the pressure falls a little faster there *in relative terms*.



(a) Linear pressure scale.

(b) Logarithmic pressure scale.

Figure 12.1: Pressure against height over St John's on 21 March 1990 (handout Table 3b), with the two layers of Question 12 shaded. (a) On a linear scale, the pressure falls steeply near the ground and very slowly high up. (b) On a logarithmic scale the profile is almost a straight line, showing that equal height steps give nearly equal *fractional* decreases in pressure: the signature of exponential decay.

13 Question 13: why doesn't the cold air aloft sink?

Question 13

Thinking of your previous answer, if cold air is supposed to sink, why is the colder air of the upper troposphere not flowing down to the surface?

What Question 13 means in ordinary words

'Cold air sinks' is only useful when two neighbouring parcels are compared at the *same pressure*. Air ten kilometres above the ground is cold partly because it has expanded in very low pressure. If that parcel is pushed downward, the increasing surrounding pressure compresses it and makes it warmer. Whether it keeps sinking depends on whether, at its new level, it is denser or lighter than the surrounding air. The correct test is therefore a **local density comparison after displacement**, not a comparison between the absolute temperatures of upper air and surface air.

13.1 Replace the slogan 'cold air sinks' with the actual buoyancy test

Absolute temperature at two very different pressures is not enough to determine which air should rise or sink. The correct comparison is local: move a parcel to the new pressure, account for adiabatic compression or expansion, and compare its density with the environment there. Potential temperature packages that comparison into one variable, so we derive it explicitly.

Things to think about

- Whether air rises or sinks depends on its *density* compared with the air around it. What two quantities does density depend on (the gas law, Section 2.2)?
- From Question 12: how quickly does pressure fall with height, compared with how quickly temperature falls?
- What happens to the temperature of air that is pushed downwards (Section 7.1)?

Model answer

The phrase 'cold air sinks' only applies when we compare neighbouring parcels at the *same pressure level*. Air high in the troposphere is cold because it has expanded in the much lower pressure aloft. It cannot simply fall to the surface while keeping that low temperature and density: as a parcel descends into higher pressure it is compressed and warms adiabatically. Whether it continues to sink or returns upward depends on its density relative to the surrounding air at the same level. In a stably stratified troposphere, potential temperature generally increases with height, so a parcel displaced downward becomes warmer and less dense than its new surroundings and experiences an upward restoring buoyancy force.

Explanation

First correct the shorthand. 'Warm air rises; cold air sinks' is not a universal rule based on absolute temperature. Buoyancy depends on a **local density difference**. At the same pressure,

the gas law

$$\rho = \frac{p}{R_{\mathrm{d}} T} \tag{13.1}$$

shows that colder air is denser than warmer air, so the colder parcel tends to sink relative to the warmer parcel. But air at 10,000 m and air at the surface are not at the same pressure, so directly comparing their actual temperatures is not the correct buoyancy test.

Why the upper atmosphere can be cold and still have low density. Pressure falls enormously with height. Temperature in kelvin falls much less. For the St John's sounding:

Table 13.1: Pressure, temperature and density over St John's on 21 March 1990. Density is calculated from $\rho = p/(R_{\mathrm{d}} T)$, using pressure in pascals and temperature in kelvin.

Height (m)	Pressure (hPa)	T (°C)	T (K)	Density ($\mathrm{kg/m^3}$)	θ (°C)
140	1005	−1.65	271.50	1.290	−2.0
3006	700	−2.85	270.30	0.902	26.1
5604	500	−15.85	257.30	0.677	40.5
10446	250	−53.45	219.70	0.396	53.3
11482	212	−62.05	211.10	0.350	55.7

At 10,446 m, for example,

$$\rho = \frac{25{,}000\,\mathrm{Pa}}{(287\,\mathrm{J/(kg\,K)})(219.70\,\mathrm{K})} = 0.396\,\mathrm{kg/m^3}, \tag{13.2}$$

whereas near the surface

$$\rho = \frac{100{,}500\,\mathrm{Pa}}{(287)(271.50)} = 1.290\,\mathrm{kg/m^3}. \tag{13.3}$$

This does *not* by itself prove stability, because those densities are being compared at different pressures. It simply shows why 'upper air is cold' does not imply 'upper air must be denser than surface air'.

Now do the correct parcel thought experiment. Take the parcel at 10,446 m, where

$$T = -53.45\,^{\circ}\mathrm{C}, \qquad p = 250\,\mathrm{hPa}. \tag{13.4}$$

If we force it downward without exchanging heat, pressure around it increases, so it is compressed and warms. A rough estimate using the dry adiabatic lapse rate gives a warming of

$$(9.8\,^{\circ}\mathrm{C/km})(10.306\,\mathrm{km}) \approx 101\,^{\circ}\mathrm{C}. \tag{13.5}$$

Its temperature after being brought near the surface would therefore be roughly

$$-53.45 + 101 \approx 48\,^{\circ}\mathrm{C}, \tag{13.6}$$

not $-53.45\,^{\circ}\mathrm{C}$. The exact dry-adiabatic calculation is better made with potential temperature and gives the same qualitative result: the parcel would be much warmer than the near-surface environment and would tend to rise again.

Potential temperature makes the comparison rigorous. The potential temperature θ is the temperature a parcel would have if moved adiabatically to 1,000 hPa. The values in the last column of Table 13.1 rise strongly with height. That means the atmosphere was statically stable: a parcel displaced downward from aloft becomes warmer and less dense than the environment

at its new level, so buoyancy pushes it back upward; a parcel displaced upward from below becomes colder and denser than its new surroundings, so it tends to sink back.

The physically precise lesson is therefore:

What determines whether air rises or sinks?

A parcel rises or sinks according to its **density relative to the surrounding air at the same pressure level after displacement**. Temperature matters because it affects density, but pressure and adiabatic compression/expansion matter too. For dry unsaturated air, an atmosphere in which potential temperature increases with height is stably stratified.

Going further: potential temperature and why it is the right stability variable

Comparing the actual temperatures of air parcels at very different pressures can be misleading because compression itself changes temperature. **Potential temperature**, θ, removes that pressure effect by asking:

> What temperature would this parcel have if it were moved adiabatically to the reference pressure $p_0 = 1{,}000\,\mathrm{hPa}$?

The result is

$$\boxed{\theta = T\left(\frac{p_0}{p}\right)^{\kappa}}, \qquad \kappa = \frac{R_\mathrm{d}}{c_p} \approx 0.286. \tag{13.7}$$

Here T and θ must be in kelvin while evaluating the power.

Derivation from the adiabatic energy equation. The dry-adiabatic derivation gave

$$c_p \mathrm{d}T = \alpha \mathrm{d}p. \tag{13.8}$$

The gas law in specific-volume form is

$$p\alpha = R_\mathrm{d} T, \tag{13.9}$$

so

$$\alpha = \frac{R_\mathrm{d} T}{p}. \tag{13.10}$$

Substitute this into the adiabatic relation:

$$c_p \mathrm{d}T = \frac{R_\mathrm{d} T}{p}\mathrm{d}p. \tag{13.11}$$

Divide by T:

$$c_p \frac{\mathrm{d}T}{T} = R_\mathrm{d}\frac{\mathrm{d}p}{p}. \tag{13.12}$$

Divide by c_p:

$$\frac{\mathrm{d}T}{T} = \frac{R_\mathrm{d}}{c_p}\frac{\mathrm{d}p}{p} = \kappa\frac{\mathrm{d}p}{p}. \tag{13.13}$$

Now integrate from the parcel's actual state (T, p) to its reference state (θ, p_0):

$$\int_T^\theta \frac{\mathrm{d}T'}{T'} = \kappa \int_p^{p_0} \frac{\mathrm{d}p'}{p'}. \tag{13.14}$$

Evaluating the logarithms,

$$\begin{aligned} \ln\theta - \ln T &= \kappa(\ln p_0 - \ln p), \\ \ln\left(\frac{\theta}{T}\right) &= \kappa \ln\left(\frac{p_0}{p}\right). \end{aligned} \tag{13.15}$$

Exponentiate both sides:

$$\begin{aligned} \frac{\theta}{T} &= \exp\left[\kappa \ln\left(\frac{p_0}{p}\right)\right] \\ &= \left(\frac{p_0}{p}\right)^{\kappa}. \end{aligned} \tag{13.16}$$

Multiply by T to obtain Equation (13.7).

Why this helps with stability. Imagine two parcels at different heights. Their actual temperatures cannot be compared fairly because the upper parcel sits at lower pressure and has expanded. Potential temperature mathematically moves both parcels to the same reference pressure without adding or removing heat. Once they are compared at the same pressure, temperature is directly related to density: the parcel with the larger θ would be warmer and less dense.

For unsaturated dry air, the stability criterion is therefore

$$\boxed{\frac{\mathrm{d}\theta}{\mathrm{d}z} > 0 \quad \Rightarrow \quad \text{statically stable.}} \tag{13.17}$$

Potential temperature increasing upward means that if an air parcel is displaced vertically and moved adiabatically, buoyancy tends to restore it toward its original level. A vertically uniform θ corresponds to neutral dry-adiabatic stratification; θ decreasing upward indicates dry convective instability.

Important refinement to the phrase 'cold air sinks'. Air does not decide whether to rise or sink by comparing its temperature with air thousands of metres away at a completely different pressure. Buoyancy is a *local comparison*: after a parcel is displaced to a new level, compare its density with the environment at the *same pressure level*. Potential temperature is the convenient variable that keeps track of that comparison under adiabatic displacement.

Common pitfalls

- **Do not say simply 'upper air floats because it is less dense'.** Density values at different pressures cannot by themselves diagnose buoyant stability. The correct comparison is local and is most cleanly expressed using potential temperature.
- **Do not imagine that descending air keeps the same temperature.** Compression during descent warms it; expansion during ascent cools it.
- **Do not treat 'cold air sinks' as an absolute rule.** It is shorthand for colder air being denser than neighbouring warmer air when they are compared at the same pressure.

14 Question 14: Hillsborough versus St John's

Question 14

Compare the surface temperatures of St John's and Hillsborough. Although Northern Ireland is located at a higher latitude, why do you think the surface temperature is higher? Do you think that this difference is the same throughout the year?

What Question 14 means in ordinary words

First compare the two measured near-surface temperatures on 21 March 1990. Then explain why latitude alone does not control them. Hillsborough is influenced by relatively mild North Atlantic water and prevailing westerly air flow, whereas St John's is beside the cold Labrador Current and can receive air that has crossed a cold North American continent. Finally, separate **one day's sounding** from **seasonal climate**: the contrast is generally strongest in the colder part of the year and much smaller in summer.

14.1 From one sounding to regional climate

The sounding gives a one-day surface contrast. To explain it, we separate altitude, latitude, ocean-current and air-mass effects, then ask how those controls change seasonally. The answer therefore distinguishes the observations on 21 March 1990 from the longer-term climatological mechanism.

Things to think about

- Read the two surface temperatures from the first row of handout Table 3a and handout Table 3b, and find the latitude of each station on a map.
- Could the difference in station height explain the temperature difference? Use a typical lapse rate to check.
- Which ocean currents flow past each place, and from which direction do the prevailing winds blow at these latitudes?
- How do land and sea temperatures change between winter and summer?

Model answer

On 21 March 1990 the surface temperature was 11.05 °C at Hillsborough and −1.65 °C at St John's, so Northern Ireland was 12.7 °C warmer even though it lies about 7° further north (54.5° N compared with 47.6° N). Both stations are close to sea level and both are near the Atlantic, so height and distance from the sea do not explain the difference. The difference comes from the ocean currents and the prevailing westerly winds. Northern Ireland is bathed by the warm North Atlantic Current, the continuation of the Gulf Stream, and the westerly winds carry mild air from over this warm water onto the land, which keeps winters mild. Newfoundland receives no warming from the Gulf Stream; quite the opposite, its coast is washed by the cold Labrador Current flowing south from the Arctic, and in winter the westerlies reach it after crossing the frozen North American continent.

The difference is not the same throughout the year. It is largest in winter and early spring, when the continent is coldest and the warm ocean matters most, and it becomes much smaller in summer, when St John's can actually be as warm as, or warmer than, Northern Ireland.

Explanation

Ruling out the obvious explanations. Latitude would lead us to expect Hillsborough to be the colder station. Height does not help either: the St John's station is only about 100 m higher, which at a typical lapse rate would make it cooler by less than 1 °C, not 12.7 °C. Both places are coastal. Something else must be supplying heat to Northern Ireland, or removing it from Newfoundland.

Ocean currents and winds. Figure 14.1 shows the setting. The Gulf Stream carries warm water from the Gulf of Mexico and the Caribbean north-eastwards across the Atlantic, where it continues as the North Atlantic Current towards Ireland, Britain and Norway. The prevailing winds at these latitudes blow from the west, so Ireland receives air that has spent days crossing this relatively warm ocean. St John's sits on the *western* side of the ocean, where the cold Labrador Current flows south from the Arctic past Newfoundland, carrying sea ice and icebergs in late winter and spring. And because the winds blow from the west, the air arriving at St John's in winter has usually just crossed the snow-covered interior of Canada. The contrast is therefore between a place downwind of a warm ocean and a place downwind of a cold continent and beside a cold current.

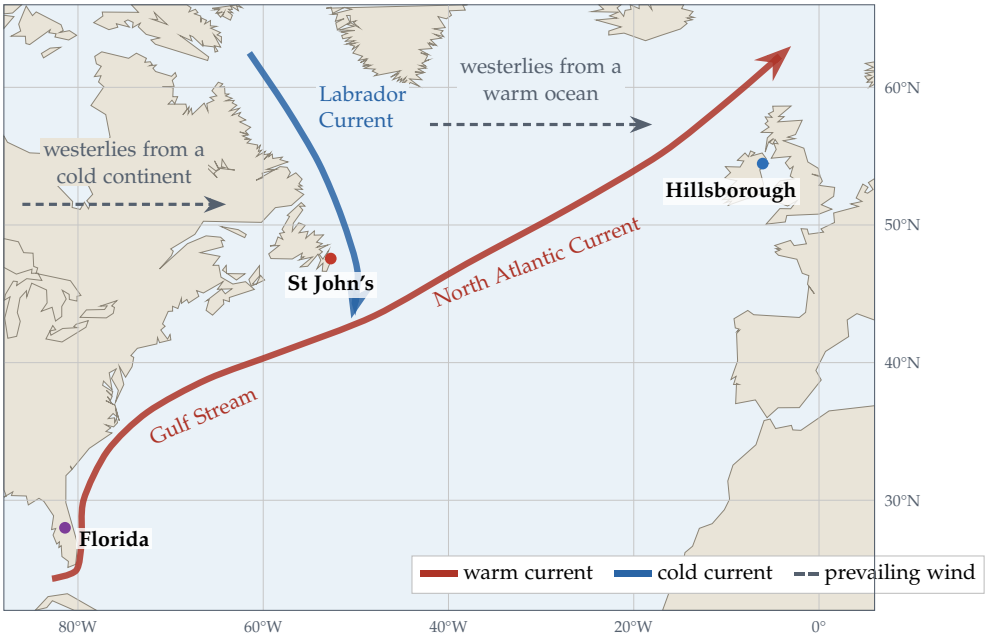


Figure 14.1: The North Atlantic setting of Question 14 (ocean currents and winds drawn schematically). Hillsborough lies downwind of the warm North Atlantic Current; St John's lies beside the cold Labrador Current and downwind of the North American continent.

Through the year. In mid-winter, average temperatures in Northern Ireland stay around 4 °C to 6 °C, while those in St John's are several degrees below freezing, a difference of roughly 10 °C. In summer the land warms up, the continental air reaching Newfoundland is warm rather than

frozen, and the difference largely disappears: both places have mid-summer averages of roughly 15 °C to 16 °C, and afternoon temperatures in St John's are often a little higher. A warm ocean has its biggest warming effect in winter, when the land would otherwise be cold; in summer the ocean can even have a slight cooling effect.

Going further: look above the surface

Your graph holds a surprise. Above about 1,700 m, the air over St John's was *warmer* than the air over Hillsborough, by 6 to 8 °C all the way up to the tropopause (Question 6). The cold at St John's was confined to a shallow layer near the ground, the nearly isothermal layer you found in Question 7, sitting over cold sea and land. This is a reminder that ocean currents act on the atmosphere from below: they warm or cool the lowest layer of air most strongly. It is also a reminder that a single sounding is a snapshot of the *weather* on one day, not the *climate*; the long-term averages quoted above are what establish the pattern.

Scientists continue to debate how much of the mildness of north-west Europe is due to the heat carried north by the ocean, and how much is due simply to the atmosphere: prevailing westerly winds that blow off a large ocean onto Europe, but off a large, cold continent onto eastern North America. Both play a part.

Common pitfalls

- **Attributing the difference to latitude or height.** Latitude points the other way, and the height difference is only about 100 m.
- **Mentioning the Gulf Stream without the winds.** Warm water on its own does not warm the land; it is the westerly winds that bring the warmth ashore. Mention both.
- **Forgetting the second part of the question.** Say how the difference changes through the year, and why.

Appendix A Technical questions that arise while doing this practical

This section collects the short questions that commonly appear while working through the worksheet. The detailed derivations remain in the question where each idea is first needed; the links below take you back to them.

What exactly does 'percentage of atmosphere above' mean?

It means the fraction of the *total atmospheric mass* lying above your chosen height, not the local density at that height. At sea level it is defined as 100 %; at the handout's half-height of 5.6 km, 50 % remains above. Under hydrostatic balance this mass fraction is approximately the pressure fraction, as shown by Equations (2.46) and (2.47).

What does density at one level mean?

Density is local mass per local volume, $\rho = m/V$. Imagine a $1\,\mathrm{m} \times 1\,\mathrm{m} \times 1\,\mathrm{m}$ cube centred at that level: density tells you how many kilograms of air are inside that cubic metre. This is different from asking how much of the *whole atmosphere* lies above the level. See Equations (1.6) and (1.8).

Why do we multiply by one half at each 5.6 km step?

Because the rule says *half of whatever remains*. Thus new amount = old amount $\times \frac{1}{2}$. Repeated multiplication creates the power law in Equations (1.15) and (1.17). Subtracting 50 percentage points would incorrectly produce $100 \to 50 \to 0$.

Where does $P = 100(0.5)^{z/5.6}$ come from?

From two statements: after n halvings, $P = 100(0.5)^n$ (Equation (1.15)); and the number of halvings at height z is $n = z/5.6$ (Equation (1.16)). Substitute the second into the first to obtain Equation (1.17).

Why does pressure decrease upward? What does 'supporting the air above' actually mean?

Pressure at a level is the weight of the overlying air per unit area. Across one thin slab, the pressure below is slightly larger than the pressure above, and that difference balances the slab's own weight: see Equations (2.44) and (2.45). In the differential limit this becomes the hydrostatic equation, Equation (2.62).

Where does scale height come from?

Combining hydrostatic balance with the ideal gas law gives $\mathrm{d}p/\mathrm{d}z = -(g/R_\mathrm{d}T)p$ (Equation (2.87)). Under the temporary isothermal assumption, the recurring combination $R_\mathrm{d}T/g$ has units of length, so we *define* $H = R_\mathrm{d}T/g$ (Equation (2.89)). This converts the equation into $\mathrm{d}p/\mathrm{d}z = -p/H$. One scale height reduces pressure to $1/e \approx 0.368$ of its starting value; a half-height is $H \ln 2$.

How do we go from $H = R_\mathrm{d}T/g$ to $1/H = g/(R_\mathrm{d}T)$?

Take the reciprocal of both sides: $1/H = 1/(R_\mathrm{d}T/g) = g/(R_\mathrm{d}T)$. The complete algebra is written in Equations (2.90) to (2.92).

Where does the 5.6 km half-height come from, and where does the 3 °C come from?

For this practical, 5.6 km is a stated rounded teaching rule. In an isothermal model the half-height is $z_{1/2} = H \ln 2$ (Equation (2.108)). Setting that equal to 5.6 km implies $H \approx 8.08$ km; then $T = gH/R_d \approx 276$ K ≈ 3 °C. The direction is therefore *5.6 km rule* → *equivalent scale height* → *effective model temperature*, not 'the atmosphere is actually 3 °C at 5.6 km'.

Why is the standard atmosphere about −21 °C at 5.6 km if the simplified model gives +3 °C?

The −21.4 °C from Equation (2.112) is the *local* standard-atmosphere temperature at that one height. The +3 °C is a single constant temperature fitted to an imaginary isothermal atmosphere so that its pressure halves at exactly 5.6 km. The real variable-temperature pressure equation is Equations (2.113) and (2.114). These are different questions, so the temperatures need not match.

Why do logarithms appear when solving the halving formula backwards?

The unknown height sits in an exponent. Logarithms are the inverse operation of exponentiation, so they bring that exponent down where ordinary algebra can isolate it. The complete route from $P = 100(1/2)^{z/5.6}$ to $z = 5.6 \log_2(100/P)$ is Equations (1.19) to (1.21) and (1.23).

When plotting the radiosonde data, which columns do I use?

For Question 6, plot $(x, y) = (\text{temperature}, \text{height})$. In Tables 3a and 3b, that means third column on the x-axis and second column on the y-axis. Pressure is not an axis in this graph. Worked coordinate examples are Equations (6.1) and (6.2) and Figure 6.1.

Why join the radiosonde points with straight lines?

The listed sounding levels are discrete measurements, including significant levels selected where the vertical profile changes slope. Straight segments between neighbouring levels represent the information actually supplied without inventing extra curvature between measurements. This differs from Question 1, where the stated halving rule defines a continuous exponential curve and a smooth curve is appropriate.

How do I infer an inversion and a tropopause from the graph?

Read upward. A segment that goes up-left is cooling with height; a segment that goes up-right is warming with height, hence an inversion (Question 7). The tropopause is the first major level where sustained tropospheric cooling stops - the first major corner where the profile turns vertical or rightward (Question 8). Use the graph to locate it, then use the table for the exact height and temperature.

Appendix B Key equations and constants

Every relationship below links directly to its numbered equation and to the question section containing the full **Where this comes from** derivation: it shows the more basic law (or laws) the equation is built from, works through the derivation step by step, lists every symbol with its meaning and unit, and explains where every number in it was measured, defined or agreed by convention. This appendix is only a *quick-reference summary* of that fuller treatment — if a value or a symbol is not self-explanatory here, the cited section has the complete story.

Relationship	Equation	Full derivation
Hydrostatic balance	$\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g$; finite form $\Delta p = -\rho g \Delta z$	Equations (2.58) and (2.62); Section 2.2
Ideal gas law	$p = \rho R_\mathrm{d} T$ or $\rho = \frac{p}{R_\mathrm{d} T}$	Equation (2.81); Section 2.2
Exponential atmosphere	$p(z) = p_0\, e^{-z/H}$, $H = \frac{R_\mathrm{d} T}{g}$	Equations (2.89) and (2.103); Section 1.2
Halving rule	$\%\text{ above} = 100\left(\frac{1}{2}\right)^{z/5.6}$, $p = 1{,}000\,\mathrm{hPa}\left(\frac{1}{2}\right)^{z/5.6}$	Equations (1.10) and (1.11); Section 1.2
Halving rule, inverted	$z = 5.6 \log_2\left(\frac{100}{P}\right)$	Equation (1.18); Section 1.2
Dalton's law	$p_\mathrm{gas} = x_\mathrm{gas}\, p$, $p_{\mathrm{O}_2} = 0.21\, p$	Equations (4.5) and (4.6); Section 4.2
Effective oxygen percentage	$21\,\% \times \frac{p}{p_0}$	Equation (4.8); Section 4.2
Lapse rate (finite/local)	$\Gamma = -\Delta T/\Delta z$; $\Gamma = -\mathrm{d}T/\mathrm{d}z$	Equations (5.1) and (5.3); Section 5.3
Dry adiabatic lapse rate	$\Gamma_\mathrm{d} = g/c_p \approx 9.8\,^\circ\mathrm{C/km}$	Equation (7.21); Section 7.1
Standard atmosphere (to 11 km)	$T(z) = 15 - 6.5z$ (z in km)	Equation (5.7); Section 5
Hypsometric equation	$z_2 - z_1 = \frac{R_\mathrm{d} \overline{T}}{g} \ln \frac{p_1}{p_2}$	Equation (11.9); Section 11
Potential temperature	$\theta = T\left(\frac{1000}{p}\right)^{0.286}$	Equation (13.7); Section 13

Symbols and abbreviations used in these equations

Symbol	SI unit	Meaning
p, p_0, p_1, p_2	Pa (hPa here)	Pressure; total, sea-level, and at two named levels.
z, z_1, z_2, Δz	m	Height above sea level, heights of two levels, and a height difference.
ρ	$\mathrm{kg/m^3}$	Density (mass per unit volume) of air.
T, $\overline{T}$, θ	K	Absolute temperature; the mean temperature of a layer; potential temperature.
g	$\mathrm{m/s^2}$	Acceleration due to gravity.
R_d	$\mathrm{J/(kg\,K)}$	Specific gas constant for dry air.
R^*	$\mathrm{J/(mol\,K)}$	Universal (molar) gas constant, the same for every gas.
M_d	kg/mol	Mean molar mass of dry air.
n, n_i, x_i	mol, mol, —	Amount of substance (moles); moles of one gas i; that gas's mole fraction (= % by volume).
c_p, c_v	$\mathrm{J/(kg\,K)}$	Specific heat of air at constant pressure, and at constant volume.
$\alpha = 1/\rho$	$\mathrm{m^3/kg}$	Specific volume: the volume of one kilogram of air.
H	m	Scale height: the height gain over which pressure falls by a factor of e.
Γ, Γ_d	K/m (°C/km here)	Lapse rate (general), and the dry adiabatic lapse rate.
$\kappa = R_\mathrm{d}/c_p$	dimensionless	Poisson's constant, the exponent in the potential-temperature formula (≈ 0.286).
e	—	Euler's number, ≈ 2.71828, the base of the natural logarithm ln.
N_A, k_B	$\mathrm{mol^{-1}}$, J/K	Avogadro constant and Boltzmann constant (fixed SI constants; $R^* = N_A k_B$).
hPa, Pa, mbar	—	Hectopascal, pascal, millibar: units of pressure (1 hPa=100 Pa=1 mbar).
K, °C	—	Kelvin (absolute temperature scale) and degrees Celsius.

Where every constant comes from

Constant or conversion	Value	Where it comes from
Acceleration due to gravity, g	$9.81\,\mathrm{m/s^2}$	Measured; rounded from the internationally agreed 'standard gravity' $9.806{,}65\,\mathrm{m/s^2}$ (Section 2.2).
Universal gas constant, R^*	$8.314\,\mathrm{J/(mol\,K)}$	Fixed SI constant, $= N_A k_B$, built from the (exactly defined) Avogadro and Boltzmann constants (Section 2.2).
Mean molar mass of dry air, M_d	$0.028{,}97\,\mathrm{kg/mol}$	Calculated from the measured composition of air, Table 4.1 (Section 2.2).
Gas constant for dry air, R_d	$287\,\mathrm{J/(kg\,K)}$	Calculated: $R_\mathrm{d} = R^*/M_\mathrm{d}$ (Section 2.2); never independently measured.
Specific heat of dry air, c_p	$1{,}005\,\mathrm{J/(kg\,K)}$	Predicted by kinetic theory as $\frac{7}{2}R_\mathrm{d}$ for a diatomic gas, confirmed by laboratory measurement (Section 7.1).
Dry adiabatic lapse rate, Γ_d	$9.8\,^\circ\mathrm{C/km}$	Calculated: $\Gamma_\mathrm{d} = g/c_p$ (Section 7.1); not independently measured.
Average (normal) environmental lapse rate	$6.5\,^\circ\mathrm{C/km}$	Empirical: the global average of many real radiosonde profiles, adopted by convention in the standard atmosphere (Section 5).
Standard sea-level pressure, temperature, density	1,013.25 hPa, 15 °C (288.15 K), $1.225\,\mathrm{kg/m^3}$	Conventional values fixed by the International Standard Atmosphere; the density follows from the other two via the gas law (Section 5).
Sea-level pressure used in this practical	1,000 hPa	A deliberately rounded version of 1,013.25 hPa, chosen by the handout to keep the arithmetic simple (Section 2.2).
Composition of dry air (78 % N_2, 21 % O_2, …)	see Table 4.1	Measured directly by chemical analysis of air samples (Section 4.2).
Half-height of the halving rule	5.6 km	A rounded teaching constant matching a scale height computed for an assumed uniform temperature of about 3 °C (Section 1.2).
Kelvin and Celsius	$T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$	Defined offset between the two temperature scales, fixed by international agreement (Section 2.2).
Pressure units	1 hPa = 100 Pa = 1 mbar	Definitions (exact conversions, not measurements).
Height units	1 km = 1,000 m	Definition (exact conversion).

Appendix C Mathematical notation and calculus reference

This appendix collects the mathematical language used throughout the notes. It is a reference, not a substitute for the worked explanations developed in context, especially in Question 2.

Notation	Meaning
x, z, p, T	Variables: quantities that may change. In this practical z is usually height, p pressure and T temperature.
$f(x)$, $p(z)$	Function notation: the value of output f at input x, or pressure evaluated at height z. It is not multiplication.
Δx	A finite change: final value minus initial value.
h	A generic finite input increment often used in the derivative definition; $h = \Delta x$.
$f(x+h) - f(x)$	The corresponding output change. It is generally not equal to h.
$\Delta y / \Delta x$	Average rate of change across a finite interval.
$\dfrac{\mathrm{d}y}{\mathrm{d}x}$	Derivative: instantaneous/local rate of change, defined by a limit of finite difference quotients.
$\lim_{h \to 0}$	Limit: asks what a quantity approaches as h is made arbitrarily close to zero; it does not instruct us to divide by zero.
$\mathrm{d}x$, $\mathrm{d}p$, $\mathrm{d}z$	Differentials used in calculus. They work naturally with derivative and integral notation.
$\int_a^b q(x)\,\mathrm{d}x$	Definite integral: accumulation of q from a to b, obtained as the limit of sums over thinner and thinner intervals.
e	Euler's number, about 2.71828; natural base of exponential functions.
$\ln x$	Natural logarithm, the inverse of e^x.
$\propto$	'Is proportional to'. If $y \propto x$, multiplying x by a factor multiplies y by the same factor when other relevant quantities are fixed.
$\approx$	Approximately equal to.
$\Rightarrow$	Implies / therefore.

The derivative definition to remember conceptually

$$\boxed{\frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}} \tag{C.1}$$

Read it as: 'the instantaneous rate of change is the value approached by the average rate of change as the input interval shrinks toward zero.'

The integration idea to remember conceptually

$$\boxed{\text{total accumulated change} = \int \text{local contribution}} \tag{C.2}$$

If $F'(x) = f(x)$, the Fundamental Theorem of Calculus gives

$$\boxed{\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).} \tag{C.3}$$

This is why integrating the hydrostatic pressure gradient reconstructs a pressure difference or the weight of a whole atmospheric column.

Dimensional checks used repeatedly

$$\begin{aligned} \mathrm{Pa} &= \mathrm{N/m^2}, \\ \mathrm{N} &= \mathrm{kg\,m/s^2}, \\ \rho A \Delta z: &\quad \frac{\mathrm{kg}}{\mathrm{m^3}}\mathrm{m^2 m} = \mathrm{kg}, \\ pA: &\quad \frac{\mathrm{N}}{\mathrm{m^2}}\mathrm{m^2} = \mathrm{N}, \\ \frac{\mathrm{d}p}{\mathrm{d}z}: &\quad \mathrm{Pa/m}, \\ \Gamma: &\quad \mathrm{K/km \text{ or } {}^\circ C/km}. \end{aligned} \tag{C.4}$$

Appendix D Glossary

Adiabatic Describes a change in which no heat is exchanged with the surroundings. Rising air cools adiabatically as it expands; sinking air warms adiabatically as it is compressed.

Altitude Height above mean sea level.

Anticyclone An area of high pressure, in which air sinks slowly and warms by compression; often brings settled weather and subsidence inversions.

Convection Vertical air motion driven by buoyancy. A parcel rises when it is less dense than the surrounding air at the same level and sinks when it is denser.

Dalton's law The total pressure of a mixture of gases is the sum of the partial pressures of the individual gases.

Density How much mass is contained in a given volume. For air, $1.2\,\mathrm{kg/m^3}$ means that a $1\,\mathrm{m^3}$ cube of that air contains 1.2 kg of mass.

Dry adiabatic lapse rate The rate at which rising unsaturated air cools, about $9.8\,^\circ\mathrm{C/km}$.

Effective oxygen percentage The percentage of oxygen that sea-level air would need to contain to have the same oxygen partial pressure as the air at a given height.

Environmental lapse rate The actual rate at which temperature decreases with height at a given place and time.

Exponential decay A decrease in which equal steps (of time, or here of height) always remove the same fraction of what remains.

Half-height The height gain over which pressure halves; 5.6 km in the halving rule.

Hectopascal (hPa) The unit of pressure used in meteorology; 1 hPa = 100 Pa = 1 mbar.

Hydrostatic balance The near-balance in which the vertical pressure-force difference across a thin layer supports that layer against gravity, giving $\mathrm{d}p/\mathrm{d}z = -\rho g$.

Hypoxia A shortage of oxygen in the body's tissues, caused at altitude by the low partial pressure of oxygen.

Ideal gas law The relationship $p = \rho R_\mathrm{d} T$ between the pressure, density and temperature of a gas.

Inversion A layer in which temperature increases with height.

Isothermal layer A layer in which temperature does not change with height.

Kármán line The height of 100 km, conventionally taken as the boundary between the atmosphere and space.

Lapse rate How many degrees the temperature falls for each unit of height gained, usually quoted in °C/km; $\Gamma = -\Delta T/\Delta z$.

Latent heat Heat released when water vapour condenses (or absorbed when water evaporates).

Mesosphere The layer from about 50 to 85 km in which temperature falls with height.

Mixed layer A layer near the ground that is well stirred by convection, with a lapse rate close to the dry adiabatic value.

Ozone A form of oxygen (O_3) that absorbs solar ultraviolet radiation, heating the stratosphere.

Partial pressure The share of the total gas pressure attributable to one gas species in the mixture; for ideal gases $p_i = x_i p_{\text{total}}$.

Potential temperature The temperature air would have if brought adiabatically to 1,000 hPa.

Pressure Force acting on each unit area. In hydrostatic balance, atmospheric pressure at a height equals the weight of the air above that height divided by horizontal area.

Radiosonde An instrument package carried by a balloon that measures pressure, temperature and humidity and radios the readings to the ground.

Saturated adiabatic lapse rate The rate at which rising saturated air cools, typically 4 to 7 °C/km.

Scale height The height gain over which pressure falls by a factor of e (about 2.718); roughly 8 km.

Significant level A level in a radiosonde report chosen where the profile changes slope.

Sounding The set of measurements from one radiosonde ascent; also called an upper-air profile.

Stability The tendency of the atmosphere to resist (stable) or encourage (unstable) vertical motion.

Standard atmosphere An agreed, idealised profile of temperature, pressure and density representing average mid-latitude conditions.

Standard (mandatory) pressure level A pressure level, such as 850 hPa or 500 hPa, reported by every radiosonde station.

Stratosphere The layer from the tropopause to about 50 km in which temperature is constant or increases with height.

Thermosphere The layer above about 85 km in which temperature rises steeply with height.

Tropopause The boundary between the troposphere and the stratosphere, where temperature stops decreasing with height.

Troposphere The lowest layer of the atmosphere, in which temperature generally decreases with height and in which almost all weather occurs.

Appendix E Drawing a good graph by hand

1. **Use a sharp pencil** for axes, points and construction lines, and coloured pens or pencils only for the final curves.
2. **Put the right quantity on each axis.** In this practical height always goes on the vertical axis.
3. **Choose scales with round numbers** (1, 2, 5 or 10 units per square, or multiples of these) that make the data fill most of the grid. Never use an uneven scale.
4. **Label each axis with the quantity and its unit**, for example ‘Temperature (°C)’.
5. **Plot each point accurately** as a small cross or dot. Double-check any value with four or five digits.
6. **Draw the right kind of line.** Use a single smooth curve for a continuous mathematical

relationship (Questions 1 and 2); join measured sounding data with straight lines between points (Questions 6 and 10).

7. **Use a different colour (and symbol) for each data set** and add a key.
8. **Give the graph a title** that says what it shows, where and when.
9. **When reading values from a graph**, draw light construction lines and read to the nearest half of a small square.

Appendix F Calculator techniques

Powers with decimal exponents. To work out $0.5^{1.580}$, key in `0.5`, the power key (x^y, y^x or `^`), then `1.580`, then `=`. You should get 0.3345. If your calculator needs brackets, write `0.5^(8.848/5.6)`.

Negative exponents. 2^{-n} gives the same result as 0.5^n. Use the `(-)` key for the minus sign in the exponent.

Logarithms. To invert the halving rule, use either key: $\log_2 x = \ln x / \ln 2 = \log_{10} x / \log_{10} 2$. For example, $\log_2 10 = 2.303/0.693 = 3.32$.

Negative temperatures. Use the `(-)` (change sign) key, not the subtraction key, and use brackets: `11.05 - (-51.05)` gives 62.1.

Rounding. Keep at least four significant figures in intermediate steps and round only the final answer, usually to three significant figures.

Checking. Always ask whether the answer is sensible: a percentage above must lie between 0 and 100, pressure must fall with height, and a tropospheric lapse rate is usually between about 4 and 10 °C/km.

Appendix G Further reading

- Stull, R. *Practical Meteorology: An Algebra-based Survey of Atmospheric Science*. An unusually accessible bridge between intuitive meteorology and the algebra used in labs like this one; particularly useful for pressure, temperature, stability and vertical structure.
- Ahrens, C.D. and Henson, R. *Meteorology Today: An Introduction to Weather, Climate, and the Environment*. Cengage Learning (any recent edition). A very readable introductory text; see the chapters on the composition and vertical structure of the atmosphere, and on atmospheric stability.
- Barry, R.G. and Chorley, R.J. (2010) *Atmosphere, Weather and Climate*, 9th edn. Routledge. A classic textbook for geography and environmental science students.
- Wallace, J.M. and Hobbs, P.V. (2006) *Atmospheric Science: An Introductory Survey*, 2nd edn. Academic Press. More mathematical; chapters 1 and 3 cover the vertical structure of the atmosphere, the hydrostatic equation, the hypsometric equation and potential temperature.
- NOAA, NASA and U.S. Air Force (1976) *U.S. Standard Atmosphere, 1976*. U.S. Government Printing Office, Washington, DC. The definitive description of the standard atmosphere used in Figure 5.1.
- Seager, R., Battisti, D.S., Yin, J., Gordon, N., Naik, N., Clement, A.C. and Cane, M.A. (2002) Is the Gulf Stream responsible for Europe's mild winters? *Quarterly Journal of the Royal Meteorological Society*, 128, 2563–2586. The research behind the discussion at the end of Question 14.